

Effective Causal Descent: Uniform Selection, Contextual Gluing, and Causal Histories

Monograph Version

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Monograph scope. This monograph develops the section-theoretic core, the presentation-relative diagnostic calculus, the physical Bell repair module, and the functorial and compositional extension theory. The finite algorithmic realization — input formats, typed certificate verifiers, width-based dynamic programs, the mechanics front end, and the prototype artifact — is delegated to the companion algorithmic monograph. Focused articles can be extracted from the selector/diagnostic core, the Bell-contextuality module, the computability module, and the causal-arena module.

Abstract

We develop effective causal descent as a section-theoretic account of decision-making under partial observation. The basic constraint is observational uniformity: an interface cannot support different responses to worlds it does not distinguish. From this arise four presentation-relative diagnostics: local validity, compatibility of a declared diagram, effective realisability, and bounded realisability. The labels are not invariants of bare capability predicates; they record where success first fails in a chosen local-to-global presentation. The static kernel is a fibrewise selector lemma. Compactness, Helly, join-tree, parity, support-contextuality, source-integration, and Bell-coupling results provide finite certificate and gluing tools. The obstruction datum records objectwise local success and the nested spaces $\Gamma_{\text{set}} \supseteq \Gamma_{\text{eff}} \supseteq \Gamma_{\mathcal{B}}$; terminal compression preserves the nested global spaces while erasing precisely the D-gap. The monograph develops the accompanying morphism theory: global-family functoriality, diagnostic and certificate morphisms, terminal compression as an idempotent functor, support-packaging functoriality, distributed/one-shot groupoid equivalence, arena patch-sheaf functoriality, and composition laws for constraints, static compilations, sequential composition, parallel composition, and guarded feedback. Computability examples yield the binary diagonally nonrecursive (DNR) Π_1^0 class, the Peano-arithmetic (PA) oracle spectrum, WKL Weihrauch degrees, and a Σ_3^0 -complete primitive-recursive capability index set. The causal tier proves backward induction, active-sensing presentation relativity, confluent causal-poset execution, global-recall action-recall redundancy, and a height-two causal DNR/PA transfer. The Bell module adds local-causal interpretation, Farkas certificates, finite-sample testing, and repair profiles.

Keywords: effective causal descent; presentation-relative diagnostics; constraint satisfaction and sheaf semantics; Bell nonlocality and contextuality; computable selection and Weihrauch reducibility; extensive-form games and causal protocols.

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Table 1: Principal notation for the monograph.

Symbol	Meaning
$\mathcal{C}, \mathcal{A}$	context category and assignment functor; arrows are restriction arrows
$\mathcal{A}_{\text{loc}, \varepsilon}(c)$	local admissibility set at context c and tolerance ε
$\Gamma_{\text{set}}^{\varepsilon}$	compatible locally admissible global families
$\Gamma_{\text{eff}}^{\varepsilon}, \Gamma_{\mathcal{B}}^{\varepsilon}$	effective and bounded realised subspaces of $\Gamma_{\text{set}}^{\varepsilon}$
λ_{ε}	local-success bit: all local admissibility sets are nonempty
$\text{L}, \text{D}, \text{K}, \text{R}, \checkmark$	local failure, compatibility failure, effectivity failure, resource failure, bounded success
$T_{\text{L}}, T_{\text{D}}, T_{\text{K}}, T_{\text{R}}$	tolerance-success spectra for the four filters
$\mathfrak{D}_{\varepsilon}$	obstruction datum: λ_{ε} plus the nested global spaces
ExtComp	terminal extensional compression
G_{D}	compatibility gap $T_{\text{L}} \setminus T_{\text{D}}$
$v_c(p; q)$	Bell critical visibility of p relative to local noise q
$\kappa_{\text{comm}}, \kappa_{\text{meas}}, \eta_c$	communication, measurement-dependence, and detector-efficiency repair coordinates

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1 Introduction

Consider a decision-maker acting from partial observations: a label on a box, a sensor reading, a measurement record, or a local transcript. The hidden world may contain more information than the record reveals. Once two worlds produce the same record, however, the decision-maker has only one response available for both. This observational uniformity is the elementary constraint from which the selector problem begins. Table 1 collects the principal notation used throughout the monograph.

The same structure appears in algorithms reading inputs through an interface, physical instruments receiving measurement outcomes, distributed agents with private signals, and sequential controllers acting through histories. In each case, the policy is a function of the information presented to it, not of the hidden world itself. Asking only whether a successful policy exists therefore hides useful structure. A failure to find such a policy can occur at several levels:

- (i) *Local impossibility*: for a fixed observation, no single response is valid for all worlds consistent with that observation.
- (ii) *Contextual incompatibility*: valid choices exist in each isolated context, but cannot be made to agree across overlapping scopes.
- (iii) *Noncomputability*: compatible choices exist as a set-theoretic family, but no computable implementation selects them.
- (iv) *Resource exhaustion*: computable implementations exist, but all of them exceed the declared time, memory, communication, or other resource bound.

Effective causal descent (ECD) separates these modes of failure inside a declared typed presentation. The guiding question is not merely whether a valid strategy exists, but at which structural layer an obstruction first appears. The same underlying decision problem can be organised in more than one mathematical way. A compressed presentation may preserve the yes–no capability predicate while hiding the difference between a missing local choice and a failure of compatibility. Conversely, a cover presentation may expose the incompatible commitments that a full global object absorbs as local emptiness. The diagnostic classification is therefore attached to a specific local-to-global representation, not to a bare capability predicate.

A running quality-control example gives the intended reading. A single inspector sees only the exterior record of a box and must choose a report–action pair. The visible label is the observation; the hidden condition of the box is the world. If the same label covers both acceptable and defective boxes, then no single response is valid behind that label: the obstruction is local. If several inspectors see different attributes and their individually consistent tables disagree on shared attributes, the obstruction lies in combining the tables: it is a compatibility failure. If the perfect table exists but encodes a noncomputable set, the obstruction is effective. If the table is computable but cannot be produced within the budget, the obstruction is resource-bounded. The example is mnemonic rather than additional structure; the formal objects are sets, maps, relations, diagrams, implementations, and loss functions.

Unless a theorem states otherwise, the set-theoretic development is carried out in ZFC. Finite constructions use only finite choice, and the computability examples provide explicit set-theoretic selectors when needed.

The monograph proceeds in four stages. First, the minimal selector kernel isolates observational uniformity. It consists of worlds, observable states, choices, a validity relation, and a policy class. The fibrewise selector lemma shows that a valid global policy is the same as a choice, at each

realised observation, of an action valid for every world in the corresponding fibre. The prediction-game instantiation then explains where validity comes from: reports, disclosures, actions, dynamics, queries, decoders, and losses.

Second, context diagrams and distributed selectors introduce overlap. Multiple observation contexts are organised as assignment spaces with restriction maps. Objectwise validity gives a constrained diagram; when validity is preserved by restriction to smaller subcontexts, it gives a descent system (Definition 5.3 below). Set-theoretic, effective, and resource-admissible global families are nested subspaces of one ambient family space. The descent kernel uses ordinary restriction agreement; arbitrary relational compatibility belongs to separate constraint extensions.

Third, the same typed language is used to locate standard quantitative, finite, computable, and causal obstructions. A quantitative interlude separates pointwise, observation-uniform, effective, and bounded worst-case optima, with a Chebyshev-radius formula in passive metric prediction. The finite gluing theory then includes compactness and Helly bounds for local certificates, join-tree gluing for acyclic context structures, a six-object parity obstruction, support-level strong contextuality, finite source integration, Bell coupling, and physical Bell repair. Constraint-datum isomorphisms preserve compatible families; affine systems have a complete cokernel obstruction; bijective constraint networks are governed by holonomy fixed points; Bell coupling infeasibility has a Farkas dual certificate; finite-sample and repair variants interpret this certificate operationally; and exact counting of global integrations is $\#P$ -complete. Past factorisation, finite propagation, and the light-cone obstruction complete the static-local tier.

The computability tier gives a separate form of failure. The selector set of the halting-stage game is a nonempty Π_1^0 class without computable members. Nested countable distributed selectors give a second, overlapping-scope K presentation of path selection. The diagonally nonrecursive (DNR) instance has the Peano-arithmetic (PA) oracle spectrum; its input-uniform path-selection operator, and uniform binary fibre selection, have Weihrauch degree WKL . Recursive capability is Σ_3^0 -complete for a concrete primitive-recursive presentation language. Computational-class separation is distinguished from deterministic-time resource separation.

The causal tier adds time while keeping the arena fixed before a strategy is chosen. Backward induction proves that perfect-information node-local nonemptiness already gives a global strategy. Activated patch semantics represents partial contingent plans and cover-relative matching. Blind-order one-shot arenas give an exact normal form for finite distributed selectors, and every finite pure-strategy imperfect-information arena has an extensional distributed compilation. On the Bell one-shot skeleton, probability measures over pure strategies are exactly couplings of deterministic response tables. Finite causal posets of universally executed cuts have unique confluent executions; each linear extension gives a policy-independent arena preserving worldwide loss. Contingent stage occurrence remains tree-indexed. A bounded-height countable arena transfers the halting-stage effectivity obstruction to the causal setting without asserting a general infinite-arena hierarchy.

Fourth, the morphism level makes the classification itself transportable. Capability, diagnostic, and certificate morphisms separate what each level of comparison preserves, and terminal compression is an idempotent functor that erases precisely the D -gap while preserving the nested global spaces. Distributed and one-shot arenas agree up to groupoid equivalence, and composition laws cover constraints, static compilations, sequential and parallel composition, and guarded feedback. Finite tabular presentations are represented as verifier games, in which certificates and refutations are winning strategies and an unfinished play decides nothing. A dependency ledger then governs source-domain placement under the topological-order development principle — dependency order is a constraint, and feasibility orders the remaining work — and the finite algorithmic theory is delegated to the companion algorithmic monograph, which realizes it in typed certificates, width-based dynamic programs, and a prototype artifact.

Figure 1 summarises the canonical presentations and the nested realisation filters.

Relation to established frameworks

The phrase “effective causal descent” denotes a presentation-level classification of selection problems subject to restriction compatibility, computability filters, and resource constraints. It is an analytical framework, not effective descent in the Barr–Beck or Grothendieck sense.

The fibrewise selector problem is closely related to uniform constraint satisfaction. Its multi-context form connects directly with relational database joins: in acyclic database schemes, join trees and separator consistency explain when local tables can be amalgamated into a single global table [3]. The contextual formulation follows the sheaf-theoretic treatment of empirical models by Abramsky and Brandenburger, where measurement contexts are local observation scopes and contextuality is the non-existence of a global section compatible with the local data [1]. The Bell presentation uses the deterministic-strategy coupling perspective: an empirical behaviour is local precisely when it is represented by a probability measure over deterministic response tables with prescribed marginals. Fine’s theorem is the historically central result for the standard $2 \times 2 \times 2$ scenario [5], while the general finite statement follows from the local polytope representation [27].

The computability arguments use standard admissible numberings, index sets, and the theory of Π_1^0 classes [10, 8, 14, 16]. Viewing policies as choices from represented multivalued maps connects the framework to computable analysis and Weihrauch complexity [21, 20], where reducibility calibrates the uniform algorithmic difficulty of mathematical choice principles. The causal tier uses the standard extensive-form distinction between nature moves, information sets, and contingent strategies [23]. The finite-poset protocol semantics is related to partial-order models of concurrency, including event structures and Mazurkiewicz traces; the specific contribution here is the loss-preserving translation from universally executed cut posets to imperfect-information arenas. The least-fixed-point appendix uses standard domain theory [2]. Past factorisation is deliberately weaker than a structural causal model; comparison with structural causal semantics concerns possible extensions rather than an identification [12].

Many component theorems used here are standard in their native settings. Their role is to supply structured reference cases for one presentation-relative selector formalism. For this reason the text cites, rather than reproves, several external results: Farkas separation, Hoeffding concentration, König’s lemma, the Low and hyperimmune-free basis theorems, standard Weihrauch equivalences for closed choice, deterministic time hierarchy theorems, and standard finite CSP complexity classifications where invoked. The contribution is the common typed comparison: local validity, compatibility, effective realisation, resource bounds, and their preservation or failure under translation are represented in one language.

2 Minimal selector kernel

At its most fundamental level, a selection problem asks whether a decision-maker can choose valid responses while observing only partial information about the underlying world. The minimal kernel contains no causality, computation, probability, or resource bound. It records only observational uniformity: a policy is a function of what is seen, so all worlds in one observation fibre must be handled by one choice.

Definition 2.1 (Selector problem). A selector problem is a tuple (W, Z, C, I, V, Π) consisting of:

- (i) a nonempty set W of underlying worlds;

single observer	$\mapsto$	fibre-indexed selector	$\mapsto$	local validity L
distributed observers	$\mapsto$	CSP incidence diagram	$\mapsto$	compatibility D
extensive arena	$\mapsto$	causal patch sheaf	$\mapsto$	backward induction / matching
		Γ_{set}	$\supseteq$	$\Gamma_{\text{eff}} \supseteq \Gamma_{\mathcal{B}}$

Figure 1: Canonical presentations of the static, distributed, and causal tiers. The first two arrows indicate how modelling data are packaged as local objects; the last column records the diagnostic level exposed by that packaging. Effectivity and resource bounds are imposed afterwards by nested realisation images in the common global-family space; the diagnostic exposed by a row is presentation-relative.

- (ii) a nonempty set Z of observable states;
- (iii) a nonempty set C of available choices;
- (iv) an observation interface $I : W \rightarrow Z$;
- (v) a validity relation $V \subseteq W \times C$, where $V(w, c)$ means that choice c is valid in world w ;
- (vi) a declared policy class $\Pi \subseteq C^Z$.

For $z \in \text{im}(I)$ define the common admissible-choice set

$$\text{Adm}_V(z) = \{c \in C : \forall w \in I^{-1}(z), V(w, c)\}.$$

The problem is capable in Π when some $\pi \in \Pi$ satisfies $V(w, \pi(I(w)))$ for every $w \in W$.

Remark 2.2 (Operational reading). Operationally, $\text{Adm}_V(z)$ is the menu of choices that work for every world that looks like z . If this menu is empty, the observation has already lost too much information. If every menu is nonempty, the next question is whether the policy class can choose one item from each menu.

Lemma 2.3 (Fibrewise selector lemma). *For every selector problem,*

$$\exists \pi \in \Pi \forall w \in W V(w, \pi(I(w))) \iff \exists \pi \in \Pi \forall z \in \text{im}(I) \pi(z) \in \text{Adm}_V(z).$$

Proof. Suppose the left side holds through π . Fix $z \in \text{im}(I)$ and $w \in I^{-1}(z)$. Then $I(w) = z$, so $V(w, \pi(z))$. Since w was arbitrary, $\pi(z) \in \text{Adm}_V(z)$. This proves the right side.

Conversely, suppose the right side holds through π , and let $w \in W$. Putting $z = I(w)$ gives $w \in I^{-1}(z)$ and $\pi(z) \in \text{Adm}_V(z)$, hence $V(w, \pi(I(w)))$. This holds for every world. $\square$

Remark 2.4. Lemma 2.3 is a representation lemma. It says that checking a policy world by world is the same as checking it fibre by fibre: at each observable state, the policy must pick a choice that works for all hidden worlds behind that state. The common-choice sets $\text{Adm}_V(z)$ are therefore the local objects on which the rest of the monograph builds.

3 Prediction-game instantiation

To model interactive systems, the minimal selector kernel is instantiated inside a typed prediction game. In this setting, validity is no longer primitive. It is computed from a report, a disclosure channel, an action, an environmental update, a query of the resulting history, a decoder, and a loss function.

Definition 3.1 (Typed prediction game). A typed prediction game is a tuple

$$G = (W, H, Z, R, A, M, Y, I, D, U, q, e, \ell),$$

where the carriers W, Z, R, A, M, H, Y are nonempty. They represent, respectively, worlds, histories, observations, reports, actions, disclosed messages, and query values. The structural maps are

$$\begin{aligned} I : W &\rightarrow Z, & D : R &\rightarrow M, & U : W \times M \times A &\rightarrow H, \\ q : H &\rightarrow Y, & e : R &\rightarrow Y, & \ell : R \times Y &\rightarrow [0, \infty]. \end{aligned}$$

Thus I observes the world, D records what a report discloses, U updates the world-history after disclosure and action, q asks the relevant question of the history, e decodes the report, and ℓ measures loss. The loss acts directly on a report and a query value; in closed loop it is evaluated as $\ell(r, q(h))$. The game is called *exact* when $\ell(r, y) = 0$ if and only if $e(r) = y$; this condition constrains only the zero level of the loss.

A policy is $\pi = (\pi_R, \pi_A)$ with

$$\pi_R : Z \rightarrow R, \quad \pi_A : Z \times R \rightarrow A.$$

Its induced choice and closed-loop history are

$$\begin{aligned} s_\pi(z) &= (\pi_R(z), \pi_A(z, \pi_R(z))), \\ h_\pi(w) &= U(w, D(\pi_R(I(w))), \pi_A(I(w), \pi_R(I(w)))). \end{aligned}$$

Only the values of π_A on the graph of π_R affect capability. The larger action type is retained because it records how an implementation would react to counterfactual reports; the selector kernel uses only s_π .

Remark 3.2 (Report, action, and observation). The closed-loop order matters. At observation z , the policy reports $r = \pi_R(z)$ and then acts by $a = \pi_A(z, r)$. The disclosure map records what the report reveals to the system, and the dynamics records how the world responds to that disclosed report and action. Exactness identifies zero loss with a correct report; positive and infinite losses record quantitative error and hard failure.

Definition 3.3 (Policy classes). Write

$$\Pi_{\text{all}} = \text{Pol}(Z; R, A) = R^Z \times A^{Z \times R}.$$

Let $\text{Impl}_{\text{eff}}(G)$ be a represented class of effective implementations and

$$\text{Realise}_{\text{eff}} : \text{Impl}_{\text{eff}}(G) \rightarrow \Pi_{\text{all}}$$

its realisation map. If $\mathcal{B}$ is a resource predicate on implementations, put

$$\text{Impl}_{\mathcal{B}} = \{\iota \in \text{Impl}_{\text{eff}}(G) : \mathcal{B}(\iota)\}, \quad \Pi_{\text{eff}} = \text{im}(\text{Realise}_{\text{eff}}), \quad \Pi_{\mathcal{B}} = \text{im}(\text{Realise}_{\text{eff}}|_{\text{Impl}_{\mathcal{B}}}).$$

Thus $\Pi_{\mathcal{B}} \subseteq \Pi_{\text{eff}} \subseteq \Pi_{\text{all}}$. Resource predicates are imposed on implementations; extensionally equal policies may have implementations with different costs. The qualifier *represented* is fixed in Definition 10.4 below; the effective and bounded classes are relative to that representation.

Remark 3.4 (Policies versus implementations). The hierarchy separates what a policy does from how it is realised. A successful input–output table may exist as a set, may be computable by some implementation, and may still fail a declared time, memory, or architectural bound. These are different filters on the same ambient space of possible policy behaviours.

Definition 3.5 (Observational policy equivalence). For policies $\pi, \pi' \in \Pi_{\text{all}}$, write $\pi \sim_{\text{obs}} \pi'$ when

$$s_\pi(z) = s_{\pi'}(z) \quad \text{for every } z \in \text{im}(I).$$

Proposition 3.6 (Observational policy quotient). *For every tolerance, the closed-loop loss and validity of an individual policy depend only on its $\sim_{\text{obs}}$ -class, and*

$$\Pi_{\text{all}} / \sim_{\text{obs}} \cong (R \times A)^{\text{im}(I)}.$$

For every policy class Π , capability in Π is determined by its image in the quotient; saturation is needed only to reconstruct Π itself as a union of complete equivalence classes.

Proof. Closed-loop loss uses a policy only through $(\pi_R(z), \pi_A(z, \pi_R(z)))$ at realised interface values, proving the first claim. The fibres of the map $\pi \mapsto s_\pi|_{\text{im}(I)}$ are exactly the $\sim_{\text{obs}}$ -classes. It is surjective: given $s(z) = (r_z, a_z)$, define $\pi_R(z) = r_z$ and $\pi_A(z, r_z) = a_z$ on the image, and use fixed elements of the nonempty sets R, A to extend the maps elsewhere. The quotient is therefore isomorphic to the displayed function space. A class contains a valid policy exactly when its quotient image contains the corresponding valid principal selector, regardless of whether the class is saturated. $\square$

Definition 3.7 (Admissible report–action pairs). For $z \in \text{im}(I)$ define

$$\text{Adm}(z) = \{(r, a) \in R \times A : \forall w \in I^{-1}(z), e(r) = q(U(w, D(r), a))\}.$$

Thus $\text{Adm}(z)$ contains report–action pairs, not reports alone. A policy is an exact valid selector when $s_\pi(z) \in \text{Adm}(z)$ for every $z \in \text{im}(I)$.

Definition 3.8 (Worst-case capability). For $\varepsilon \in [0, \infty)$ and $\Pi \subseteq \Pi_{\text{all}}$, write $\text{Cap}^{wc}(G, \Pi; q, \varepsilon)$ when some $\pi \in \Pi$ satisfies

$$\ell(\pi_R(I(w)), q(h_\pi(w))) \leq \varepsilon \quad \text{for every } w \in W.$$

Proposition 3.9 (Prediction-game selector representation). *If G is exact,*

$$\text{Cap}^{wc}(G, \Pi; q, 0) \iff \exists \pi \in \Pi \forall z \in \text{im}(I) s_\pi(z) \in \text{Adm}(z).$$

Proof. Let

$$S_\Pi = \{s_\pi : Z \rightarrow R \times A : \pi \in \Pi\} \subseteq (R \times A)^Z$$

be the principal-selector image of the policy class. Apply Lemma 2.3 with $C = R \times A$, policy class S_Π , and

$$V(w, (r, a)) \iff e(r) = q(U(w, D(r), a)).$$

A witness in S_Π is, by definition, s_π for some $\pi \in \Pi$. Exactness identifies this relation with zero closed-loop loss. $\square$

3.1 Passive factorisation

Proposition 3.10 (Set-theoretic fibre factorisation). *Let $\bar{I} : W \rightarrow \text{im}(I)$ be the corestriction of I . For a map $f : W \rightarrow Y$, the following are equivalent:*

- (i) *there is $g : \text{im}(I) \rightarrow Y$ with $f = g \circ \bar{I}$;*
- (ii) *f is constant on every fibre of I .*

The factor g is unique.

Proof. If $f = g \circ \bar{I}$, equal interface values give equal values of f . Conversely, if f is constant on fibres, define $g(z)$ to be $f(w)$ for any $w \in I^{-1}(z)$. Fibre constancy makes this independent of the chosen world, and $f = g \circ \bar{I}$. Every $z \in \text{im}(I)$ has a preimage, so this equation also forces uniqueness. $\square$

Proposition 3.11 (Passive prediction). *Assume exact loss. Suppose $U(w, m, a) = U_0(w)$ for all m, a , and put $f = q \circ U_0$. Let $g : \text{im}(I) \rightarrow Y$ be the factor of Proposition 3.10, when it exists. Exact capability in Π holds if and only if f is constant on the fibres of I and there is $\pi \in \Pi$ such that*

$$e(\pi_R(z)) = g(z) \quad \text{for every } z \in \text{im}(I).$$

Proof. In the passive game the action and disclosure do not change the queried value, so exact validity of π is $e(\pi_R(I(w))) = f(w)$ for every w . If it holds, f is constant on fibres and the displayed lifting equation follows. Conversely, fibre constancy and the lifting equation give $e(\pi_R(I(w))) = g(I(w)) = f(w)$ for every world; exact loss then gives capability. Exact validity imposes no further condition on the action component in this passive game, although membership in the declared policy class Π may still constrain it. $\square$

Remark 3.12 (Passive prediction). In a passive game the action does not change what is being predicted. Exact prediction is possible precisely when the target value is already determined by the observation, up to the decoder. Thus two indistinguishable worlds with different target values are the simplest information-theoretic obstruction.

3.2 Anti-validity and the uniform diagonal

Definition 3.13 (Anti-validity). A relevant fibre $I^{-1}(z)$ is *anti-valid* when for every (r, a) there is $w \in I^{-1}(z)$ such that

$$e(r) \neq q(U(w, D(r), a)).$$

Equivalently, $\text{Adm}(z) = \emptyset$.

Definition 3.14 (Uniform diagonal). A query has *uniform diagonal form* when there is a fixed-point-free $\delta : Y \rightarrow Y$ such that

$$q(U(w, D(r), a)) = \delta(e(r))$$

for every w, r, a .

Proposition 3.15 (Elementary diagonal obstruction). *Uniform diagonal form makes every relevant fibre anti-valid. Consequently exact capability fails for every policy class.*

Proof. For any $z \in \text{im}(I)$, choose $w \in I^{-1}(z)$. For every (r, a) , $q(U(w, D(r), a)) = \delta(e(r)) \neq e(r)$. Thus $\text{Adm}(z) = \emptyset$ and Proposition 3.9 precludes capability. $\square$

Remark 3.16 (Diagonal form). A uniform diagonal is the extreme local obstruction: every possible report is turned into a different queried value. The obstruction is local because it already empties each relevant fibre. The computability examples below use a diagonal idea differently: their fibres remain nonempty, but no computable selector can choose valid values for all fibres at once.

Proposition 3.17 (Distinct local failure shapes). *There is a two-world obstruction that is not a uniform diagonal, and a uniform diagonal obstruction with no two-world witness.*

Proof. For the first, take two worlds with one interface value, binary reports, singleton actions, passive dynamics, and distinct query values 0 and 1 on the worlds. They form the pair obstruction of Corollary 4.9 below, but the query cannot be one fixed function of the report on both worlds. For the second, take one world and set $q(U(w, D(r), *)) = 1 - r$ with identity decoder. This is a uniform diagonal, while a two-world witness is impossible. $\square$

3.3 Robust validity and display-map semantics

Definition 3.18 (Robust admissibility). For $\varepsilon \in [0, \infty)$ and $z \in \text{im}(I)$ put

$$\text{Adm}_\varepsilon(z) = \{(r, a) : \forall w \in I^{-1}(z), \ell(r, q(U(w, D(r), a))) \leq \varepsilon\}.$$

Lemma 3.19 (Robust selector representation). *For every $\varepsilon \geq 0$,*

$$\text{Cap}^{wc}(G, \Pi; q, \varepsilon) \iff \exists \pi \in \Pi \forall z \in \text{im}(I) s_\pi(z) \in \text{Adm}_\varepsilon(z).$$

Proof. If capability holds, restrict its inequality to each fibre and substitute $I(w) = z$. This shows $s_\pi(z) \in \text{Adm}_\varepsilon(z)$. Conversely, if these memberships hold and $w \in W$, then $w \in I^{-1}(I(w))$, so the defining inequality for $\text{Adm}_\varepsilon(I(w))$ is exactly the capability inequality at w . $\square$

Definition 3.20 (Validity family and realised sections). The *validity family* at tolerance ε has display map

$$p^\varepsilon : \text{Tot}^\varepsilon \rightarrow \text{im}(I), \quad \text{Tot}^\varepsilon = \{(z, r, a) : z \in \text{im}(I), (r, a) \in \text{Adm}_\varepsilon(z)\},$$

with $p^\varepsilon(z, r, a) = z$. No topology or local triviality is intended. A section is a map $s : \text{im}(I) \rightarrow \text{Tot}^\varepsilon$ with $p^\varepsilon \circ s = \text{id}$, equivalently a principal part $\tilde{s} : \text{im}(I) \rightarrow R \times A$ satisfying $\tilde{s}(z) \in \text{Adm}_\varepsilon(z)$. It is Π -realised when $\tilde{s} = s_\pi|_{\text{im}(I)}$ for some $\pi \in \Pi$. Write $\Gamma_\Pi(p^\varepsilon)$ for these realised sections.

Lemma 3.21 (Section representation).

$$\text{Cap}^{wc}(G, \Pi; q, \varepsilon) \iff \Gamma_\Pi(p^\varepsilon) \neq \emptyset.$$

Proof. By definition, $\Gamma_\Pi(p^\varepsilon)$ is nonempty exactly when some $\pi \in \Pi$ has $s_\pi(z) \in \text{Adm}_\varepsilon(z)$ for every $z \in \text{im}(I)$. Apply Lemma 3.19. $\square$

Remark 3.22 (Section language). The display map turns the admissible choices into a fibre bundle in the bare set-theoretic sense: over each observation sits the set of choices that work there. Capability asks for a section of this family, and realised capability asks that the section come from the declared policy class. This is the single-observer prototype of the multi-context gluing problems.

3.4 Monotonicity and interface refinement

Proposition 3.23 (Monotonicity). *For a fixed typed game:*

- (i) *enlarging the policy class preserves capability;*
- (ii) *enlarging the tolerance preserves capability;*
- (iii) *if $\emptyset \neq W' \subseteq W$, the same policy that is valid on W is valid after restricting the game to W' .*

The third claim concerns a fixed policy class. Transport of an implementation class under world restriction is separate modelling data.

Proof. The first claim keeps the same witness in the larger policy class. The second keeps the same witness and uses $\varepsilon \leq \varepsilon'$. For the third, every fibre of the restricted interface is a subset of the corresponding original fibre, so every report–action pair valid on the original fibre remains valid on the restricted one. The policy itself is unchanged. $\square$

Proposition 3.24 (Interface refinement). *Suppose $I = k \circ I'$ with $I' : W \rightarrow Z'$. For a policy π on Z , define*

$$k^* \pi = (\pi_R \circ k, \pi'_A), \quad \pi'_A(z', r) = \pi_A(k(z'), r).$$

Let $\Pi_Z \subseteq \text{Pol}(Z; R, A)$ and $\Pi_{Z'} \subseteq \text{Pol}(Z'; R, A)$. If $k^(\Pi_Z) \subseteq \Pi_{Z'}$, capability for (I, Π_Z) implies capability for $(I', \Pi_{Z'})$.*

Proof. For each $z' \in \text{im}(I')$, $I'^{-1}(z') \subseteq I^{-1}(k(z'))$. Hence every pair admissible on the latter fibre is admissible on the former. Pulling a witnessing policy back by k^* therefore yields a valid policy for the refined interface, and the image condition places it in $\Pi_{Z'}$. $\square$

Remark 3.25 (Refinement). A refined interface distinguishes worlds that the coarse interface may have identified. Since each refined fibre is contained in a coarse fibre, refinement cannot destroy an already valid observation-indexed policy once the policy class admits the corresponding pullback.

4 Static local obstructions and past factorisation

This section examines how local obstructions can be certified. A fibre is locally impossible when the worlds behind one observation admit no common valid choice. This is expressed as emptiness of an intersection of worldwise validity sets. Compactness and convexity turn such failures into finite certificates; past factorisation and finite propagation explain how causal limits generate instances of the same obstruction.

Definition 4.1 (Obstructing family). For $w \in W$ let

$$\text{Val}(w) = \{(r, a) : e(r) = q(U(w, D(r), a))\}.$$

A subset $S \subseteq I^{-1}(z)$ is obstructing when $\bigcap_{w \in S} \text{Val}(w) = \emptyset$.

Lemma 4.2 (Fibre intersection identity). *For every $z \in \text{im}(I)$,*

$$\text{Adm}(z) = \emptyset \iff I^{-1}(z) \text{ contains an obstructing family.}$$

Proof. By definition, $\text{Adm}(z) = \bigcap_{w \in I^{-1}(z)} \text{Val}(w)$. If it is empty, the whole fibre is an obstructing family. If an obstructing $S \subseteq I^{-1}(z)$ exists, the intersection over the full fibre is contained in the empty intersection over S , and is therefore empty. $\square$

Definition 4.3 (Robust singleton-validity sets). For $w \in W$ and $\varepsilon \geq 0$ put

$$\text{Val}_\varepsilon(w) = \{(r, a) \in R \times A : \ell(r, q(U(w, D(r), a))) \leq \varepsilon\}.$$

Lemma 4.4 (Robust fibre intersection).

$$\text{Adm}_\varepsilon(z) = \bigcap_{w \in I^{-1}(z)} \text{Val}_\varepsilon(w).$$

Proof. Both sides consist exactly of the report–action pairs whose loss is at most ε for every world in the fibre. $\square$

Theorem 4.5 (Compact local-obstruction certificate). *Suppose $R \times A$ is compact and every sublevel set $\text{Val}_\varepsilon(w)$ is closed. This holds, for example, if for every world the map*

$$(r, a) \mapsto \ell(r, q(U(w, D(r), a)))$$

is lower semicontinuous. Then $\text{Adm}_\varepsilon(z) = \emptyset$ if and only if some finite $S \subseteq I^{-1}(z)$ satisfies

$$\bigcap_{w \in S} \text{Val}_\varepsilon(w) = \emptyset.$$

Proof. The reverse implication follows because the full intersection is contained in the intersection over S . Conversely, if every finite subfamily had nonempty intersection, the closed sets would have the finite-intersection property. Compactness of $R \times A$ would then make their total intersection nonempty, contrary to $\text{Adm}_\varepsilon(z) = \emptyset$. $\square$

Remark 4.6 (Finite certificates). The theorem turns an infinite-looking local failure into finite evidence. If no choice works for all worlds behind one observation, then compactness supplies a finite subfamily of worlds already witnessing the failure. Under convexity, Helly’s theorem bounds the size of such a certificate.

Corollary 4.7 (Helly bound). *Suppose $R \times A$ is identified with a compact convex set of affine dimension d and every $\text{Val}_\varepsilon(w)$ is closed and convex. If $\text{Adm}_\varepsilon(z) = \emptyset$, there is an obstructing $S \subseteq I^{-1}(z)$ with $|S| \leq d + 1$.*

Proof. Theorem 4.5 gives a finite obstructing subfamily. Helly’s theorem applied in the affine hull yields an empty subfamily of size at most $d + 1$ [18]. $\square$

Remark 4.8 (Randomised-choice instance). In the selector-kernel specialisation, or in the typed game after taking the report component singleton and $A = \Delta(C_0)$, let a finite primitive choice set C_0 be replaced by the simplex of randomised choices. If primitive losses $L_0(w, c)$ are finite and expected loss is $L(w, p) = \sum_{c \in C_0} p(c) L_0(w, c)$, then the robust validity sets $\{p \in \Delta(C_0) : L(w, p) \leq \varepsilon\}$ are closed convex polytopes in a simplex of affine dimension $|C_0| - 1$. Corollary 4.7 gives an obstruction certificate involving at most $|C_0|$ worlds.

Corollary 4.9 (World-pair obstruction). *Suppose $I(w_0) = I(w_1)$ and*

$$q(U(w_0, D(r), a)) \neq q(U(w_1, D(r), a))$$

for every (r, a) . Then their common fibre has empty admissibility and exact capability fails for every policy class.

Proof. No pair (r, a) can have decoded report equal to both distinct query values, so $\{w_0, w_1\}$ is obstructing. Apply Lemma 4.2 and Proposition 3.9. $\square$

A cut τ is a declared boundary between accessible records and future evolution. The following factorisation is additional structure. In the causal tier, Proposition 11.73 derives it from independently supplied exogenous records after a causal predecessor trace is fixed.

Definition 4.10 (Past-factored observation). At a cut τ , a past-factored observation for a cut interface $I_\tau : W \rightarrow Z$ consists of independently specified accessible past records P_τ and maps

$$P_\tau : W \rightarrow P_\tau, \quad J_\tau : P_\tau \rightarrow Z$$

with $I_\tau = J_\tau \circ P_\tau$. When this is meant to factor the original game interface, one assumes $I_\tau = I$. Thus equal accessible past records imply equal interface values; the converse is not assumed. The factorisation is causally informative only because P_τ is supplied independently of I_τ ; the extensive-form construction of Section 11 provides such an independent history structure.

Corollary 4.11 (Past-cone obstruction). *For the prediction game whose interface is the cut interface I_τ of Definition 4.10 (or for the original game when $I_\tau = I$), if $P_\tau(w_0) = P_\tau(w_1)$ and the separation condition of Corollary 4.9 holds, exact capability fails for every policy class.*

Proof. Past factorisation gives $I_\tau(w_0) = I_\tau(w_1)$, so Corollary 4.9 applies. $\square$

Proposition 4.12 (Finite-propagation factorisation). *Consider a deterministic cellular automaton of interaction radius ρ , an observer window V , and horizon T . Let $V^{+\rho T}$ be the cells within graph distance ρT of V . Any interface computed from the states of V through time T factors through the initial restriction*

$$P_T(w) = w|_{V^{+\rho T}}.$$

Proof. By induction on time, the state of a cell after t updates depends only on initial cells within distance ρt . Therefore every observer-window state through time T , and every function of that record, is determined by $w|_{V^{+\rho T}}$. $\square$

Corollary 4.13 (Light-cone prediction obstruction). *For a passive target $f : W \rightarrow Y$, suppose two worlds agree on $V^{+\rho T}$ but satisfy $f(w_0) \neq f(w_1)$. Then exact capability fails for every policy class whose interface is confined to the observer record through T .*

Proof. Proposition 4.12 gives the same interface value in both worlds, while the passive query separates them under every report–action pair. Corollary 4.9 applies. $\square$

Remark 4.14 (Causal reach). The light-cone obstruction is a fibre obstruction with causal content supplied by the factorisation. If the observer record through time T is determined by the initial data in $V^{+\rho T}$, then targets depending on differences outside that region cannot be exactly predicted from that record.

5 Context diagrams and effective descent

The discussion now broadens from one observation channel to many overlapping contexts. A local assignment is a partial description of a possible global policy, and compatibility means that two partial descriptions agree wherever they overlap. Category theory supplies the bookkeeping: contexts are objects, restriction maps are arrows, and a global assignment is a compatible family of

local choices. Relational CSP constraints can be added as a separate extension, but the descent kernel itself uses ordinary restriction agreement. Arrows are oriented from larger contexts to smaller contexts, so restriction is covariant; reversing all arrows gives the usual presheaf convention.

Definition 5.1 (Constrained assignment diagram). A constrained assignment diagram consists of a small category $\mathcal{C}$, a functor

$$\mathcal{A} : \mathcal{C} \rightarrow \mathbf{Set},$$

and objectwise admissibility sets $\mathcal{A}_{\text{loc}}(c) \subseteq \mathcal{A}(c)$. Standard compatibility of a family $s \in \prod_c \mathcal{A}(c)$ is

$$\mathcal{A}(f)(s_c) = s_{c'} \quad (f : c \rightarrow c').$$

The set of compatible admissible families is

$$\Gamma_{\text{set}}(\mathcal{A}_{\text{loc}}) = \left\{ s \in \prod_c \mathcal{A}(c) : s_c \in \mathcal{A}_{\text{loc}}(c) \ \forall c, \ \mathcal{A}(f)(s_c) = s_{c'} \ \forall f : c \rightarrow c' \right\}.$$

Remark 5.2 (Gluing interpretation). A family s chooses one local assignment at every context. It is compatible when every overlap check passes. The diagnostic distinction between **L** and **D** is exactly the distinction between an empty local menu and a collection of nonempty local menus whose choices cannot be stitched into one consistent global assignment.

Definition 5.3 (Descent system). A constrained assignment diagram is a *descent system* when the admissibility sets form a subfunctor: for every $f : c \rightarrow c'$,

$$s_c \in \mathcal{A}_{\text{loc}}(c) \implies \mathcal{A}(f)(s_c) \in \mathcal{A}_{\text{loc}}(c').$$

Proposition 5.4 (Limit semantics of a descent system). *For every descent system,*

$$\Gamma_{\text{set}}(\mathcal{A}_{\text{loc}}) = \varprojlim_{\mathcal{C}} \mathcal{A}_{\text{loc}} = \left(\varprojlim_{\mathcal{C}} \mathcal{A} \right) \cap \prod_c \mathcal{A}_{\text{loc}}(c),$$

where all sets are viewed in the ambient product.

Proof. A family belongs to the limit of the subfunctor exactly when each component lies in $\mathcal{A}_{\text{loc}}(c)$ and the same restriction equalities that define the limit of $\mathcal{A}$ hold. These are precisely the conditions defining the displayed intersection. $\square$

Theorem 5.5 (Initial-object collapse). *Let $\mathcal{C}$ have an initial object c_0 , with unique arrow $!_c : c_0 \rightarrow c$. Then*

$$\Gamma_{\text{set}}(\mathcal{A}_{\text{loc}}) \cong \{u \in \mathcal{A}_{\text{loc}}(c_0) : \mathcal{A}(!_c)(u) \in \mathcal{A}_{\text{loc}}(c) \ \forall c\}.$$

If $\mathcal{A}_{\text{loc}}$ is a subfunctor, this reduces to

$$\Gamma_{\text{set}}(\mathcal{A}_{\text{loc}}) \cong \mathcal{A}_{\text{loc}}(c_0).$$

Consequently a subfunctorial diagram with an initial object cannot have outcome D_ε while all local sets are nonempty.

Proof. Compatibility forces $s_c = \mathcal{A}(!_c)(s_{c_0})$ for every object, so a family is determined by its initial component. Conversely, any initial component whose images are locally admissible defines a compatible family: for $f : c \rightarrow c'$, uniqueness gives $f \circ !_c = !_c'$, and functoriality gives the required restriction equality. If admissibility is a subfunctor, every image of an admissible initial component is automatically admissible. $\square$

Remark 5.6 (Why initial objects remove compatibility failure). An initial object acts as a complete context whose restrictions determine all others. If validity is stable under restriction, any admissible assignment at that complete context automatically supplies the matching local assignments. Thus compatibility failure can appear only when the chosen presentation lacks such a determining object or when admissibility is not restriction-stable.

Definition 5.7 (Realised family spaces). Let

$$\sigma_{\text{eff}} : \text{Impl}_{\text{eff}} \rightarrow \prod_c \mathcal{A}(c)$$

be a represented realisation map, and let $\sigma_{\mathcal{B}}$ be its restriction to a declared bounded implementation class. Define

$$\Gamma_{\text{eff}} = \text{im}(\sigma_{\text{eff}}) \cap \Gamma_{\text{set}}, \quad \Gamma_{\mathcal{B}} = \text{im}(\sigma_{\mathcal{B}}) \cap \Gamma_{\text{set}}.$$

Proposition 5.8 (Fibre-indexed static diagram). *For a typed prediction game at tolerance ε , take the discrete category with objects $z \in \text{im}(I)$, put*

$$\mathcal{A}(z) = R \times A, \quad \mathcal{A}_{\text{loc}, \varepsilon}(z) = \text{Adm}_{\varepsilon}(z),$$

and send an implementation to the family $(s_{\text{Realise}_{\text{eff}}(\iota)}(z))_z$. Then effective, respectively bounded, global-family existence is equivalent to capability in Π_{eff} , respectively $\Pi_{\mathcal{B}}$.

Proof. A global family on the discrete category is a map $s : \text{im}(I) \rightarrow R \times A$ with $s(z) \in \text{Adm}_{\varepsilon}(z)$. Requiring that family to be realised gives exactly the selector condition of Lemma 3.19. $\square$

Proposition 5.9 (Compressed terminal-category diagram). *For a typed prediction game at tolerance ε , take the terminal category, with one object and only its identity arrow, and put*

$$\mathcal{A}(c) = \text{Maps}(\text{im}(I), R \times A),$$

$$\mathcal{A}_{\text{loc}, \varepsilon}(c) = \{s : s(z) \in \text{Adm}_{\varepsilon}(z) \text{ for every } z \in \text{im}(I)\}.$$

Define $\sigma_{\text{eff}}(\iota) = s_{\text{Realise}_{\text{eff}}(\iota)}|_{\text{im}(I)}$ and restrict it for $\sigma_{\mathcal{B}}$. Then effective and bounded family existence are respectively equivalent to capability in Π_{eff} and $\Pi_{\mathcal{B}}$.

Proof. The terminal-category compatibility condition is vacuous. A realised ambient map lies in $\mathcal{A}_{\text{loc}, \varepsilon}(c)$ exactly when its policy is an ε -valid selector. Apply Lemma 3.19. $\square$

Proposition 5.10 (Discrete compatibility and choice). *On a discrete context category,*

$$\Gamma_{\text{set}}(\mathcal{A}_{\text{loc}}) = \prod_c \mathcal{A}_{\text{loc}}(c).$$

For finitely many objects, nonempty local sets give a global family by finite choice. For an arbitrary small discrete category, the corresponding implication uses the axiom of choice in the ambient ZFC foundation.

Proof. There are no nonidentity restriction equations, so a global family is exactly a choice of one local element at every object. $\square$

Remark 5.11 (Presentation relativity). The labels local, compatibility, effective, and bounded are relative to the declared diagram and are not invariants of a bare capability predicate. Table 4 summarises the structural cases used in the monograph; several rows list constructions appearing in Sections 7, 9, and 11. Repackaging all constraints into one object can move a failure between descriptions. The terminal-category construction of Proposition 5.9 and the extensive-form construction of Section 11 are canonical relative to their input data; externally supplied contextual diagrams carry their stated packaging as part of the theorem.

Remark 5.12 (Principal specialisations). The fibre-indexed discrete diagram gives the primary static prediction specialisation; the terminal-category diagram is its compressed form. Allowing all set-theoretic families with standard restriction agreement gives the contextual specialisation. Intersecting global families with a represented implementation image gives the effective specialisation. The downward-closed incidence diagram of Definition 9.5 gives the empirical-support specialisation. These are uses of one ambient section formalism, not claims that their native mathematical theories are identical.

Definition 5.13 (Tolerance-filtered constrained diagram). A tolerance-filtered constrained diagram has subsets $\mathcal{A}_{\text{loc},\varepsilon}(c) \subseteq \mathcal{A}(c)$ satisfying, for every object,

$$\varepsilon \leq \varepsilon' \implies \mathcal{A}_{\text{loc},\varepsilon}(c) \subseteq \mathcal{A}_{\text{loc},\varepsilon'}(c).$$

It is a *filtered descent system* when, in addition, every tolerance level is restriction-stable:

$$\mathcal{A}(f)(\mathcal{A}_{\text{loc},\varepsilon}(c)) \subseteq \mathcal{A}_{\text{loc},\varepsilon}(c') \quad (f : c \rightarrow c').$$

Use the same realisation maps at every tolerance and define

$$\Gamma_{\text{set}}^\varepsilon = \Gamma_{\text{set}}(\mathcal{A}_{\text{loc},\varepsilon}), \quad \Gamma_{\text{eff}}^\varepsilon = \text{im}(\sigma_{\text{eff}}) \cap \Gamma_{\text{set}}^\varepsilon, \quad \Gamma_{\mathcal{B}}^\varepsilon = \text{im}(\sigma_{\mathcal{B}}) \cap \Gamma_{\text{set}}^\varepsilon.$$

For a static prediction game also define the fibre-local success set

$$T_{\text{fib}} = \{\varepsilon : \forall z \in \text{im}(I), \text{Adm}_\varepsilon(z) \neq \emptyset\}.$$

The diagram-level success sets are

$$\begin{aligned} T_{\text{L}} &= \{\varepsilon : \forall c \mathcal{A}_{\text{loc},\varepsilon}(c) \neq \emptyset\}, \\ T_{\text{D}} &= \{\varepsilon : \Gamma_{\text{set}}^\varepsilon \neq \emptyset\}, \\ T_{\text{K}} &= \{\varepsilon : \Gamma_{\text{eff}}^\varepsilon \neq \emptyset\}, \\ T_{\text{R}} &= \{\varepsilon : \Gamma_{\mathcal{B}}^\varepsilon \neq \emptyset\}. \end{aligned}$$

Put $\varepsilon_\chi = \inf T_\chi \in [0, \infty]$, with the convention $\inf \emptyset = +\infty$.

Remark 5.14 (Success-set notation). The notation $T_{\text{L}}, T_{\text{D}}, T_{\text{K}}, T_{\text{R}}$ records success of the successive filters, not the tolerances at which the corresponding failure labels occur. For example, $\varepsilon \in T_{\text{L}}$ means all local sets are nonempty; the failure outcome L_ε occurs when $\varepsilon \notin T_{\text{L}}$. For the terminal-category static compression, $T_{\text{fib}} = T_{\text{L}}$ in ZFC; the equality uses choice when the observation image is infinite. In the fibre-indexed discrete presentation, T_{fib} is objectwise local success by definition.

6 Well-formedness, semantic outcomes, and transport

This section turns the preceding distinctions into a diagnostic partition. The questions are asked in order. First, does every declared local object have an admissible element? Second, do these local elements glue to a compatible global family? Third, is some global family effectively realised? Fourth, is some effective realisation within the resource bound? The local-success bit separates local failure from compatibility failure; the nested global spaces separate set-theoretic, effective, and bounded realisation.

Definition 6.1 (Well-formed presentation). For each presentation language used in the monograph, a raw presentation is a code for the corresponding carrier sets, maps, category, admissibility predicates, and realisation maps. It is *well formed*, written $\text{WF}(P)$, when the schema-specific type checks, category laws, functor laws, admissibility predicates, and realisation maps have the required types. The diagnostic code

$$\text{U}(P) \iff \neg \text{WF}(P)$$

records malformed input. Semantic outcomes are defined only for $\text{WF}(P)$.

Definition 6.2 (Tolerance-relative semantic outcomes). Fix a well-formed presentation P , a tolerance ε , an effective realisation image, and a bounded implementation model. Use the filtered local and global spaces of Definition 5.13, and define

$$\begin{aligned} \text{L}_\varepsilon(P) &: \exists c [\mathcal{A}_{\text{loc},\varepsilon}(c) = \emptyset], \\ \text{D}_\varepsilon(P) &: (\forall c \mathcal{A}_{\text{loc},\varepsilon}(c) \neq \emptyset) \wedge \Gamma_{\text{set}}^\varepsilon = \emptyset, \\ \text{K}_\varepsilon(P) &: \Gamma_{\text{set}}^\varepsilon \neq \emptyset \wedge \Gamma_{\text{eff}}^\varepsilon = \emptyset, \\ \text{R}_\varepsilon(P) &: \Gamma_{\text{eff}}^\varepsilon \neq \emptyset \wedge \Gamma_{\mathcal{B}}^\varepsilon = \emptyset, \\ \checkmark_\varepsilon(P) &: \Gamma_{\mathcal{B}}^\varepsilon \neq \emptyset. \end{aligned}$$

Proposition 6.3 (Invariance under represented natural isomorphism). *Fix a tolerance and let two constrained diagrams on the same context category be related by a natural isomorphism $\alpha : \mathcal{A} \Rightarrow \mathcal{B}$ satisfying $\alpha_c(\mathcal{A}_{\text{loc},\mathcal{A}}(c)) = \mathcal{A}_{\text{loc},\mathcal{B}}(c)$ for every object. Then α induces a bijection of compatible admissible-family spaces and preserves the L and D outcomes. If α and its inverse are computable and intertwine the effective realisation images, they also preserve K . If they intertwine the bounded images, they preserve R .*

Proof. Apply α componentwise. Naturality preserves every restriction equation, and the local-set hypothesis preserves objectwise admissibility. The inverse natural transformation gives the inverse map on global families. The additional computability and image hypotheses carry membership in the effective and bounded subsets in both directions. $\square$

Proposition 6.4 (Functoriality of global-family spaces). *Fix a context category and a tolerance. Let P and Q be constrained diagrams with assignment functors $\mathcal{A}^P$ and $\mathcal{A}^Q$. If $\alpha : \mathcal{A}^P \Rightarrow \mathcal{A}^Q$ is a natural transformation such that*

$$\alpha_c(\mathcal{A}_{\text{loc}}^P(c)) \subseteq \mathcal{A}_{\text{loc}}^Q(c) \quad (c \in \mathcal{C}),$$

then componentwise application of α defines a map

$$\Gamma_{\text{set}}(P) \longrightarrow \Gamma_{\text{set}}(Q).$$

If α also carries the effective, respectively bounded, realised images of P into those of Q , this map restricts to

$$\Gamma_{\text{eff}}(P) \rightarrow \Gamma_{\text{eff}}(Q), \quad \Gamma_{\mathcal{B}}(P) \rightarrow \Gamma_{\mathcal{B}}(Q).$$

Proof. For $s \in \Gamma_{\text{set}}(P)$, put $(\alpha_* s)_c = \alpha_c(s_c)$. Local admissibility is preserved by the displayed hypothesis. For an arrow $f : c \rightarrow c'$, naturality gives

$$\mathcal{A}^Q(f)(\alpha_c(s_c)) = \alpha_{c'}(\mathcal{A}^P(f)(s_c)) = \alpha_{c'}(s_{c'}),$$

so $\alpha_* s$ is compatible. The effective and bounded restrictions are exactly the stated image-preservation hypotheses. $\square$

Definition 6.5 (Extensional obstruction datum). For a fixed well-formed presentation P at tolerance ε , put

$$\lambda_\varepsilon(P) = \begin{cases} 1, & \mathcal{A}_{\text{loc},\varepsilon}(c) \neq \emptyset \text{ for every } c, \\ 0, & \text{otherwise,} \end{cases}$$

and define

$$\mathfrak{D}_\varepsilon(P) = (\lambda_\varepsilon(P), \Gamma_{\text{set}}^\varepsilon(P) \supseteq \Gamma_{\text{eff}}^\varepsilon(P) \supseteq \Gamma_{\mathcal{B}}^\varepsilon(P)).$$

An isomorphism $\mathfrak{D}_\varepsilon(P) \cong \mathfrak{D}_\varepsilon(Q)$ is a bijection of the set-theoretic global-family spaces carrying the effective and bounded subsets onto their counterparts and preserving λ_ε .

Definition 6.6 (Capability, global-family, and diagnostic equivalence). A *capability equivalence* between two presentations preserves the relevant policy or strategy space and the worldwide loss functional, and therefore preserves the capability predicate at every tolerance. A *global-family equivalence* preserves the nested global spaces $\Gamma_{\text{set}} \supseteq \Gamma_{\text{eff}} \supseteq \Gamma_{\mathcal{B}}$. A *diagnostic equivalence* at tolerance ε preserves the full obstruction datum $\mathfrak{D}_\varepsilon$, including the objectwise local-success bit. Capability equivalence need not imply diagnostic equivalence.

Proposition 6.7 (Obstruction-data transport). *An isomorphism $\mathfrak{D}_\varepsilon(P) \cong \mathfrak{D}_\varepsilon(Q)$ preserves and reflects each outcome*

$$\mathsf{L}_\varepsilon, \quad \mathsf{D}_\varepsilon, \quad \mathsf{K}_\varepsilon, \quad \mathsf{R}_\varepsilon, \quad \checkmark_\varepsilon.$$

If only the three nested global spaces are preserved, then K_ε , R_ε , and $\checkmark_\varepsilon$ are preserved and reflected, because they are Boolean combinations of emptiness facts about those spaces. The distinction between L_ε and D_ε need not be preserved without the local-success bit.

Proof. The local outcome is exactly $\lambda_\varepsilon = 0$. When $\lambda_\varepsilon = 1$, the compatibility outcome is emptiness of $\Gamma_{\text{set}}^\varepsilon$. After that space is nonempty, the effectivity outcome is emptiness of $\Gamma_{\text{eff}}^\varepsilon$; after the effective space is nonempty, the resource outcome is emptiness of $\Gamma_{\mathcal{B}}^\varepsilon$. A bijection carrying each distinguished subset onto its counterpart preserves all these emptiness and nonemptiness conditions. Without the local-success bit, emptiness of the set-theoretic global space does not record whether some local set was already empty. $\square$

Remark 6.8 (Diagnostic data and certificate data). The obstruction datum $\mathfrak{D}_\varepsilon$ is sufficient for transporting the five diagnostic outcomes because the outcomes use only local nonemptiness and the emptiness pattern of the three nested global spaces. It is not enough to transport the shape of a certificate. For that one must also remember the object set, the local admissibility sets, the restriction maps, and the way local elements occur as restrictions of compatible global families. Minimal diagnostic transport and certificate transport are therefore distinct requirements.

Definition 6.9 (Terminal extensional compression). For a constrained diagram P at tolerance ε , let $\text{ExtComp}_\varepsilon(P)$ be the terminal-category diagram with

$$\mathcal{A}(*) = \prod_c \mathcal{A}(c), \quad \mathcal{A}_{\text{loc},\varepsilon}(*) = \Gamma_{\text{set}}^\varepsilon(P).$$

Use as effective and bounded realisation images the original images in the same ambient product.

Theorem 6.10 (Compression boundary for L and D). *For every constrained diagram P ,*

$$\Gamma_{\chi}^{\varepsilon}(\text{ExtComp}_{\varepsilon}(P)) = \Gamma_{\chi}^{\varepsilon}(P) \quad (\chi \in \{\text{set}, \text{eff}, \mathcal{B}\}).$$

If P has outcome D_{ε} , its extensional compression has outcome L_{ε} . Consequently no equivalence that retains only the three nested global-family spaces can preserve the distinction between local and compatibility failure.

Proof. A terminal-category diagram has no nontrivial compatibility equations, so its global set is its sole local set, namely $\Gamma_{\text{set}}^{\varepsilon}(P)$. Intersecting with the unchanged effective and bounded images gives the other two displayed equalities. If P has outcome D_{ε} , all its local sets are nonempty but $\Gamma_{\text{set}}^{\varepsilon}(P) = \emptyset$. The sole local set of the compression is therefore empty, which is outcome L_{ε} . $\square$

Definition 6.11 (Obstruction spectrum and D-gap). For a tolerance-filtered presentation, its obstruction spectrum is

$$\text{Spec}(P) = (T_L(P), T_D(P), T_K(P), T_R(P)),$$

with the success sets of Definition 5.13. The compatibility gap is

$$G_D(P) = T_L(P) \setminus T_D(P).$$

Thus $\varepsilon \in G_D(P)$ exactly when the presentation has outcome D_{ε} .

Corollary 6.12 (Compression spectrum). *For a tolerance-filtered presentation, define $\text{ExtComp}(P)$ levelwise by applying Definition 6.9 at every ε . Then*

$$\text{Spec}(\text{ExtComp}(P)) = (T_D(P), T_D(P), T_K(P), T_R(P)).$$

Consequently $G_D(\text{ExtComp}(P)) = \emptyset$, and for each tolerance the only diagnostic state changed by terminal compression is D_{ε} , which becomes L_{ε} .

Proof. The compressed presentation has one local set at tolerance ε , namely $\Gamma_{\text{set}}^{\varepsilon}(P)$; hence its local-success set is $T_D(P)$. The terminal category has no nontrivial compatibility equations, so its set-theoretic global-family space is the same set $\Gamma_{\text{set}}^{\varepsilon}(P)$. The effective and bounded images are unchanged by construction, giving the T_K and T_R identities. The statement about G_D and the fixed-tolerance outcomes follows from the definitions of the five semantic outcomes. $\square$

Remark 6.13. The compression boundary says exactly what is lost when all constraints are packed into one object. The compression-invariant part of the spectrum is (T_D, T_K, T_R) . The part erased by compression is G_D , the tolerance region where local validity holds but gluing fails. Recovering that region requires local incidence and admissibility data in addition to the nested global-family spaces.

Corollary 6.14 (Compression thresholds and idempotence). *For levelwise terminal compression,*

$$\varepsilon_L(\text{ExtComp}(P)) = \varepsilon_D(P), \quad \varepsilon_D(\text{ExtComp}(P)) = \varepsilon_D(P), \quad \varepsilon_K(\text{ExtComp}(P)) = \varepsilon_K(P), \quad \varepsilon_R(\text{ExtComp}(P)) = \varepsilon_R(P).$$

Moreover, terminal compression is idempotent at the level of spectra and at each fixed tolerance:

$$\text{Spec}(\text{ExtComp}(\text{ExtComp}(P))) = \text{Spec}(\text{ExtComp}(P)).$$

If the finite difference $\delta_D(P) = \varepsilon_D(P) - \varepsilon_L(P)$ is defined, then the compressed presentation has $\delta_D = 0$.

Proof. The threshold identities follow by taking infima in the four success-set identities of Corollary 6.12. Applying Corollary 6.12 a second time leaves the spectrum unchanged because the first two components of the compressed spectrum are both $T_D(P)$. The final statement follows from the first two threshold identities. $\square$

Remark 6.15 (Fixed-tolerance compression signature). At a fixed tolerance, write

$$\lambda = \mathbf{1}[\text{all local sets are nonempty}], \quad s = \mathbf{1}[\Gamma_{\text{set}} \neq \emptyset], \quad e = \mathbf{1}[\Gamma_{\text{eff}} \neq \emptyset], \quad b = \mathbf{1}[\Gamma_{\mathcal{B}} \neq \emptyset].$$

Nestedness gives $s \geq e \geq b$. Terminal compression acts on this Boolean signature by

$$(\lambda, s, e, b) \mapsto (s, s, e, b).$$

Thus the unique changed diagnostic state is $(1, 0, 0, 0)$, the D state, which is sent to the L state $(0, 0, 0, 0)$. Table 2 tabulates this action on the well-formed signatures.

Table 2: Fixed-tolerance action of terminal compression on diagnostic signatures.

Original signature (λ, s, e, b)	Original outcome	Compressed outcome
$(0, 0, 0, 0)$	L	L
$(1, 0, 0, 0)$	D	L
$(1, 1, 0, 0)$	K	K
$(1, 1, 1, 0)$	R	R
$(1, 1, 1, 1)$	$\checkmark$	$\checkmark$

Proposition 6.16 (Diagnostic partition). *Every raw presentation has exactly one outcome among U and, when well formed, $L_\varepsilon, D_\varepsilon, K_\varepsilon, R_\varepsilon, \checkmark_\varepsilon$.*

Proof. A malformed presentation has outcome U. Assume $\text{WF}(P)$. If some local set is empty, L_ε holds. Otherwise all local sets are nonempty. Then either $\Gamma_{\text{set}}^\varepsilon$ is empty, giving D_ε , or it is nonempty. In the latter case either $\Gamma_{\text{eff}}^\varepsilon$ is empty, giving K_ε , or it is nonempty. Finally either $\Gamma_{\mathcal{B}}^\varepsilon$ is empty, giving R_ε , or it is nonempty, giving $\checkmark_\varepsilon$. The alternatives are pairwise disjoint at the first condition on which they differ. $\square$

Remark 6.17. The sequence is a diagnostic partition of a presentation at fixed tolerance and fixed realisation models, not an intrinsic order on physical mechanisms. The letter records the first failed filter in the chosen presentation; different mechanism-specific certificates may witness the same failed filter. Table 3 summarises the outcomes, certificates, and representative inhabitants; representative results are cited in the table. Figure 2 displays the same cascade as a decision flow, and the canonical presentations collected in Table 4 return throughout Sections 7–11.

7 Distributed selectors and canonical compatibility presentations

A single interface generates independent fibre constraints and cannot produce a compatibility obstruction in its canonical discrete presentation in ZFC. Distributed observation changes the situation. Several decision-makers may see different slices of the same world, and the same local response rule must be reused whenever the same local observation reappears. The obstruction is no longer that one fibre has no admissible value; it is that locally admissible response rules cannot be made to agree on their shared variables.

Table 3: Tolerance-relative outcomes and representative certificates.

Code	Defining condition	Certificate form	Representative result
U	malformed presentation	failed typing or functor law	well-formedness check
L_ε	some local set empty	finite or Helly witness	Theorem 4.5
D_ε	locals nonempty; no matching family	incompatible scope constraints	Corollary 7.14
K_ε	global families exist; no effective one	nonempty Π_1^0 class without computable point	Theorems 10.11 and 10.16
R_ε	effective families exist; no bounded one	cost-class separation	Theorems 10.27 and 10.29
$\checkmark_\varepsilon$	bounded effective family exists	realised bounded section	capability

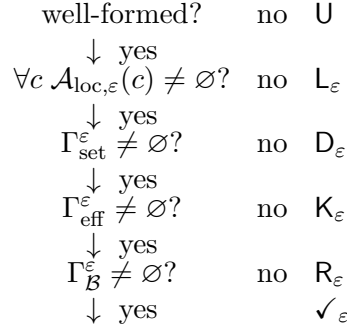


Figure 2: Diagnostic decision flow at fixed tolerance. The labels identify the first failed filter in the chosen presentation, not an invariant physical cause.

Table 4: Structural effect of canonical presentations on compatibility failure.

Presentation	Initial object?	Admissibility functor?	sub-	Compatibility outcome
fibre-indexed single observer	generally no	yes		no D in ZFC
terminal-category compression	yes	yes		no D
full causal patch site	yes	yes		no D
sensing cover incidence	no	yes		cover-relative D possible
distributed CSP incidence	generally no	not generally		D possible
nested reduced prefix chain	no	yes		set-theoretic gluing; K possible
empirical support presheaf	generally no	yes		D possible
Bell star diagram	yes	no		D possible

Definition 7.1 (Distributed selector problem). A distributed selector problem consists of a nonempty world set W , a finite agent set J , interfaces $I_j : W \rightarrow Z_j$, nonempty choice sets C_j , a nonempty active scope $S(w) \subseteq J$ for each world, and relations

$$V_w \subseteq \prod_{j \in S(w)} C_j.$$

A strategy is a tuple $\pi = (\pi_j)_{j \in J}$ with $\pi_j : Z_j \rightarrow C_j$. It is valid when

$$(\pi_j(I_j(w)))_{j \in S(w)} \in V_w \quad \text{for every } w \in W.$$

Definition 7.2 (Canonical distributed CSP and incidence diagram). For each agent define its active observation set

$$Z_j^{\text{act}} = \{I_j(w) : w \in W, j \in S(w)\}, \quad \mathcal{X}_{\text{act}} = \{(j, z) : j \in J, z \in Z_j^{\text{act}}\}.$$

The domain at (j, z) is C_j , and the scope of w is

$$\text{sc}(w) = \{(j, I_j(w)) : j \in S(w)\}.$$

Let the incidence category have as objects exactly the nonempty subsets of realised scopes, with a unique arrow from D to E precisely when $E \subseteq D$, and put

$$\mathcal{A}(D) = \prod_{(j,z) \in D} C_j.$$

For w with $\text{sc}(w) = D$, let

$$\theta_w : \prod_{j \in S(w)} C_j \rightarrow \mathcal{A}(D)$$

be the coordinate relabelling $(c_j)_{j \in S(w)} \mapsto (a_{(j, I_j(w))})_{(j, I_j(w)) \in D}$ with $a_{(j, I_j(w))} = c_j$. For each realised scope define

$$R_D = \bigcap_{\text{sc}(w)=D} \theta_w(V_w) \subseteq \mathcal{A}(D),$$

and set

$$\mathcal{A}_{\text{loc}}(D) = \begin{cases} R_D, & D \text{ is a realised scope,} \\ \mathcal{A}(D), & \text{otherwise.} \end{cases}$$

Compatibility is coordinate restriction agreement. Every unrealised object has $\mathcal{A}_{\text{loc}}(D) = \mathcal{A}(D) \neq \emptyset$; hence objectwise local nonemptiness is equivalent to $R_D \neq \emptyset$ for every realised scope. This is generally a constrained diagram rather than a subfunctor.

Remark 7.3 (Team interpretation). In the distributed setting, each agent writes one table from observations to choices. That table is reused across all worlds in which the agent sees the same observation. Each world then tests a tuple of table entries. Compatibility can fail because the same table entry is pulled into several tests and the tests force contradictory values for it.

Theorem 7.4 (Distributed selector–CSP–section equivalence). *For unrestricted local policy classes, the following are equivalent:*

- (i) *a valid distributed strategy exists;*

(ii) the canonical CSP has a solution;

(iii) the incidence diagram has a compatible admissible family.

If some realised-scope relation R_D is empty, the presentation has local outcome L_0 . If all objectwise sets are nonempty but no global family exists, it has compatibility outcome D_0 .

Proof. A strategy tuple restricts to an assignment on active variables by $\alpha(j, z) = \pi_j(z)$. Conversely, since every C_j is nonempty, an assignment on $\mathcal{X}_{\text{act}}$ extends by arbitrary values to observations never used while agent j is active, producing a full policy tuple. Validity at w is exactly the constraint $\alpha|_{\text{sc}(w)} \in R_{\text{sc}(w)}$, proving (i) $\Leftrightarrow$ (ii).

A compatible family is determined by singleton components on realised scopes, which define an assignment on $\mathcal{X}_{\text{act}}$. Compatibility forces every scope component to be its restriction, and local admissibility forces membership in R_D . Conversely, a CSP solution restricts to a compatible admissible family. This proves (ii) $\Leftrightarrow$ (iii). The outcome statements are the definitions at tolerance zero. $\square$

Definition 7.5 (Relational constraint datum). The relational constraint datum of a finite distributed selector problem is

$$\mathfrak{R}(P) = (\mathcal{X}_{\text{act}}, (C_x)_x, \mathcal{D}, (R_D)_{D \in \mathcal{D}}),$$

where $\mathcal{D}$ is the set of realised scopes and $C_{(j,z)} = C_j$. An isomorphism $\mathfrak{R}(P) \rightarrow \mathfrak{R}(P')$ consists of a bijection $\eta : \mathcal{X}_{\text{act}} \rightarrow \mathcal{X}'_{\text{act}}$ and domain bijections $\delta_x : C_x \rightarrow C'_{\eta(x)}$ such that $D \mapsto \eta(D)$ bijects the realised scopes and the product bijection $\delta_D = \prod_{x \in D} \delta_x$ satisfies

$$\delta_D(R_D) = R'_{\eta(D)} \quad (D \in \mathcal{D}).$$

Thus the datum includes the relations, not merely the scope hypergraph or its nerve.

Proposition 7.6 (Constraint-datum invariance). An isomorphism $\mathfrak{R}(P) \cong \mathfrak{R}(P')$ induces canonical bijections between the two CSP solution sets and between the compatible-family spaces of their canonical incidence diagrams. It preserves and reflects D_0 whenever the realised-scope relations are nonempty.

Proof. For a CSP assignment a , define

$$(\eta_* a)(\eta(x)) = \delta_x(a(x)).$$

For every realised scope D ,

$$a|_D \in R_D \iff (\eta_* a)|_{\eta(D)} = \delta_D(a|_D) \in R'_{\eta(D)}.$$

Hence $a \mapsto \eta_* a$ bijects the solution sets, with inverse induced by η^{-1} and the inverse domain bijections. Theorem 7.4 identifies each solution set with the compatible-family space of the canonical incidence diagram. Objectwise nonemptiness is preserved by the same domain bijections, so absence of a solution is a D_0 outcome on one side exactly when it is on the other. $\square$

Remark 7.7. Proposition 7.6 is an isomorphism theorem for the full relational constraint datum. Scope incidence or homotopy type alone does not determine compatibility, since changing the relations while retaining the same scopes can change solvability.

Definition 7.8 (Solution-saturated constraint datum). Let P be a finite distributed selector problem with nonempty CSP solution set $S(P)$. For every active variable x , put

$$C_x^S = \{a(x) : a \in S(P)\},$$

and, for every realised scope D , put

$$R_D^S = \{a|_D : a \in S(P)\} \subseteq \prod_{x \in D} C_x^S.$$

The *solution-saturated constraint datum* $\mathfrak{R}^S(P)$ retains the active variables and realised-scope family of $\mathfrak{R}(P)$ and replaces its domains and relations by $(C_x^S)_x$ and $(R_D^S)_D$.

A coordinate isomorphism from $S(P)$ to $S(P')$ consists of a variable bijection η that bijects the realised-scope family onto the realised-scope family, domain bijections $\delta_x : C_x^S \rightarrow C_{\eta(x)}^{S'}$, and a bijection $\Phi : S(P) \rightarrow S(P')$ satisfying

$$\Phi(a)(\eta(x)) = \delta_x(a(x)) \quad (a \in S(P), x \in \mathcal{X}_{\text{act}}).$$

Proposition 7.9 (Recovery of saturated constraint data). *For finite distributed selector problems with nonempty solution sets, the following are equivalent:*

- (i) *their solution sets are coordinate-isomorphic in the sense of Definition 7.8;*
- (ii) *their solution-saturated relational constraint data are isomorphic.*

Moreover, replacing $\mathfrak{R}(P)$ by $\mathfrak{R}^S(P)$ gives a solution set canonically identified with the original one under the inclusions $C_x^S \subseteq C_x$.

Proof. First consider the final assertion. Every $a \in S(P)$ has $a|_D \in R_D^S$ for each realised scope, so it solves the saturated system. Conversely, $R_D^S \subseteq R_D$ for every D . Hence every assignment solving the saturated system solves the original system and belongs to $S(P)$. The inclusion of the support-reduced product into the original product therefore canonically identifies the two solution sets.

Suppose (i) holds. For every realised scope D , coordinate preservation gives

$$\delta_D(R_D^S) = \{\Phi(a)|_{\eta(D)} : a \in S(P)\} = R_{\eta(D)}^{S'},$$

where $\delta_D = \prod_{x \in D} \delta_x$. Thus $(\eta, (\delta_x)_x)$ is an isomorphism of the saturated data, proving (ii).

Conversely, an isomorphism of the saturated data sends assignments coordinatewise to assignments and preserves every saturated scope relation. Proposition 7.6 therefore gives a bijection of the saturated solution sets. By the first paragraph these are $S(P)$ and $S(P')$, and the induced bijection is coordinate-preserving. This proves (i). $\square$

Remark 7.10. Proposition 7.9 is a recovery theorem for nonempty, support-reduced solution data. It is not a complete invariant of arbitrary presentations or of unsolvable systems: an empty solution space cannot recover unused domain elements or the local relations responsible for failure.

Corollary 7.11 (One-observer recovery). *If $J = \{j\}$, nonemptiness of $S(w)$ forces $S(w) = \{j\}$ for every world. Hence every realised scope is a singleton, the incidence category is discrete, and Theorem 7.4 reduces to fibrewise selection. Nonempty common-choice sets $\text{Adm}_V(z)$ give a selector in ZFC, so $\mathbf{D}$ is absent.*

Proof. Variables are the realised observations and constraints are precisely the common-choice sets $\text{Adm}_V(z)$. Apply discrete compatibility and choice. $\square$

Theorem 7.12 (Finite distributed capability is NP-complete). *For finite tabular distributed selector problems, deciding capability is NP-complete.*

Proof. Membership in NP follows by guessing one value for every variable (j, z) and checking every tabulated world relation.

For hardness, reduce 3SAT, which is NP-complete [9]. Use one agent per Boolean variable, a singleton observation set, and choice set $\{0, 1\}$. For each clause create a world whose active scope is the set of its variables and whose relation consists of the assignments satisfying that clause (with signs interpreted in the relation table). A distributed strategy is then a truth assignment, valid exactly when it satisfies every clause. $\square$

Remark 7.13 (Compatibility is not effectivity). Nonempty local relations do not imply a global distributed strategy. The team problem of Corollary 7.14 has nonempty local relations but no global member, so its outcome is D, not K. Section 10 supplies a distinct infinite represented K example with nested distributed scopes.

Corollary 7.14 (Private-information team obstruction). *Let two agents receive private bits x, y and respond with $u = f(x), v = g(y)$. The four relations*

$$u \oplus v = x \wedge y$$

are nonempty, but no distributed strategy satisfies all four. The induced incidence diagram has outcome D_0 .

Proof. The inputs $(0, 0), (0, 1), (1, 0)$ imply $f(0) = g(0) = g(1) = f(1)$, while $(1, 1)$ requires $f(1) \neq g(1)$. $\square$

Remark 7.15 (Deterministic CHSH core and Bell coupling). Identifying x, y with Clauser–Horne–Shimony–Holt (CHSH) settings and $f(x), g(y)$ with deterministic local responses makes these the CHSH-game winning equations. A deterministic response table is a pair (f, g) , and Corollary 7.14 proves that no such table wins all four inputs. This is the standard XOR-game form of CHSH, up to the conventional relabelling between Boolean outputs and $\{\pm 1\}$ outcomes. The Bell coupling theorem allows probability measures over the same deterministic tables. Thus the team theorem is the deterministic compatibility obstruction, while Bell locality is its distributional coupling relaxation. The one-shot extensive-form realisation is Theorem 11.40. Theorem 11.47 gives an extensional repeated-move compilation; preservation of temporal structure beyond the one-shot normal form remains conjectural.

The deterministic strategies of a finite Bell scenario are likewise distributed strategies with parties as agents and settings as local observations. The XOR team example is therefore the support-level deterministic core of the CHSH game: it has locally satisfiable input relations, and, as noted after Corollary 7.14, no single deterministic response table satisfies all of them, the probability-valued coupling diagram of Definition 9.23 being their convex relaxation. The join-tree theorem gives the acyclic, separator-consistent positive case for the scope hypergraph.

8 Quantitative decomposition

A single gap between unrestricted and realised policies conflates information, effectivity, and resources. Four optima are separated. Let $C = R \times A$ and, for $c = (r, a)$, put

$$L(w, c) = \ell(r, q(U(w, D(r), a))).$$

Definition 8.1 (Four worst-case degrees). Define

$$\begin{aligned} d_{\text{pt}} &= \inf_{c: W \rightarrow C} \sup_{w \in W} L(w, c(w)), \\ d_{\text{obs}} &= \inf_{s: \text{im}(I) \rightarrow C} \sup_{w \in W} L(w, s(I(w))), \\ d_{\text{eff}} &= \inf_{\pi \in \Pi_{\text{eff}}} \sup_{w \in W} L(w, s_{\pi}(I(w))), \\ d_{\mathcal{B}} &= \inf_{\pi \in \Pi_{\mathcal{B}}} \sup_{w \in W} L(w, s_{\pi}(I(w))), \end{aligned}$$

with $\inf \emptyset = +\infty$.

Proposition 8.2 (Nested degrees).

$$d_{\text{pt}} \leq d_{\text{obs}} \leq d_{\text{eff}} \leq d_{\mathcal{B}}.$$

Proof. A policy depending only on the observation is a special case of a world-dependent choice after precomposition with I . Effective policy selectors form a subset of all observation-indexed selectors, and bounded effective policies form a subset of effective policies. Taking infima over successively smaller classes gives the inequalities. $\square$

Proposition 8.3 (Chebyshev-radius formula). *Assume passive dynamics, singleton actions, report and target space equal to a metric space (Y, d) , identity decoder, and loss d . Put $f = q \circ U_0$ and*

$$\text{rad}(S) = \inf_{y \in Y} \sup_{x \in S} d(y, x) \in [0, \infty],$$

with value ∞ for unbounded S . Then

$$d_{\text{pt}} = 0, \quad d_{\text{obs}} = \sup_{z \in \text{im}(I)} \text{rad}(f(I^{-1}(z))).$$

Moreover

$$\frac{1}{2} \text{diam}(S) \leq \text{rad}(S) \leq \text{diam}(S)$$

for every nonempty bounded $S \subseteq Y$; for bounded subsets of $\mathbb{R}$, the first inequality is an equality. If $I = k \circ I'$, then

$$d_{\text{obs}}(I') \leq d_{\text{obs}}(I).$$

Proof. World-dependent reports choose $f(w)$ and give zero pointwise loss. For an observation-indexed report s , the worst loss on fibre z is $\sup_{w \in I^{-1}(z)} d(s(z), f(w))$. Taking the infimum independently at each fibre gives the displayed supremum of radii; in the declared ZFC foundation, choose a δ -approximate centre simultaneously in every fibre to establish the upper direction when centres are not attained. Finite observation images use only finite choice. The triangle inequality gives $\text{diam}(S) \leq 2 \text{rad}(S)$, while choosing a centre in S gives the upper bound. On $\mathbb{R}$, the midpoint of the infimum and supremum is a centre. Refinement replaces each fibre image by a subset, whose radius cannot increase. $\square$

Definition 8.4 (Adjacent gaps). When the adjacent values are finite, define

$$\Delta_{\text{uniform}} = d_{\text{obs}} - d_{\text{pt}}, \quad \Delta_{\text{effective}} = d_{\text{eff}} - d_{\text{obs}}, \quad \Delta_{\text{resource}} = d_{\mathcal{B}} - d_{\text{eff}}.$$

These are respectively the observation-uniformity, effectivity, and resource gaps.

Remark 8.5 (Quantitative gaps and diagnostic outcomes). The numerical gaps in Definition 8.4 are not the same data as the semantic outcomes L, D, K, R. The gaps compare optimal worst-case losses over nested classes of selectors, whereas the outcomes classify emptiness of presentation-level local and global spaces at a fixed tolerance. In particular, a single-observer discrete presentation can have a positive observation-uniformity gap while D is impossible in ZFC.

Proposition 8.6 (Nested upward-closed success sets). *Each T_χ is upward closed, and*

$$T_{\mathbf{R}} \subseteq T_{\mathbf{K}} \subseteq T_{\mathbf{D}} \subseteq T_{\mathbf{L}}, \quad \varepsilon_{\mathbf{L}} \leq \varepsilon_{\mathbf{D}} \leq \varepsilon_{\mathbf{K}} \leq \varepsilon_{\mathbf{R}}.$$

Status at an endpoint is determined by membership, not merely by comparison with the infimum.

Proof. Increasing tolerance preserves every local validity inequality and hence every witness at each successive level. The class inclusions give the nesting. Infima reverse the displayed success-set inclusions in the stated order. An infimum need not be attained, so endpoint membership must be checked separately. $\square$

Corollary 8.7 (Outcome at a fixed tolerance). *For a well-formed problem and fixed ε :*

- (i) $\varepsilon \notin T_{\mathbf{L}}$ gives outcome $\mathbf{L}_\varepsilon$;
- (ii) $\varepsilon \in T_{\mathbf{L}} \setminus T_{\mathbf{D}}$ gives $\mathbf{D}_\varepsilon$;
- (iii) $\varepsilon \in T_{\mathbf{D}} \setminus T_{\mathbf{K}}$ gives $\mathbf{K}_\varepsilon$;
- (iv) $\varepsilon \in T_{\mathbf{K}} \setminus T_{\mathbf{R}}$ gives $\mathbf{R}_\varepsilon$;
- (v) $\varepsilon \in T_{\mathbf{R}}$ gives outcome $\checkmark_\varepsilon$, i.e. bounded capability.

Proof. Apply the diagnostic partition of Proposition 6.16 to the four nested success conditions at tolerance ε . Moving down the nested chain $T_{\mathbf{L}} \supseteq T_{\mathbf{D}} \supseteq T_{\mathbf{K}} \supseteq T_{\mathbf{R}}$, each case records the first failed success test; if none fails, the outcome is $\checkmark_\varepsilon$. $\square$

Proposition 8.8 (Robustness under class enlargement). *Local emptiness and set-theoretic compatibility failure are invariant under enlargement of the implementation class. Effectivity and resource outcomes are relative to the corresponding realisation images.*

Proof. The local sets $\mathcal{A}_{\text{loc},\varepsilon}(c)$ and the set-theoretic global space $\Gamma_{\text{set}}^\varepsilon$ are defined before an implementation class is imposed. Changing that class cannot populate an empty local set and cannot change whether locally admissible families glue set-theoretically. By contrast, $\Gamma_{\text{eff}}^\varepsilon$ and $\Gamma_{\mathcal{B}}^\varepsilon$ are intersections with class-dependent images; enlarging either image can turn the corresponding empty intersection into a nonempty one. $\square$

9 Finite gluing, contextuality, and coupling

This section studies finite gluing from both sides. Some local data cannot be assembled, and the obstruction can be seen in a parity cycle or in a strongly contextual support model. Other local data glue for structural reasons: join trees give the acyclic positive case, affine systems reduce to linear algebra, and bijective networks reduce to holonomy fixed points. At the probabilistic level, Bell locality becomes a coupling problem: the local probability tables must be marginals of one probability measure on complete response tables.

9.1 Parity as a compatibility obstruction

Definition 9.1 (Triangle assignment diagram). Let $\mathcal{C}^\Delta$ have as objects the nonempty proper subsets of $\{0, 1, 2\}$ and an arrow $S \rightarrow T$ when $T \subseteq S$. Thus arrows point from edges to their vertices. Put $\mathcal{A}(S) = \{0, 1\}^S$ with ordinary restriction, and define

$$\mathcal{A}_{\text{loc}}(\{i\}) = \{0, 1\}^{\{i\}}, \quad \mathcal{A}_{\text{loc}}(\{i, j\}) = \{v : v(i) + v(j) \equiv 1 \pmod{2}\}.$$

Use standard compatibility.

Theorem 9.2 (Triangle compatibility obstruction). *Every local admissibility set in the triangle is nonempty, but $\Gamma_{\text{set}}(\mathcal{A}_{\text{loc}}) = \emptyset$. Hence its semantic obstruction is D.*

Proof. Each singleton admits two assignments and each edge admits the two odd-parity assignments. Suppose a compatible family existed. Agreement along edge-to-vertex arrows would determine bits $v(0), v(1), v(2)$ satisfying

$$v(0) + v(1) \equiv v(1) + v(2) \equiv v(2) + v(0) \equiv 1 \pmod{2}.$$

Adding the three equations gives $2(v(0) + v(1) + v(2)) \equiv 1 \pmod{2}$, while the left side is zero modulo two. This contradiction excludes a compatible family. $\square$

Remark 9.3. As a constraint-satisfaction instance, the example is an inconsistent odd-parity cycle. The six-object diagram adds explicit overlap objects and restriction arrows, thereby exhibiting which local assignments must agree and why the failure is one of gluing rather than local satisfiability. The Lean artifact checks the parity core and local non-emptiness, not the full categorical packaging.

9.2 Finite empirical support models

Definition 9.4 (Finite empirical model). Let X be finite and let M be a finite antichain of nonempty maximal contexts with $X = \bigcup M$. Each measurement m has a nonempty finite outcome set O_m . A possibilistic empirical model assigns a nonempty support $e_C \subseteq \prod_{m \in C} O_m$ to each $C \in M$ and satisfies overlap compatibility:

$$\{s|_{C \cap C'} : s \in e_C\} = \{s'|_{C \cap C'} : s' \in e_{C'}\} \quad (C, C' \in M)$$

[1]. Strong contextuality means that no $g \in \prod_{m \in X} O_m$ has $g|_C \in e_C$ for every $C \in M$.

Definition 9.5 (Downward-closed packaging). Let

$$\downarrow M = \{D \subseteq X : D \subseteq C \text{ for some } C \in M\}.$$

Use objects $D \in \downarrow M$, arrows $D \rightarrow D'$ for $D' \subseteq D$, and

$$\mathcal{A}(D) = \prod_{m \in D} O_m$$

with coordinate restriction. For any maximal $C \supseteq D$, define the restricted support

$$e_D = \{s|_D : s \in e_C\}.$$

Overlap compatibility makes this independent of the choice of maximal C . Set $\mathcal{A}_{\text{loc}}(D) = e_D$ for every $D \in \downarrow M$. For $D = \emptyset$ the assignment space and support are the singleton empty assignment. Standard compatibility defines $\text{Pack}(e)$.

Proposition 9.6 (Restriction stability of empirical admissibility). *The admissibility sets of $\text{Pack}(e)$ form a subfunctor of the assignment functor.*

Proof. Let $D' \subseteq D$ and $s \in e_D$. Choose a maximal context $C \supseteq D$ and $t \in e_C$ with $t|_D = s$. Then $s|_{D'} = t|_{D'} \in e_{D'}$. Thus coordinate restriction sends $\mathcal{A}_{\text{loc}}(D)$ into $\mathcal{A}_{\text{loc}}(D')$, and identity and composition laws are inherited from ordinary restriction. $\square$

Theorem 9.7 (Strong contextuality as gluing failure). *A finite empirical model is strongly contextual if and only if $\text{Pack}(e)$ has nonempty local admissibility sets and no compatible global family.*

Proof. A compatible family determines one value at every singleton context. Coverage combines these values into a global assignment g on X . Compatibility then forces the component at every D to be $g|_D$, and maximal-context validity gives $g|_C \in e_C$ for every $C \in M$. Conversely, any such global assignment defines a compatible family by restriction. Thus a compatible family exists exactly when the model is not strongly contextual. Since every local set is nonempty, its absence is a D obstruction. $\square$

Proposition 9.8 (Empirical supports as distributed selectors). *A finite empirical support model is the distributed selector problem whose agents are measurements with singleton interfaces $Z_m = \{*\}$, whose world set is the set of maximal contexts, whose active scope at context C is C , whose outcome choices are O_m , and whose world relation is e_C . The restriction-stable support packaging of Definition 9.5 refines its canonical incidence diagram and has the same global families. Thus distributed capability holds exactly when the model is not strongly contextual.*

Proof. The variables are measurement–outcome coordinates, and the scope relation at C is exactly e_C . The canonical diagram leaves proper subsopes unconstrained, while the support packaging records their projected supports. Every global assignment satisfying the maximal supports automatically satisfies those projections, and conversely maximal-context validity is unchanged. Thus the two global-family spaces coincide. Apply Theorem 7.4. $\square$

Remark 9.9. The result represents the established global-section criterion in the same typed assignment spaces used for effectivity and resources. The restriction-stability proved in Proposition 9.6 is internal to a fixed empirical model; functoriality of the construction $e \mapsto \text{Pack}(e)$ on noninvertible morphisms of empirical models remains Conjecture 13.2.

9.3 Acyclic gluing and finite solution count

Theorem 9.10 (Join-tree gluing). *Let V be a finite tree vertex set, let every C_i be a finite context, and let every outcome set O_x be nonempty. Assume the running-intersection property: for every variable x , the vertices $\{i : x \in C_i\}$ induce a connected subtree. Let $R_i \subseteq \prod_{x \in C_i} O_x$ be nonempty finite relations and define*

$$\bowtie_i R_i = \left\{ t \in \prod_{x \in \bigcup_i C_i} O_x : t|_{C_i} \in R_i \ \forall i \right\}.$$

If for every tree edge $i-j$,

$$\text{proj}_{C_i \cap C_j}(R_i) = \text{proj}_{C_i \cap C_j}(R_j),$$

then the natural join $\bowtie_i R_i$ is nonempty.

Proof. Root the tree and process vertices in root-to-leaf order. Choose a tuple in the root relation. Suppose a tuple has been chosen at a parent i and let j be a child. Its value on $C_i \cap C_j$ lies in the parent projection and therefore, by projection equality, in the child projection. Choose a tuple in R_j with that separator value. Continue recursively through the finite tree. The induction invariant is that the union of tuples chosen at processed vertices is a well-defined partial assignment. When a child j is added, any variable shared with an earlier processed vertex lies, by running intersection, in the parent separator $C_i \cap C_j$, where agreement was enforced.

If a variable occurs in two chosen contexts, the running-intersection property places it in every context and separator on the path between them. The recursive choices therefore give it the same value throughout. The chosen tuples hence have a well-defined union, and that union restricts into every R_i . $\square$

The theorem is the positive acyclic counterpart of the parity cycle and is the local-to-global principle behind join-tree database schemes [3].

Theorem 9.11 (Finite solution count). *Suppose a finite category has passed the well-formedness check, and all objects, arrows, assignment spaces, admissibility predicates, equality tests, and restriction maps are given by finite effective tables satisfying identity, composition, and functor laws. Define*

$$N(P) = \#\{s \in \prod_c \mathcal{A}(c) : s_c \in \mathcal{A}_{\text{loc}}(c) \ \forall c, \ \mathcal{A}(f)(s_c) = s_{c'} \ \forall f : c \rightarrow c'\}.$$

Call $N(P)$ the finite solution count. It is computable, and a well-formed problem has obstruction D exactly when every $\mathcal{A}_{\text{loc}}(c)$ is nonempty and $N(P) = 0$.

Proof. Enumerate the finite product of assignment spaces. The supplied decision tables determine local admissibility and every compatibility equation, so counting the families that pass all tests computes $N(P)$. Local non-emptiness excludes L , and $N(P) = 0$ is precisely absence of a compatible admissible family. $\square$

The count is presentation-dependent and is not called an invariant. Under an explicit relational encoding, existence is a finite CSP problem and counting is a finite counting-CSP problem.

9.4 Application: finite source integration and counting

Finite source integration provides a concrete tabular instance of the compatibility diagnostic. A source-integration instance consists of a finite attribute set X , nonempty finite domains $(O_x)_{x \in X}$, and local sources $(U_i, R_i)_{i=1}^m$ with $U_i \subseteq X$ and

$$R_i \subseteq \prod_{x \in U_i} O_x.$$

A global record is an element $t \in \prod_{x \in X} O_x$, and it integrates the sources when $t|_{U_i} \in R_i$ for every i . The integration count is

$$N_{\text{int}} = \#\{t \in \prod_{x \in X} O_x : t|_{U_i} \in R_i \text{ for all } i\}.$$

For each realised source scope $U \subseteq X$, let

$$\bar{R}_U = \bigcap_{i: U_i = U} R_i$$

be the aggregated relation for all sources with that scope. Then global records integrate the original sources exactly when their restrictions lie in $\bar{R}_U$ for every realised source scope U .

The aggregated source relations are the local tables of the canonical incidence presentation. An integrating record is a global row whose projections lie in all of those tables. The D outcome captures the situation in which every aggregated local table is nonempty but the natural join is empty; if some aggregated relation is empty, the obstruction is local in this presentation.

Proposition 9.12 (Source integration as a distributed selector). *Every finite source-integration instance has a canonical distributed-selector presentation whose valid distributed strategies are exactly the integrating global records.*

Proof. Use one distributed agent for each attribute $x \in X$, give it singleton observation set $\{*\}$ and choice set O_x , and create one world w_i for each source. The active scope at w_i is U_i , and the world relation is R_i under the evident coordinate relabelling. A distributed strategy is a global record $(a_x)_{x \in X}$, and validity at w_i is exactly $(a_x)_{x \in U_i} \in R_i$. Theorem 7.4 identifies these strategies with compatible admissible families of the canonical incidence diagram. $\square$

Corollary 9.13 (Source-integration diagnostics). *For the canonical integration presentation at tolerance zero, if at least one aggregated realised-source-scope relation $\bar{R}_U$ is empty, the outcome is L_0 ; nonempty aggregated relations with no integrating global record give outcome D_0 ; and a global record gives set-theoretic capability. Effective and bounded diagnostics are obtained by intersecting the global-record space with the declared realisation images.*

Proof. In the distributed presentation of Proposition 9.12, all worlds with the same realised source scope have the same realised scope. The local relation attached to that scope in Definition 7.2 is therefore the intersection of the corresponding source relations, namely $\bar{R}_U$. The three cases are then exactly the local, compatibility, and set-theoretic-success cases of the diagnostic partition. $\square$

Theorem 9.14 (Counting source integrations is $\#P$ -complete). *For explicitly tabulated finite source-integration instances, the function N_{int} is $\#P$ -complete. Hardness holds with Boolean domains, source scopes of size at most three, and nonempty source relations.*

Proof. Membership in $\#P$ follows by nondeterministically guessing one value for each attribute and checking all source tables in polynomial time.

For hardness, reduce parsimoniously from $\#3\text{SAT}$, which is $\#P$ -complete [19]. Given a 3CNF formula, use its variables as Boolean attributes. For each clause, make a source whose scope is the set of variables appearing in the clause and whose relation is the set of satisfying assignments to those variables. Each relation is nonempty and has arity at most three. A global record satisfies all sources exactly when the corresponding truth assignment satisfies all clauses, so the number of integrating records is the number of satisfying assignments. Thus exact counting is $\#P$ -hard, and membership gives completeness. $\square$

Corollary 9.15 (Counting compatible families). *Counting compatible admissible families in canonical incidence diagrams of finite distributed selector problems is $\#P$ -complete, even for Boolean choice sets and ternary realised scopes.*

Proof. Membership in $\#P$ follows by nondeterministically guessing one value for each active variable and checking all explicitly tabulated realised-scope relations. Hardness follows from the source-integration subclass: by Theorem 7.4, valid distributed strategies, CSP solutions, and compatible admissible families of the canonical incidence diagram are cardinality-preservingly identified. Apply Theorem 9.14. $\square$

Remark 9.16 (Counting and presentation). The number of compatible families is invariant under isomorphism of the explicit relational constraint datum, and under the canonical CSP–incidence bijection of Theorem 7.4. It is not an invariant of an arbitrary repackaging: terminal compression and support saturation can preserve existence while changing local incidence and certificate structure.

Remark 9.17 (Compression of source integration). Terminal compression replaces the source-level incidence diagram by a single object whose local set is the global join. When the join is empty, the compressed presentation has outcome L_0 regardless of whether the original source relations were nonempty. Thus source integration instantiates the compression boundary between local inconsistency and compatibility failure.

Theorem 9.18 (Affine compatibility certificate). *Let X be a finite global variable set over a fixed finite field. Suppose each context is a subset $c \subseteq X$, its assignment space is $\mathbb{F}^c$, arrows are coordinate projections, and every local admissibility relation is specified by affine linear equations in those coordinates. Assume that a compatible family is induced by a single global assignment $x \in \mathbb{F}^X$, as holds for ordinary agreement on overlaps when the context diagram enforces all shared variable identifications. Pad each local matrix with zero columns outside its context and stack the equations into one global system $Ax = b$. Such a compatible global assignment exists iff*

$$\text{rank}(A) = \text{rank}([A \mid b]).$$

Consequently compatibility on this subclass is decidable in polynomial time by Gaussian elimination.

Proof. A global assignment is compatible and locally admissible exactly when it satisfies every local affine equation, hence exactly when it solves $Ax = b$. The standard rank criterion characterises consistency of a finite linear system, and Gaussian elimination computes both ranks in polynomial time. $\square$

Proposition 9.19 (Affine cokernel obstruction). *In the setting of Theorem 9.18, regard the stacked matrix as a linear map*

$$A : \mathbb{F}^X \longrightarrow \mathbb{F}^m$$

and put $H_A^1 = \text{coker}(A)$. Then

$$Ax = b \text{ is solvable} \iff [b] = 0 \text{ in } H_A^1.$$

If every local affine relation is nonempty, the canonical compatibility presentation has outcome D_0 exactly when $[b] \neq 0$.

Proof. The class $[b]$ vanishes in the cokernel precisely when $b \in \text{im}(A)$, which is equivalent to the existence of $x \in \mathbb{F}^X$ with $Ax = b$. By Theorem 9.18, such an x is exactly a compatible globally admissible assignment. Under the additional local-nonemptiness hypothesis, failure of this global equation is D_0 rather than L_0 . $\square$

Theorem 9.20 (Bijective-network holonomy). *Let $G = (X, E)$ be a finite connected graph. Give each vertex x a nonempty finite domain C_x . For every oriented edge $e : u \rightarrow v$, let its constraint be the graph of a bijection $\rho_e : C_u \rightarrow C_v$, with $\rho_{\bar{e}} = \rho_e^{-1}$. Fix a base vertex x_0 and a spanning tree K . For each vertex v , let $\theta_v : C_{x_0} \rightarrow C_v$ be transport along the unique path in K . Choose one orientation $e : u \rightarrow v$ for each non-tree edge. For such an oriented non-tree edge, put*

$$h_e = \theta_v^{-1} \circ \rho_e \circ \theta_u \in \text{Aut}(C_{x_0}).$$

Then the global solution set is canonically in bijection with

$$\bigcap_{e \in E \setminus K} \text{Fix}(h_e),$$

with the empty intersection understood as C_{x_0} . The result is independent of the chosen orientations because a permutation and its inverse have the same fixed-point set. Equivalently, it is the common fixed-point set of all closed-walk holonomies based at x_0 .

Proof. A value $c_0 \in C_{x_0}$ determines values $c_v = \theta_v(c_0)$ satisfying every tree-edge constraint. A non-tree edge $e : u \rightarrow v$ is satisfied exactly when

$$\rho_e(\theta_u(c_0)) = \theta_v(c_0),$$

which is equivalent to $h_e(c_0) = c_0$. Hence common fixed points give, and are given by, global solutions. The fundamental closed walks determined by the non-tree edges generate every closed walk, and transport respects concatenation and reversal, giving the equivalent formulation. $\square$

Corollary 9.21 (Bijective-cycle fixed-point certificate). *For a cycle with edge bijections $\rho_i : C_i \rightarrow C_{i+1}$, with indices read modulo n so that $C_n = C_0$, the global solution set is in bijection with*

$$\text{Fix}(\rho_{n-1} \circ \cdots \circ \rho_0).$$

Its canonical incidence diagram has outcome D_0 exactly when this fixed-point set is empty.

Proof. Choose the cycle with one edge deleted as the spanning tree in Theorem 9.20. The sole non-tree edge produces the displayed cycle composite. Every edge relation and singleton projection is nonempty, so absence of a global solution is a compatibility outcome. $\square$

Remark 9.22. For binary odd parity on a cycle, each edge transports a bit either by the identity or by complementation; an odd number of complementations gives the fixed-point-free bit flip. The deterministic CHSH equations similarly form a four-cycle with three equality transports and one complementation. These results concern graphs of bijective binary constraints. General relations need not define transports, a nonidentity permutation may have fixed points, and no higher-arity, homotopy, or cohomological completeness statement is inferred.

The cokernel class of Proposition 9.19 is complete for the stated affine subclass. General non-linear contextual relations require additional certificate theories beyond the affine cokernel case.

9.5 Finite Bell coupling

The Bell presentation uses the same local-to-global idea with probabilities. The local data are observed distributions for different measurement settings. Compatibility asks whether these distributions can be coupled into a single probability measure over deterministic response tables. This is the finite local polytope viewpoint, expressed in the same diagrammatic language as the set-valued examples.

Definition 9.23 (Distributional coupling diagram). For a finite Bell scenario, let X_i and O_i be the settings and outcomes of party i , and fix a behaviour $p = (p_{\mathbf{x}})_{\mathbf{x}}$ with $p_{\mathbf{x}} \in \Delta(\prod_i O_i)$ for every setting tuple. Write $p_{\mathbf{x}}(\mathbf{o})$ equivalently as $p(\mathbf{o} \mid \mathbf{x})$. Put $\Omega = \prod_i O_i^{X_i}$, and for a setting tuple $\mathbf{x} = (x_i)_i$ define

$$\pi_{\mathbf{x}} : \Omega \rightarrow \prod_i O_i, \quad \pi_{\mathbf{x}}(\omega) = (\omega_i(x_i))_i.$$

Let $\mathcal{C}_{\text{Bell}}$ be the star-shaped restriction category with a global object $\top$, one object for each setting tuple $\mathbf{x}$, and arrows $\top \rightarrow \mathbf{x}$. Define

$$\mathcal{A}(\top) = \Delta(\Omega), \quad \mathcal{A}(\mathbf{x}) = \Delta\left(\prod_i O_i\right),$$

with arrow map $(\pi_{\mathbf{x}})_*$. Put

$$\mathcal{A}_{\text{loc}}(\top) = \Delta(\Omega), \quad \mathcal{A}_{\text{loc}}(\mathbf{x}) = \{p_{\mathbf{x}}\}.$$

This is a constrained assignment diagram, not generally a descent subfunctor: a measure at $\top$ restricts into the singleton local set at every setting object exactly when it lies in the coupling set below. Its compatible-family space is determined by

$$\text{Coupl}(p) = \{\mu \in \Delta(\Omega) : (\pi_{\mathbf{x}})_*\mu = p_{\mathbf{x}} \text{ for every } \mathbf{x}\}.$$

Theorem 9.24 (Finite local-behaviour coupling). *For a finite Bell behaviour, the following are equivalent:*

- (i) *the behaviour has a local hidden-variable model, equivalently lies in the finite convex hull of deterministic response tables;*
- (ii) $\text{Coupl}(p) \neq \emptyset$;
- (iii) *its distributional coupling diagram has a compatible global family.*

Thus a nonlocal finite behaviour gives a distributional D obstruction in this presentation. No nonsignalling assumption is needed for this equivalence; signalling behaviours are simply outside the local convex hull as well.

Proof. The proof treats either standard formulation of locality.

If locality is defined by deterministic tables, then for some $\mu \in \Delta(\Omega)$,

$$p(\mathbf{o}|\mathbf{x}) = \sum_{\omega \in \Omega} \mu(\omega) \prod_i \mathbf{1}_{\{o_i = \omega_i(x_i)\}},$$

which is exactly $(\pi_{\mathbf{x}})_*\mu(\mathbf{o})$.

For the stochastic-kernel formulation, suppose the maps $\lambda \mapsto p_i(o|x, \lambda)$ are measurable and

$$p(\mathbf{o}|\mathbf{x}) = \int_{\Lambda} \prod_i p_i(o_i|x_i, \lambda) d\nu(\lambda).$$

For each λ define

$$\mu_{\lambda}(\omega) = \prod_i \prod_{x \in X_i} p_i(\omega_i(x)|x, \lambda), \quad \mu(\omega) = \int_{\Lambda} \mu_{\lambda}(\omega) d\nu(\lambda).$$

The coordinate sets are finite, and summing μ_{λ} over all tables gives one, so $\mu \in \Delta(\Omega)$. For fixed $\mathbf{x}, \mathbf{o}$, interchange the finite sum with the integral. Summation over unconstrained table coordinates contributes normalized factors one, leaving

$$(\pi_{\mathbf{x}})_*\mu(\mathbf{o}) = \int_{\Lambda} \prod_i p_i(o_i|x_i, \lambda) d\nu(\lambda) = p(\mathbf{o}|\mathbf{x}).$$

Thus every stochastic local model gives a coupling.

Conversely, if $\mu \in \text{Coupl}(p)$, take hidden variable $\omega \in \Omega$ with law μ and deterministic responses

$$p_i(o|x, \omega) = \begin{cases} 1, & o = \omega_i(x), \\ 0, & \text{otherwise.} \end{cases}$$

Their induced distributions are $(\pi_{\mathbf{x}})_*\mu = p_{\mathbf{x}}$. Finally, a compatible family in the star diagram is a measure at $\top$ whose pushforward at every setting object is $p_{\mathbf{x}}$, exactly the coupling condition. $\square$

Remark 9.25 (Presentation of Bell nonlocality). The D classification in Theorem 9.24 belongs to the star presentation of Definition 9.23. In that presentation the top object allows all probability measures on deterministic response tables, and the setting objects prescribe the observed marginals. If the top local admissibility set is instead replaced by $\text{Coupl}(p)$, coupling failure becomes local emptiness at that object. The invariant content is coupling feasibility, equivalently membership in the finite local polytope, not the diagnostic letter after arbitrary repackaging.

Definition 9.26 (Bell coupling matrix). Let

$$Q = \left\{ (\mathbf{x}, \mathbf{o}) : \mathbf{x} \in \prod_i X_i, \mathbf{o} \in \prod_i O_i \right\}.$$

For $(\mathbf{x}, \mathbf{o}) \in Q$ and $\omega \in \Omega$, define

$$A_{(\mathbf{x}, \mathbf{o}), \omega} = \mathbf{1}_{\{\pi_{\mathbf{x}}(\omega) = \mathbf{o}\}},$$

and regard a behaviour as the vector $p = (p_{\mathbf{x}}(\mathbf{o}))_{(\mathbf{x}, \mathbf{o}) \in Q}$.

Theorem 9.27 (Bell coupling dual certificate). *For every finite Bell behaviour p , exactly one of the following alternatives holds:*

- (i) *there is $\mu \in \Delta(\Omega)$ with $(\pi_{\mathbf{x}})_*\mu = p_{\mathbf{x}}$ for every $\mathbf{x}$;*
- (ii) *there is $\gamma \in \mathbb{R}^Q$ such that*

$$A^\top \gamma \geq 0, \quad \langle \gamma, p \rangle < 0.$$

Equivalently, coupling failure has a linear inequality certificate which is nonnegative on every finite local behaviour and negative on p . If p is rational, γ may be chosen rational.

Proof. For a nonnegative vector $\mu \in \mathbb{R}^\Omega$,

$$(A\mu)_{(\mathbf{x}, \mathbf{o})} = \sum_{\omega: \pi_{\mathbf{x}}(\omega) = \mathbf{o}} \mu(\omega) = (\pi_{\mathbf{x}})_*\mu(\mathbf{o}).$$

Thus the coupling equations are $A\mu = p$ with $\mu \geq 0$. Summing these equations over $\mathbf{o}$ for any fixed $\mathbf{x}$ gives $\sum_{\omega} \mu(\omega) = \sum_{\mathbf{o}} p_{\mathbf{x}}(\mathbf{o}) = 1$, so normalisation follows automatically.

Farkas' lemma gives exactly one of

$$\exists \mu \geq 0 (A\mu = p) \quad \text{and} \quad \exists \gamma (A^\top \gamma \geq 0, \langle \gamma, p \rangle < 0)$$

[28]. If $q = A\nu$ is local, where $\nu \in \Delta(\Omega)$, then

$$\langle \gamma, q \rangle = \langle A^\top \gamma, \nu \rangle \geq 0.$$

Conversely, nonnegativity on every deterministic local table is precisely the coordinate inequality $A^\top \gamma \geq 0$. For rational A and p , the rational form of Farkas' lemma supplies a rational separating vector. $\square$

Corollary 9.28 (Finite Bell certificates). *A finite nonlocal behaviour has a finite Bell-inequality certificate. For rational explicit input tables, coupling feasibility is decidable by linear programming, and an infeasible instance has a certificate verifiable by finite rational arithmetic. The polynomial bound is measured in the encoded size of the explicit matrix A ; that matrix may be exponentially larger than a compressed scenario description.*

Proof. Apply Theorems 9.24 and 9.27. Standard rational linear programming decides feasibility in time polynomial in the explicit matrix and bit lengths, and the displayed inequalities verify the dual witness. $\square$

9.6 Physical Bell obstruction, robustness, and repair

The preceding coupling theorem answers a structural question: do the observed setting-indexed probability tables have a common hidden-variable coupling? This subsection gives that question its physical reading. The global object is a hidden-variable space of complete response tables; the setting objects are the experimentally observed marginals. A D outcome in this presentation means that the observed behaviour cannot be embedded in the local-causal model without relaxing one of its assumptions. The same formal obstruction then has three operational faces: a Bell inequality, a finite-sample statistical test, and a family of possible physical repairs.

Definition 9.29 (Local-causal Bell model). For a finite Bell scenario, a local-causal model of a behaviour p consists of a finite hidden-variable space Λ , a probability distribution $\mu \in \Delta(\Lambda)$ independent of the setting tuple, and local response kernels

$$p_i(o_i \mid x_i, \lambda) \quad (i = 1, \dots, n)$$

such that

$$p(\mathbf{o} \mid \mathbf{x}) = \sum_{\lambda \in \Lambda} \mu(\lambda) \prod_i p_i(o_i \mid x_i, \lambda).$$

The independence of μ from $\mathbf{x}$ is the measurement-independence, or freedom-of-choice, assumption. For finite scenarios this stochastic formulation is equivalent to a probability measure on deterministic response tables $\Omega = \prod_i O_i^{X_i}$, by the product-table construction in the proof of Theorem 9.24.

Proposition 9.30 (Local causality implies no-signalling). *Every finite local-causal Bell behaviour is nonsignalling. That is, for every party i , setting x_i , local outcome o_i , and settings $\mathbf{x}_{-i}, \mathbf{x}'_{-i}$ of the other parties,*

$$\sum_{\mathbf{o}_{-i}} p(o_i, \mathbf{o}_{-i} \mid x_i, \mathbf{x}_{-i}) = \sum_{\mathbf{o}_{-i}} p(o_i, \mathbf{o}_{-i} \mid x_i, \mathbf{x}'_{-i}).$$

Proof. Using the local-causal representation and summing over $\mathbf{o}_{-i}$, all factors belonging to parties other than i sum to one. The marginal becomes

$$\sum_{\lambda \in \Lambda} \mu(\lambda) p_i(o_i \mid x_i, \lambda),$$

which is independent of $\mathbf{x}_{-i}$. This proves the displayed equality. $\square$

Remark 9.31 (Scope of no-signalling). The conclusion is the operational no-signalling condition. It follows from measurement independence together with local response factorisation. The converse fails: in the binary-input, binary-output Popescu–Rohrlich (PR) box behaviour [13], outputs satisfy $a \oplus b = xy$ with probability $1/2$ on each satisfying pair. This behaviour is nonsignalling but has no local coupling. Bell nonlocality is therefore a stronger obstruction than signalling.

Definition 9.32 (Bell functionals, local bounds, and visibility). A Bell functional is a real family $\beta = (\beta_{\mathbf{x}}(\mathbf{o}))_{\mathbf{x}, \mathbf{o}}$. Its unweighted value on a behaviour is

$$\beta(p) = \sum_{\mathbf{x}, \mathbf{o}} \beta_{\mathbf{x}}(\mathbf{o}) p_{\mathbf{x}}(\mathbf{o}),$$

and its local bound is

$$B(\beta) = \max_{\ell \in \mathcal{L}} \beta(\ell),$$

where $\mathcal{L}$ is the finite local polytope. Thus $\beta(q) \leq B(\beta)$ is valid for every local behaviour q . For a local noise behaviour $q \in \mathcal{L}$, define the critical visibility of p along the segment from q to p by

$$v_c(p; q) = \sup\{v \in [0, 1] : vp + (1 - v)q \in \mathcal{L}\}.$$

Proposition 9.33 (Farkas certificates as Bell inequalities). *A Farkas certificate in Theorem 9.27 is equivalently a Bell inequality separating p from the local polytope, up to sign and affine terms vanishing on the normalisation constraints of the behaviour space.*

Proof. The condition $A^T \gamma \geq 0$ and $\langle \gamma, p \rangle < 0$ says that the linear functional γ is nonnegative on every deterministic local behaviour and negative on p . Equivalently, $\beta = -\gamma$ has value larger on p than on every deterministic local behaviour, hence larger than on every convex combination of deterministic behaviours. This is a Bell inequality with local bound $B(\beta) = \max_{\ell \in \mathcal{L}} \beta(\ell)$. Adding a functional that is constant on the affine normalisation subspace changes neither the validity of the inequality nor its violation by p . $\square$

Theorem 9.34 (Visibility from Bell inequalities). *For $q \in \mathcal{L}$,*

$$v_c(p; q) = \min \left(1, \inf_{\beta(p) > \beta(q)} \frac{B(\beta) - \beta(q)}{\beta(p) - \beta(q)} \right),$$

where the infimum ranges over Bell functionals valid on $\mathcal{L}$ and is understood as $+\infty$ if the indexing set is empty. If $p \in \mathcal{L}$, then $v_c(p; q) = 1$; if $p \notin \mathcal{L}$, then $v_c(p; q) < 1$.

Proof. The feasible set

$$\{v \in [0, 1] : vp + (1 - v)q \in \mathcal{L}\}$$

is a closed interval containing 0, because $\mathcal{L}$ is a compact convex polytope and $q \in \mathcal{L}$. Every valid Bell inequality gives

$$\beta(vp + (1 - v)q) = v\beta(p) + (1 - v)\beta(q) \leq B(\beta).$$

When $\beta(p) > \beta(q)$, this is exactly the displayed upper bound on v . Taking the infimum over such inequalities gives an upper bound for v_c .

If $p \in \mathcal{L}$, convexity gives feasibility for all $v \in [0, 1]$. If $p \notin \mathcal{L}$, closedness gives a boundary point $r = v_c p + (1 - v_c)q \in \mathcal{L}$ with $v_c < 1$. Since the segment immediately leaves the closed convex set at r in the direction $p - q$, that direction is not in the tangent cone of $\mathcal{L}$ at r . The separating-hyperplane theorem in this pointed direction supplies a supporting Bell functional β at r whose local bound is attained at r and whose value increases in the direction from q to p . In particular, $\beta(p) > \beta(q)$, so the infimum ranges over a nonempty class of functionals whenever $p \notin \mathcal{L}$. For this β , the displayed ratio is v_c , so the upper bound is tight. $\square$

Example 9.35 (CHSH visibility). In the two-party, two-setting, binary-outcome CHSH scenario, write outcomes as ± 1 and put

$$S(p) = E_{00}(p) + E_{01}(p) + E_{10}(p) - E_{11}(p), \quad E_{xy}(p) = \sum_{a,b=\pm 1} ab p(a, b \mid x, y).$$

The local bound is $S \leq 2$, and the uniformly random local behaviour q has $S(q) = 0$. For the standard isotropic CHSH family, where this CHSH facet is the first local facet met along the ray from q to p ,

$$v_c(p; q) = \frac{2}{S(p)} \quad (S(p) > 2).$$

Thus the PR box, with $S = 4$, has threshold $v_c = 1/2$. A maximal quantum CHSH correlation has $S = 2\sqrt{2}$, giving $v_c = 1/\sqrt{2}$. This equality is a statement about the isotropic CHSH ray, not about arbitrary behaviours with $S(p) > 2$.

Theorem 9.36 (Finite-sample Bell certification). *Fix a setting distribution ν and a Bell functional β . Define the weighted value and local bound by*

$$\beta_\nu(p) = \sum_{\mathbf{x}, \mathbf{o}} \nu(\mathbf{x}) \beta_{\mathbf{x}}(\mathbf{o}) p(\mathbf{o} \mid \mathbf{x}), \quad B_\nu(\beta) = \max_{d \text{ deterministic}} \sum_{\mathbf{x}} \nu(\mathbf{x}) \beta_{\mathbf{x}}(d(\mathbf{x})).$$

Assume N independent trials, settings sampled independently from ν , and single-trial local behaviour. If $|\beta_{\mathbf{x}}(\mathbf{o})| \leq M$ for all $\mathbf{x}, \mathbf{o}$ and

$$\widehat{\beta}_N = \frac{1}{N} \sum_{k=1}^N \beta_{\mathbf{x}_k}(\mathbf{o}_k),$$

then every i.i.d. local model satisfies

$$\Pr \left[\widehat{\beta}_N \geq B_\nu(\beta) + \delta \right] \leq \exp \left(-\frac{N\delta^2}{2M^2} \right).$$

Consequently, the event $\widehat{\beta}_N \geq B_\nu(\beta) + M\sqrt{2\log(1/\alpha)/N}$ has probability at most α under every such local model.

Proof. Under the stated sampling convention, the expectation of the single-trial summand is at most $B_\nu(\beta)$ for every local model. The summands are independent and lie in an interval of length at most $2M$. Hoeffding's inequality [6] gives

$$\Pr \left[\widehat{\beta}_N - \mathbb{E} \widehat{\beta}_N \geq \delta \right] \leq \exp \left(-\frac{2N\delta^2}{(2M)^2} \right),$$

and the bound follows from $\mathbb{E} \widehat{\beta}_N \leq B_\nu(\beta)$. Solving for δ gives the stated level- α threshold. $\square$

Remark 9.37 (Physical scope of finite-sample certification). The finite-sample statement assumes independent trials, exogenous setting choice, and a fixed local model for the trial distribution. Memory, locality, freedom-of-choice, and detection loopholes require additional presentation assumptions or martingale-style analyses. Under the stated assumptions, a Farkas certificate is an experimentally testable Bell inequality with a finite-sample violation bound.

Definition 9.38 (Bell repair profile). A repair of a nonlocal Bell behaviour relaxes a specified assumption of the local-causal presentation. The basic repair coordinates used here are: communication, where parties exchange messages; measurement dependence, where λ may depend on $\mathbf{x}$; and detector inefficiency, where local models are allowed no-detection outcomes and the observed behaviour is obtained by postselection. For fixed quantitative conventions, the corresponding repair profile records optimal costs such as

$$(v_c(p; q), \kappa_{\text{comm}}(p), \kappa_{\text{meas}}(p), \eta_c(p)).$$

Here v_c is the visibility of Definition 9.32, κ_{comm} is the minimum communication cost in a declared communication model, κ_{meas} is the minimum dependence cost for a chosen distance from measurement independence, and η_c is the maximum detector efficiency still compatible with a local postselected explanation. The coordinates are model-relative: changing the repair model changes the numerical profile. Table 5 collects the obstruction, certificate, and repair coordinates for the components used here.

Table 5: Bell obstruction–certificate–repair coordinates.

Component	Fixed local-causal assumption	Relaxation or quantity	Certificate / optimisation
Local coupling	measurement independence and local response factorisation	none; require $\text{Coupl}(p) \neq \emptyset$	Farkas/Bell inequality
Noise robustness	fixed local noise $q \in \mathcal{L}$	critical visibility $v_c(p; q)$	supporting Bell functional
Communication repair	no inter-party messages	one-way alphabet cost κ_{comm}	message lower/upper bounds
Measurement-dependence repair	λ independent of settings	dependence cost κ_{meas}	finite dependence optimisation after a distance is fixed
Detection repair	observed outcomes without postselection	no-detection outcome $\perp$ and threshold η_c	efficiency Bell inequality
Contextual fraction	noncontextual convex decomposition	$\text{CF} = 1 - \text{NCF}$	linear programme and dual noncontextuality inequality

Proposition 9.39 (One-way communication repair). *For a finite two-party behaviour $p(a, b \mid x, y)$ whose first-party marginal is independent of y , define $\kappa_{\text{comm}}(p)$ as the minimum value of $\log |M|$ among exact worst-case shared-randomness-assisted one-way simulations from the first party to the second, with $\kappa_{\text{comm}}(p) = +\infty$ if none exists. The following hold:*

- (i) *p has a zero-bit communication repair, equivalently a repair with $|M| = 1$, if and only if p is local;*
- (ii) *if p is nonlocal, every such repair has message alphabet size at least 2;*
- (iii) *every such behaviour has a repair with $|M| \leq |X_1| |O_1|$.*

Proof. A zero-bit communication repair, with shared randomness and $|M| = 1$, is exactly a local hidden-variable model, giving (i), and (ii) follows immediately. For (iii), sample a from the marginal $p(a \mid x)$, which is well defined by the no-signalling condition from the second party to the first. Send (x, a) to the second party. If $p(a \mid x) > 0$, the second party samples b from $p(b \mid x, y, a)$; for zero-probability pairs (x, a) , choose an arbitrary conditional distribution. Local randomness used in these sampling steps can be absorbed into an enlarged hidden variable. The resulting joint distribution is $p(a, b \mid x, y)$, and the message alphabet has size at most $|X_1||O_1|$. $\square$

Proposition 9.40 (Measurement-dependent repair). *Every finite Bell behaviour has a local repair if measurement independence is removed.*

Proof. Let $\lambda = (\mathbf{x}, \mathbf{o})$ and define

$$p(\lambda = (\mathbf{x}, \mathbf{o}) \mid \mathbf{x}) = p(\mathbf{o} \mid \mathbf{x}).$$

The deterministic output rule returns the stored component o_i at party i . This reproduces p , but the hidden variable is maximally correlated with the settings. $\square$

Remark 9.41. The measurement-dependent repair proves sufficiency of unrestricted setting dependence, not a minimal dependence cost. Such costs can be introduced by optimising a distance between $p(\lambda \mid \mathbf{x})$ and a setting-independent hidden-variable distribution.

Definition 9.42 (Detector-efficiency repair). Extend each outcome set to $O_i^\perp = O_i \cup \{\perp\}$, where $\perp$ denotes no detection. For fixed η , an efficiency- η local extension of a behaviour p is a local behaviour $\tilde{p}$ on the extended outcomes such that, for every setting tuple, the probability that all parties detect is at least η , and conditioning on the all-detected event yields $p(\cdot \mid \mathbf{x})$. This is an all-detection efficiency convention; party-specific efficiency conventions give related but different linear programmes. Define

$$\eta_c(p) = \sup\{\eta \in [0, 1] : p \text{ has an efficiency-}\eta \text{ local extension}\}.$$

Proposition 9.43 (Detector-efficiency feasibility and threshold). *For every finite Bell behaviour p and fixed η , existence of an efficiency- η local extension is a finite linear feasibility problem. Moreover, $\eta_c(p)$ is the optimum of a finite linear programme. Infeasibility at a specified efficiency has a Farkas dual certificate, an efficiency Bell inequality.*

Proof. Write the extended local behaviour as a convex combination of deterministic response tables with outcomes in $O_i^\perp$. Detection probabilities are linear marginals of this extended behaviour. For each setting tuple $\mathbf{x}$, introduce a variable $d_{\mathbf{x}}$ for the probability that all parties detect, and impose $d_{\mathbf{x}} \geq \eta$. The postselection equations are

$$\tilde{p}(\mathbf{o} \mid \mathbf{x}) = d_{\mathbf{x}} p(\mathbf{o} \mid \mathbf{x}) \quad (\mathbf{o} \in \prod_i O_i),$$

where the left side denotes the extended probability of the all-detected outcome $\mathbf{o}$, with no $\perp$ coordinate. Since p is fixed, these equations are linear in the unknown convex weights and the variables $d_{\mathbf{x}}$. Together with normalisation and the lower bounds $d_{\mathbf{x}} \geq \eta$, they form a finite linear feasibility problem. Maximising η subject to the same linear constraints gives the linear programme for $\eta_c(p)$. Farkas' lemma gives a dual certificate when the system is infeasible at a specified efficiency. $\square$

Remark 9.44 (Contextuality counterpart). Finite contextuality admits the same obstruction–robustness pattern after the relevant support or probabilistic presentation is fixed. Strong contextuality is the support-level case: all local supports are nonempty, but no global assignment lies in all of them. For a finite probabilistic empirical model e , one common quantitative convention is the noncontextual fraction

$$\text{NCF}(e) = \max\{\lambda \in [0, 1] : e = \lambda e_{\text{nc}} + (1-\lambda)e' \text{ for some noncontextual } e_{\text{nc}} \text{ and empirical model } e'\},$$

with contextual fraction $\text{CF}(e) = 1 - \text{NCF}(e)$ [24]. In finite scenarios this is a linear programme, and dual solutions give noncontextuality inequalities. These are presentation-relative analogues of the Bell coupling and repair problems above.

Remark 9.45 (Stochastic kernels and deterministic tables). For finite Bell scenarios, stochastic local response kernels and the finite convex hull of deterministic response tables define the same local polytope. The product-table construction in the proof gives the conversion. Since the local behaviour set is a finite-dimensional polytope, Carathéodory’s theorem gives an equivalent finite convex decomposition into deterministic response tables.

Remark 9.46. Fine’s theorem gives the celebrated equivalences for the standard $2 \times 2 \times 2$ Bell scenario [5]. Theorem 9.24 uses the standard finite local-polytope representation for arbitrary finite scenarios [27]. A CHSH behaviour violating the local bound has no global coupling [4].

Remark 9.47 (Support versus probability). Strong contextuality excludes every deterministic global assignment from the support. Bell nonlocality excludes a probability measure on deterministic response tables with the required marginals. A behaviour may have support-level global assignments and nevertheless have no coupling reproducing its probabilities.

10 Effectivity, computability, and resources

The higher layers of the diagnostic distinguish mathematical existence from realisation. A compatible valid section may exist in the set-theoretic sense and still be unavailable to any implementation in the declared class. Abstract effectivity is therefore relative to a realisation map. Ordinary computability is the special case in which the domains carry effective representations and the realisation image is a class of computable maps. The central distinction is between nonemptiness of the valid-section space and the existence of a uniform procedure selecting a valid section.

Definition 10.1 (Uniform binary fibre selection). Throughout the Weihrauch statements in this section, finite spaces carry their standard discrete representations, countable products carry the product representation, and closed subsets of Cantor space carry negative information. The symbol WKL denotes path selection for nonempty infinite prefix-closed binary trees under the corresponding negative tree representation, equivalently closed choice on nonempty effectively closed subsets of Cantor space. Give a closed subset of the discrete space $2 = \{0, 1\}$ by negative information, namely by an enumeration of its complement. The multivalued problem FibSel_2 takes a sequence $(A_n)_{n \in \mathbb{N}}$ of nonempty closed subsets $A_n \subseteq 2$, uniformly given by negative information, and returns any $f \in 2^{\mathbb{N}}$ satisfying

$$f(n) \in A_n \quad \text{for every } n.$$

The domain representation includes the promise that no A_n is empty.

Theorem 10.2 (Weihrauch degree of uniform binary fibre selection). *With the representation of Definition 10.1,*

$$\text{FibSel}_2 \equiv_{\text{W}} \widehat{\text{C}}_2 \equiv_{\text{W}} \text{C}_{2^{\mathbb{N}}} \equiv_{\text{W}} \text{WKL},$$

where C_X denotes closed choice on the represented space X and the hat denotes countable parallelisation.

Proof. A negatively named nonempty subset of 2 is precisely an instance of C_2 . The input to FibSel_2 is a sequence of such instances, and its output selects one point from every instance. Thus FibSel_2 is, under the stated product representation, exactly the countable parallelisation $\widehat{C}_2$.

The standard represented-space equivalences

$$\widehat{C}_2 \equiv_W C_{2^\mathbb{N}} \equiv_W \text{WKL}$$

are the parallel binary-choice and closed-choice formulation of Brattka and Gherardi [21]. They are nontrivial: an arbitrary closed subset of Cantor space need not be a coordinate product. Applying that cited theorem to the exact parallel-choice problem identified in the first paragraph proves the displayed chain. $\square$

Remark 10.3 (Scope of the classification). Theorem 10.2 classifies the input-uniform operator on negatively represented binary fibres. It does not classify arbitrary validity representations or countably valued fibres. The halting-stage family of Definition 10.8 is a single computable instance of the same selection pattern. Its solution set has the PA-degree spectrum of Corollary 10.12, but this nonuniform degree statement is different from Weihrauch-completeness of the input-uniform problem.

Definition 10.4 (Represented realisation classes). Fix representations of the implementation and policy spaces and a realisation map

$$\text{Realise}_{\text{eff}} : \text{Impl}_{\text{eff}} \rightarrow \Pi_{\text{all}}.$$

The effective and bounded policy classes are its full image and the image of its restriction to $\text{Impl}_{\mathcal{B}}$.

Definition 10.5 (Cost-bounded implementation classes). A quantitative resource model consists of a cost map

$$\text{cost} : \text{Impl}_{\text{eff}}(G) \times W \rightarrow \overline{\mathbb{N}} = \mathbb{N} \cup \{\infty\}.$$

For an input-dependent bound $B : W \rightarrow \overline{\mathbb{N}}$ define

$$\text{Impl}_{\leq B} = \{\iota : \forall w \in W, \text{cost}(\iota, w) \leq B(w)\}.$$

Bounds are ordered pointwise; $B \leq B'$ implies $\text{Impl}_{\leq B} \subseteq \text{Impl}_{\leq B'}$. A policy is bounded when it has at least one realising implementation in $\text{Impl}_{\leq B}$.

Remark 10.6 (Dependence on the implementation model). Set-theoretically, any $P \subseteq \Pi_{\text{all}}$ can be presented as an image by taking implementation set P and the inclusion map. Thus “effective” has mathematical content only after representations and an implementation model are fixed. The computability results in this section use standard acceptable numberings.

Proposition 10.7 (External-policy special case). *Let $\Pi_{\text{ext}} \subseteq \Pi_{\text{all}}$ be a specified external policy class, and recall that the bounded realised policy class is defined as the image*

$$\text{im}(\text{Realise}_{\text{eff}}|_{\text{Impl}_{\mathcal{B}}}).$$

When this defining image equals Π_{ext} , the identification $\Pi_{\mathcal{B}} = \Pi_{\text{ext}}$ is definitional rather than an additional conclusion, and every selector statement for the internal presentation reduces to the corresponding external-policy problem.

Proof. Only the reduction requires argument: all selector statements depend on implementations only through the defining image of $\Pi_{\mathcal{B}}$. $\square$

10.1 A recursive co-c.e. effectivity obstruction

The first effectivity example has no local failure and no compatibility failure. There is a valid selector. The problem is that every computable candidate can be turned against itself at its own index. This is the standard diagonally nonrecursive construction, expressed as a selector game.

Fix an admissible numbering (φ_n) of all partial binary-valued recursive functions, so every total computable binary function has an index in the numbering [14]. Let $\text{Comp}(n, s, b)$ be a fixed primitive-recursive predicate saying that $\varphi_n(n)$ halts within s steps with output $b \in \{0, 1\}$.

Definition 10.8 (Halting-stage selector game). Take

$$W = \mathbb{N} \times \mathbb{N}, \quad Z = \mathbb{N}, \quad I(n, s) = n, \quad R = M = H = Y = \{0, 1\}, \quad A = \{*\},$$

with D, q, e the identity maps and zero-one loss. Define the total recursive dynamics

$$U((n, s), m, *) = \begin{cases} 1 - m, & \text{Comp}(n, s, m), \\ m, & \text{otherwise.} \end{cases}$$

Let Π_{eff} be the total computable binary-valued policies.

Lemma 10.9 (Uniform co-c.e. validity). *Under the standard pairing code for (n, r) ,*

$$(r, *) \in \text{Adm}(n) \iff \forall s \neg \text{Comp}(n, s, r).$$

*Hence the complement of $\{(n, r) : (r, *) \in \text{Adm}(n)\}$ is computably enumerable uniformly in (n, r) , and every fibre is nonempty.*

Proof. If a computation witness with output r exists, the corresponding witness world makes the dynamics return $1 - r$, so report r is invalid. If no such witness exists, the dynamics returns r on every world in the fibre. The complement is therefore defined by the existential primitive-recursive predicate $\exists s \text{Comp}(n, s, r)$. A partial function has at most one binary output, so at most one report is excluded. $\square$

Theorem 10.10 (Co-c.e. effectivity obstruction). *The halting-stage selector game has a set-theoretic valid selector but no total computable valid selector. Its semantic obstruction is $\mathcal{K}$.*

Proof. Define set-theoretically

$$f(n) = \begin{cases} 1 - b, & \varphi_n(n) \downarrow = b \in \{0, 1\}, \\ 0, & \varphi_n(n) \uparrow. \end{cases}$$

If the computation diverges, both reports are valid; if it halts with value b , Lemma 10.9 excludes only b , so $1 - b$ is valid. Thus f is a valid selector. This explicit definition uses classical excluded middle for the halting statement but no choice principle.

Suppose a total computable valid selector g existed. Choose an index k in the fixed binary numbering with $g = \varphi_k$, and put $b = g(k)$. A computation witness shows $b \notin \text{Adm}(k)$ by Lemma 10.9, contradicting validity. Local sets and the set-theoretic global family are nonempty, so the first semantic failure is effectivity. $\square$

Theorem 10.11 (Π_1^0 selector class). *Let*

$$\mathcal{S} = \{g \in 2^{\mathbb{N}} : \forall n (g(n), *) \in \text{Adm}(n)\}.$$

Then $\mathcal{S}$ is a nonempty Π_1^0 class with no computable member.

Proof. The complement of $\mathcal{S}$ is the c.e. open set

$$\bigcup_{\text{Comp}(n,s,b)} \{g \in 2^{\mathbb{N}} : g(n) = b\}.$$

Equivalently, $\mathcal{S}$ is the path set of the computable tree

$$T = \left\{ \sigma \in 2^{<\mathbb{N}} : \forall n < |\sigma| \forall s < |\sigma|, \neg \text{Comp}(n, s, \sigma(n)) \right\}.$$

Hence $\mathcal{S}$ is effectively closed. The explicit selector constructed in Theorem 10.10 proves nonemptiness. If a computable g had index k , a computation witness for $\varphi_k(k) = g(k)$ would violate membership at $n = k$. $\square$

Corollary 10.12 (Oracle degrees of capability). *For the fixed acceptable numbering (φ_n) introduced in Section 10, let Π_X be the policies computable relative to oracle X . The halting-stage selector game is capable in Π_X exactly when X has PA degree, equivalently when X computes a binary diagonally nonrecursive function. The resulting oracle-degree class is invariant under replacement by another acceptable numbering. In particular, the game has a low noncomputable selector and a selector of hyperimmune-free degree.*

Proof. The class $\mathcal{S}$ is exactly the class of binary diagonally nonrecursive functions for the fixed acceptable numbering. The degrees computing such a function are the PA degrees for every acceptable numbering, and the Low Basis and Hyperimmune-Free Basis Theorems supply low and hyperimmune-free members of every nonempty Π_1^0 class [7, 8, 14]. $\square$

The index argument is not the local uniform diagonal of Definition 3.14: no admissibility fibre is empty.

10.2 Nested distributed effectivity

The finite distributed theory separates local and compatibility failure. The following countable finitary extension supplies an overlapping-scope effectivity obstruction.

Definition 10.13 (Countable finitary distributed selector). A countable finitary distributed selector problem has the data of Definition 7.1, except that W and J may be countable, every active scope $S(w)$ is finite, and all carriers and relations have fixed effective representations when computability is discussed.

Definition 10.14 (Nested-prefix distributed selector). Let $T \subseteq 2^{<\mathbb{N}}$ be a prefix-closed infinite binary tree. Define a countable finitary distributed selector P_T by

$$J = W = \mathbb{N}, \quad Z_j = \{*\}, \quad C_j = \{0, 1\},$$

$$S(n) = D_n = \{0, \dots, n\}, \quad V_n = T \cap 2^{n+1}.$$

Its reduced prefix diagram has objects D_n , arrows $D_n \rightarrow D_m$ for $m \leq n$, assignment sets 2^{n+1} , local sets V_n , and prefix-restriction maps. The full canonical incidence category additionally contains every nonempty finite subset of a realised prefix. When T is computable, take Impl_{eff} to be indices of total computable binary functions, use the standard evaluation realisation map into $2^{\mathbb{N}}$, and define the effective global-family space by intersecting its image with $[T]$.

Remark 10.15 (Variance convention for prefix diagrams). In the reduced prefix diagram, arrows go from longer prefixes to shorter prefixes, $D_n \rightarrow D_m$ for $m \leq n$, because they are restriction maps. Later causal patch categories use the same restriction direction, from larger patches to smaller patches; the difference is only that a full finite tree can be a largest patch and hence an initial object in the restriction category, while the chain of finite prefixes has no largest finite prefix. Thus the nested-prefix “no initial object” statement is a consequence of unbounded length, not a change in variance.

Theorem 10.16 (Nested distributed path representation). *For every prefix-closed infinite binary tree T :*

- (i) *every local set in the reduced prefix diagram is finite and nonempty;*
- (ii) *the set-theoretic global-family space and the valid distributed-strategy space are both canonically $[T]$;*
- (iii) *the reduced prefix category has no initial object.*

If membership in T is decidable, then $[T]$ is a nonempty effectively closed subset of Cantor space, hence a Π_1^0 class. If it has no computable member, computable strategies as the effective image give outcome K_0 . For the binary DNR tree of Theorem 10.11, an oracle computes a global strategy exactly when it has PA degree.

Proof. Since T is infinite, finitely branching, and prefix closed, König’s lemma [11] gives a path through T ; in particular, T has nodes of every finite length, and every V_n is nonempty. A distributed strategy is a binary sequence x , and validity at world n is precisely $x|_{n+1} \in T$. Thus validity for every world is equivalent to $x \in [T]$. Likewise, a compatible family $(\sigma_n)_n$ in the reduced prefix diagram satisfies $\sigma_m = \sigma_n|_{m+1}$ for $m \leq n$, so its union is a path in T ; restrictions of a path give the inverse construction.

An initial object would require some finite prefix D_n to have an arrow to every D_m , which would require $n \geq m$ for every m , impossible. If membership in T is computable, its path set is effectively closed; absence of a computable path then gives the K_0 classification. For the stated DNR tree, Corollary 10.12 gives the final oracle characterisation. $\square$

Corollary 10.17 (Pointwise versus uniform prefix sections). *If T has no computable path, there is a matching family on the reduced prefix diagram such that every finite component has a computable name, but no single computable procedure produces these names uniformly in n .*

Proof. Choose a path $x \in [T]$ set-theoretically and take $\sigma_n = x|_{n+1}$. Every finite string σ_n is computable, and the family is matching. A uniform procedure computing σ_n from n would compute their union x , contrary to the absence of computable paths. $\square$

The preceding effective-outcome classification concerns a fixed tree with decidable membership. In Theorem 10.18, the tree itself is represented input and may be noncomputable.

Theorem 10.18 (Uniform degree of nested-prefix selection). *Let the input be a nonempty path set represented by a prefix-closed binary tree T given by negative information for its complement, equivalently by a negative name for the closed set $[T] \subseteq 2^{\mathbb{N}}$. The multivalued operator taking T to a valid strategy of P_T is Weihrauch-equivalent to WKL. The input need not be computable.*

Proof. By Theorem 10.16(ii), the valid strategies of P_T are exactly the paths through the input tree T , and both translations are the identity on the tree code and on the output binary sequence. The represented problem is therefore the standard path-selection formulation of WKL. $\square$

Remark 10.19. The full canonical incidence category of P_T is generally not a chain, and adjacent projection equality can fail when T has dead ends; the finite join-tree theorem therefore does not apply. The exact Weihrauch statement concerns the input-uniform operator $T \mapsto [T]$, not any fixed DNR instance.

10.3 A master represented section-selection problem

The preceding Weihrauch results are instances of one represented selection problem. The input is not a fixed computable presentation, but a negative description of the finite obstructions to a total section.

Definition 10.20 (Binary forbidden-pattern section selection). Let ForbSec_2 be the multivalued problem whose input is an enumeration of finite binary strings $\sigma \in 2^{<\mathbb{N}}$, interpreted as forbidden cylinders $[\sigma] \subseteq 2^{\mathbb{N}}$, together with the promise that

$$F = 2^{\mathbb{N}} \setminus \bigcup_{\sigma \text{ enumerated}} [\sigma]$$

is nonempty. An output is any $x \in F$.

Theorem 10.21 (Forbidden-pattern selection has Weihrauch degree WKL).

$$\text{ForbSec}_2 \equiv_{\text{W}} \text{WKL}.$$

Proof. The enumeration of forbidden cylinders is exactly a negative name for the closed set $F \subseteq 2^{\mathbb{N}}$. Hence ForbSec_2 is closed choice on Cantor space, which is Weihrauch-equivalent to WKL by Theorem 10.2.

Conversely, an infinite binary tree T determines the closed set $[T]$ by enumerating every finite string not in T as a forbidden cylinder. A point of the resulting closed set is precisely an infinite path through T . These translations are computable in the standard negative representations. $\square$

Corollary 10.22 (Uniform section-selection templates). *The following represented problems are Weihrauch-reducible to ForbSec_2 , and hence to WKL:*

- (i) *selection from any nonempty effectively closed subset of $2^{\mathbb{N}}$;*
- (ii) *countable binary section selection with uniformly c.e. forbidden finite patterns;*
- (iii) *represented patch- or arena-section selection whenever invalidity is enumerated by finite binary witnesses and nonemptiness is promised.*

Each template is Weihrauch-equivalent to WKL whenever arbitrary infinite binary trees, equivalently arbitrary nonempty effectively closed subsets of Cantor space, can be encoded into the corresponding presentation class.

Proof. Each input supplies an enumeration of basic cylinders covering exactly the invalid total assignments, so it computes an instance of ForbSec_2 . A selected point is a valid section. The hardness statement is the reverse construction in the proof of Theorem 10.21 whenever the presentation class can realise arbitrary tree-prefix constraints. $\square$

Remark 10.23 (Scope of the WKL classification). The theorem separates three cases. Finite explicit tabular presentations are solved by finite search. A fixed computable infinite presentation may have a particular degree spectrum, such as the PA degrees in the DNR example, without being an input-uniform Weihrauch-complete problem. A represented family whose invalid total sections are given by c.e. finite patterns has uniform selection bounded by WKL, and it is exactly WKL when it can encode arbitrary binary trees. Thus the represented-space content lies in the input representation and in the class of finite-pattern constraints allowed.

Theorem 10.24 (Π_1^0 realisation and oracle spectrum). *For every nonempty Π_1^0 class $P \subseteq 2^\mathbb{N}$, there is a computable nested-prefix ECD presentation E_P whose set-theoretic global strategy space is computably homeomorphic to P . Consequently, for every oracle X ,*

$$E_P \text{ has an } X\text{-computable global strategy} \iff X \text{ computes a member of } P.$$

Thus the oracle degrees capable for E_P are exactly the Muchnik upper cone of P .

Proof. Write $P = [T]$ for a computable infinite prefix-closed binary tree T . Construct the nested-prefix presentation P_T of Definition 10.14. By Theorem 10.16, its global strategies are exactly the paths through T . The correspondence sends a global strategy to the binary sequence obtained by taking the union of its coherent finite prefixes, and sends a path to its finite-prefix family. Both maps are computable and inverse to one another. Relativising the same maps to an oracle X gives the stated oracle spectrum. $\square$

Theorem 10.25 (Reverse-mathematical form of nested-prefix selection). *Over RCA_0 , the following are equivalent:*

- (i) WKL_0 ;
- (ii) every binary nested-prefix presentation P_T arising from a prefix-closed tree T with nonempty local set $T \cap 2^{n+1}$ at every level n has a global section;
- (iii) every countable binary forbidden-pattern presentation whose every finite stage is satisfiable has a total section.

Proof. Weak König's lemma [11, 15] gives a path through the tree of coherent finite choices, proving (ii) and (iii). Conversely, given an infinite binary tree T , form the nested-prefix presentation with local set $T \cap 2^{n+1}$ at level n . A global section is exactly a path through T , so (ii) implies WKL_0 . A tree can also be represented by forbidding finite strings not in T ; then (iii) again gives a path, proving WKL_0 . The codings are primitive recursive and hence available over RCA_0 . $\square$

Remark 10.26 (Degree-theoretic contribution). Table 6 separates input-uniform Weihrauch classification from nonuniform oracle spectra. The realisation theorem above is the nonuniform counterpart of the WKL classification: every nonempty Π_1^0 path class, not only the DNR example, can occur as the global strategy space of a computable ECD presentation. The represented forbidden-pattern theorem gives the corresponding uniform operator.

10.4 Computational-class and resource separations

Theorem 10.27 (Primitive-recursive class separation). *Let $a : \mathbb{N} \rightarrow \mathbb{N}$ be a total recursive function that is not primitive recursive, for example a standard diagonal Ackermann function [14]. There is a passive recursive game whose unique valid principal selector has report component a . With total*

Table 6: Representative Weihrauch degrees of ECD selection variants.

Selection variant	Representation	Weihrauch degree
finite explicit tabular section selection	finite tables	computable
finite alphabet, c.e. forbidden finite patterns, nonempty promise	negative closed-set name in a computably compact product	WKL when arbitrary binary trees encode
binary fibre selection	negative information for each nonempty fibre	WKL
nested-prefix path selection	represented infinite binary tree	WKL
fixed computable DNR instance	no varying input; oracle spectrum question	PA degrees, not a Weihrauch degree computation
countable discrete alphabets	closed subsets of Baire space	closed choice on $\mathbb{N}^{\mathbb{N}}$ under the standard representation
unique valid section	singleton closed sets	unique closed choice for the underlying represented space

recursive policies as Π_{eff} and primitive-recursive policies as $\Pi_{\mathcal{B}}$, the bounded-class obstruction is $\mathbf{R}$ relative to that declared class. This is a computational-class separation rather than a quantitative time or space lower bound. The presentation here is recursive, not an instance of the primitive-recursive syntax $\mathcal{L}_{\text{PR}}$ used in Theorem 10.32.

Proof. Take $W = Z = R = M = H = Y = \mathbb{N}$, $I = D = e = q = \text{id}$, singleton actions, zero-one loss, and passive total recursive dynamics $U(n, m, *) = a(n)$. At interface value n , exact validity is $r = a(n)$, so $\text{Adm}(n) = \{(a(n), *)\}$. Consequently every valid selector has report component a . The function a is total recursive, hence gives an effective admissible section, but by hypothesis no primitive-recursive policy can realise it. Thus $\Gamma_{\text{eff}} \neq \emptyset$ and $\Gamma_{\mathcal{B}} = \emptyset$. $\square$

Definition 10.28 (Deterministic-time policy class). For the binary-string specialisation and a time-constructible bound t , let implementations be deterministic Turing machines, let $\text{cost}(\iota, x)$ be running time, put $B_c(x) = ct(|x|)$ for $c \geq 1$, and define

$$\Pi_{\text{DTIME}(t)} = \bigcup_{c \geq 1} \text{im} \left(\text{Realise}_{\text{eff}} \mid \text{Impl}_{\leq B_c} \right).$$

Theorem 10.29 (Deterministic-time resource separation). *Let $t, g : \mathbb{N} \rightarrow \mathbb{N}$ be time-constructible bounds with $t(n) \geq n$ and $t(n) \log t(n) = o(g(n))$. There is a passive exact game on binary-string inputs with a computable valid selector but no valid selector in $\Pi_{\text{DTIME}(t)}$.*

Proof. By the deterministic time-hierarchy theorem, since $ct(n) \log(ct(n)) = o(g(n))$ for every fixed c , choose a language $L \in \text{DTIME}(g) \setminus \bigcup_{c \geq 1} \text{DTIME}(ct)$ [17]. Take

$$W = Z = \{0, 1\}^*, \quad I = \text{id}, \quad R = H = Y = \{0, 1\},$$

with singleton message and action sets, identity decoder, zero-one loss, and passive queried value $q(U_0(x)) = \chi_L(x)$. The unique valid report selector is χ_L . Take Π_{eff} to be the total computable report policies and $\Pi_{\mathcal{B}} = \Pi_{\text{DTIME}(t)}$ from Definition 10.28. A valid bounded report policy would decide L in $\text{DTIME}(t)$, contradicting the choice of L . Thus an effective section exists but no section lies in the declared time class. $\square$

Proposition 10.30 (Degrees of the computability examples). *For the halting-stage selector game with zero-one loss,*

$$d_{\text{pt}} = d_{\text{obs}} = 0, \quad d_{\text{eff}} = 1.$$

For the deterministic-time hierarchy game with $\Pi_{\mathcal{B}} = \text{DTIME}(t)$,

$$d_{\text{pt}} = d_{\text{obs}} = d_{\text{eff}} = 0, \quad d_{\mathcal{B}} = 1.$$

Proof. The explicit set-theoretic selector for the halting-stage selector game gives $d_{\text{obs}} = 0$, and world-dependent choices give $d_{\text{pt}} = 0$. Every computable policy fails on at least one world, so its worst-case zero-one loss is 1; the constant-zero policy is computable, so the class is nonempty and $d_{\text{eff}} = 1$.

For the time-hierarchy game, the characteristic function of L is an effective zero-loss selector, so the first three degrees vanish. Every policy in $\text{DTIME}(t)$ fails on some input, hence has worst-case loss 1; constant machines belong to the class, so it is nonempty. Thus $d_{\mathcal{B}} = 1$. $\square$

10.5 Arithmetic complexity of recursive capability

Definition 10.31 (Concrete primitive-recursive game language). Let $\mathcal{L}_{\text{PR}}$ be a fixed effective syntax of well-typed game descriptions. A description contains a finite list of carrier sorts, each equal to $\mathbb{N}$ or to a fixed finite tagged set, together with indices from a standard effective enumeration of primitive-recursive structural maps. Every index has explicit source and target sort annotations, and the grammar permits only sort-compatible composition and application. Syntactic well-formedness is decidable. The index set CAP_{PR} is viewed as a subset of $\mathbb{N}$ by declaring malformed strings false; equivalently, it is the subset of valid syntax codes satisfying the capability predicate. Exact zero-one loss uses decidable equality. A policy is coded by a fixed primitive-recursive pairing of a pair of ordinary partial-recursive program indices, and membership in CAP_{PR} requires both policy components to be total on their entire declared domains and to form an exact valid selector. Product carriers such as $\mathbb{N} \times \mathbb{N}_{\geq 1}$ are represented in the $\mathbb{N}$ sort by a fixed primitive-recursive pairing, with positive coordinates encoded by a successor. Totality on counterfactual report-action pairs is part of this language; requiring action totality only on the realised report graph would define a different index set. The universal variable tags and all finite tuples of computation stages are encoded by fixed primitive-recursive pairing functions.

Theorem 10.32 (Recursive capability is Σ_3^0 -complete). *The index set CAP_{PR} is Σ_3^0 -complete.*

Proof. Membership. Encode the report and action programs by one index p and let the universal variable x carry one of three tags:

- (a) $x = \langle 0, z \rangle$ asks for a stage at which the report program halts on z ;
- (b) $x = \langle 1, z, r \rangle$ asks for a stage at which the action program halts on (z, r) ;
- (c) $x = \langle 2, w \rangle$ asks for stages computing the selected report at $I(w)$ and the corresponding action, followed by verification of the exact validity equality at w .

The well-formedness predicate for G is decidable. Primitive-recursive structural maps are total by syntax, and their values and the final equality are decidable once the policy computations have converged. A single stage code s can pair all finite computation witnesses needed by the tag. Hence a recursive predicate R satisfies

$$G \in \text{CAP}_{\text{PR}} \iff \text{WF}(G) \wedge \exists p \forall x \exists s R(G, p, x, s).$$

Since WF is decidable, this places the index set in Σ_3^0 .

Hardness. Let W_k be the k th c.e. set. The set $\text{REC} = \{k : W_k \text{ is recursive}\}$ is Σ_3^0 -complete [16]. Let $E(k, n, t)$ be the primitive-recursive predicate saying that n first enters W_k at stage $t \geq 1$. Uniformly from k , construct

$$W = \mathbb{N} \times \mathbb{N}_{\geq 1}, \quad Z = \mathbb{N}, \quad I(n, t) = n, \quad R = M = \mathbb{N}, \quad A = \{*\}, \quad H = Y = \{0, 1\},$$

with $D(r) = r$, $q = \text{id}$, constant decoder $e(r) = 0$, and

$$\ell(r, y) = \begin{cases} 0, & e(r) = y, \\ 1, & e(r) \neq y. \end{cases}$$

Define the primitive-recursive dynamics

$$U_k((n, t), r, *) = \begin{cases} 1, & r = 0 \text{ and } E(k, n, t), \\ 1, & r > 0 \text{ and } \neg E(k, n, r), \\ 0, & \text{otherwise.} \end{cases}$$

For $r > 0$, the condition $\neg E(k, n, r)$ is evaluated at the proposed report r , not at the world stage t ; this makes a positive report valid uniformly throughout the fibre exactly when it is the first-entry stage. Report 0 is valid throughout the fibre over n exactly when $n \notin W_k$; a positive report r is valid exactly when $E(k, n, r)$. Thus

$$\text{Adm}_k(n) = \begin{cases} \{(0, *)\}, & n \notin W_k, \\ \{(r, *) : r > 0 \wedge E(k, n, r)\}, & n \in W_k. \end{cases}$$

If W_k is recursive, decide membership, output 0 on nonmembers, and search for the unique positive first-entry stage on members. Conversely, a total computable selector ρ yields the total characteristic function

$$\chi_{W_k}(n) = \begin{cases} 1, & \rho(n) > 0, \\ 0, & \rho(n) = 0, \end{cases}$$

so W_k is recursive. The map from k to the displayed tuple of primitive-recursive structural indices is computable. Therefore $k \in \text{REC}$ iff $G(k) \in \text{CAP}_{\text{PR}}$, proving many-one hardness. $\square$

11 Extensive-form causal descent

The causal tier adds temporal structure. The arena is fixed before the strategy is chosen: nature moves, information sets, choice sets, and losses are part of the input. A strategy then selects actions at the predictor information states. In perfect-information arenas, backward induction characterises capability because each reached decision node can be solved using the information available there. Compatibility obstructions arise only when choices are shared across information-limited contexts; Corollary 7.14 supplies that imperfect-information case. Declared covers may still exhibit presentation-relative matching failure, as active sensing shows.

Definition 11.1 (Finite extensive prediction arena). A finite extensive prediction arena consists of a finite rooted tree $\mathcal{T}$ with root ε . Every non-root node has a unique predecessor. Terminal nodes have no children. The nonterminal nodes are partitioned as $\mathcal{T}_N \sqcup \mathcal{T}_P$ into nature and predictor nodes. Every nonterminal node h has a finite nonempty set $\text{Ch}(h)$ of outgoing labels, with one child $h \cdot d$ for each $d \in \text{Ch}(h)$. Write $\text{Ch}(h) = M_N(h)$ at nature nodes and $\text{Ch}(h) = C_h$ at predictor nodes.

Let $W \neq \emptyset$. Define world-consistency sets recursively by

$$W_\varepsilon = W,$$

$$W_{h \cdot m} = \{w \in W_h : \nu_h(w) = m\} \quad (h \text{ nature}), \quad W_{h \cdot c} = W_h \quad (h \text{ predictor}),$$

where $\nu_h : W_h \rightarrow M_N(h)$. At each predictor node there is a loss $\lambda_h : W_h \times C_h \rightarrow [0, \infty]$. Empty W_h are allowed; such nodes are not reached by any world. Predictor nodes are their own information states, so the proved arena has perfect information. Arena losses may take the value $+\infty$, as do the static prediction-game losses of Definition 3.1.

Definition 11.2 (Strategies, reachability, and play loss). A strategy is $\pi \in \text{Strat}(\mathcal{T}) = \prod_{h \in \mathcal{T}_P} C_h$. Define $\text{Reach}(w, h; \pi)$ recursively: the root is reached; a nature child $h \cdot m$ is reached iff h is reached and $\nu_h(w) = m$; a predictor child $h \cdot c$ is reached iff h is reached and $\pi(h) = c$.

For a partial strategy s on a prefix-closed patch containing h , define $\text{Reach}(w, h; s)$ by the same recursion. Every predictor ancestor of h belongs to the patch, so all required choices are defined. A node h is π -consistent when π selects every predictor-edge label on the unique root-to- h path.

The aggregate finite-play loss is

$$L_{\mathcal{T}}(\pi, w) = \max(\{\lambda_h(w, \pi(h)) : h \in \mathcal{T}_P, \text{Reach}(w, h; \pi)\} \cup \{0\}).$$

A strategy is ε -capable iff $L_{\mathcal{T}}(\pi, w) \leq \varepsilon$ for every world.

Lemma 11.3 (Path characterisation of reachability). *For a total strategy π , world w , and node h , $\text{Reach}(w, h; \pi)$ holds iff $w \in W_h$ and π selects every predictor-edge label on the unique root-to- h path. The same equivalence holds for a partial strategy on a prefix-closed patch containing h .*

Proof. Induct along the unique path. At a nature edge, the recursive definition of $W_{h \cdot m}$ adds exactly the condition $\nu_h(w) = m$. At a predictor edge, reachability adds exactly the condition that the strategy selects its label. Prefix closure supplies every ancestor coordinate in the partial case. $\square$

Proposition 11.4 (Finite causal unfolding). *Every strategy and world determine a unique finite play.*

Proof. At a reached nature node, $\nu_h(w)$ selects one child; at a reached predictor node, $\pi(h)$ selects one child. Finiteness and acyclicity force termination, and deterministic selection at each node gives uniqueness. $\square$

Definition 11.5 (Backward solvability). For fixed ε , define $\text{Good}_\varepsilon(h)$ at a predictor node by

$$\text{Good}_\varepsilon(h) = \{c \in C_h : \forall w \in W_h, \lambda_h(w, c) \leq \varepsilon\}.$$

Define $\text{Solv}_\varepsilon(h)$ by backward recursion. It holds at terminal nodes and at every node with $W_h = \emptyset$. At a nonempty nature node it holds iff it holds at every child with nonempty world-consistency set. At a nonempty predictor node it holds iff

$$\exists c \in \text{Good}_\varepsilon(h), \quad \text{Solv}_\varepsilon(h \cdot c).$$

Theorem 11.6 (Perfect-information backward induction). *A finite perfect-information arena has an ε -capable strategy iff $\text{Solv}_\varepsilon(\varepsilon)$ holds.*

Proof. Assume root solvability. Since the arena is finite, use finite choice to select, for every solvable predictor node h with nonempty W_h , a witness $c_h \in \text{Good}_\varepsilon(h)$ whose child is solvable. Set $\pi(h) = c_h$ at those nodes and choose an arbitrary element of C_h at all remaining predictor nodes. Induction down any π -consistent path shows that every node on the path with nonempty world set is solvable. Hence each reached predictor choice is good, and the strategy is capable.

Conversely, let π be capable. Use backward induction on height for nodes that are π -consistent and have nonempty W_h . At a nature node, each child with nonempty world set is reached by the worlds selecting that move and is solvable by induction. At a predictor node, Lemma 11.3 shows that every world in W_h reaches h ; capability therefore gives $\pi(h) \in \text{Good}_\varepsilon(h)$, and induction gives solvability of $h \cdot \pi(h)$. The root is consequently solvable. $\square$

Corollary 11.7 (Node-local sufficiency in perfect-information arenas). *If $\text{Good}_\varepsilon(h) \neq \emptyset$ at every predictor node with $W_h \neq \emptyset$, then the arena is ε -capable.*

Proof. Backward induction on height shows every node solvable: all relevant nature children are solvable by induction, and at a predictor node any good choice has a solvable child by the induction hypothesis. Apply Theorem 11.6. $\square$

Remark 11.8 (Perfect-information case). In a perfect-information arena, node-local nonemptiness is enough. Once every reached predictor node has at least one choice that is good for all worlds still consistent with that node, backward induction assembles those choices into a global strategy. Compatibility becomes a separate obstruction only when choices are shared across information-limited nodes or external cover objects.

Definition 11.9 (Backward-induction congruence). Let $\mathcal{T}$ be a finite perfect-information arena. A *backward-induction congruence* is an equivalence relation $\sim$ on nodes together with a rank $r : \mathcal{T} \rightarrow \mathbb{N}$ which strictly decreases along every edge, such that equivalent nodes have equal rank and:

- (i) they have the same status: terminal, nature, or predictor;
- (ii) their world-consistency sets are equal;
- (iii) equivalent predictor nodes have the same choice set and identical loss tables;
- (iv) equivalent nature nodes have the same nature alphabet and identical nature maps on their common world set;
- (v) children reached under any common outgoing label are equivalent.

The quotient graph has one vertex per equivalence class and the induced labelled edges.

Theorem 11.10 (Quotient backward induction). *For a backward-induction congruence, Good_ε at predictor nodes and Solv_ε at all nodes are constant on equivalence classes. The quotient graph is acyclic, and backward induction on it decides capability. For explicit tables, a safe running-time bound is*

$$O\left(\sum_{[h] \text{ predictor}} |W_{[h]}| |C_{[h]}| + \sum_{[h] \text{ nature}} (|W_{[h]}| + |M_N([h])|) + |E_{\text{quot}}|\right).$$

Proof. At equivalent predictor nodes, equality of world sets, choice sets, and loss tables gives equality of Good_ε directly. Equal rank and strict rank decrease along edges make the quotient acyclic. Prove constancy of Solv_ε by induction on rank. Terminal status and empty-world status agree. At equivalent nature nodes, equality of nature maps identifies the nonempty-world children label by label, and successor congruence plus the induction hypothesis gives the same universal solvability condition. At equivalent predictor nodes, equality of good choices and successor classes gives the same existential condition. Root solvability therefore can be evaluated on the quotient and is equivalent to capability by Theorem 11.6.

Computing each predictor good set uses the first displayed sum. Nature-world routing and nonempty-child tests are bounded by the second, and propagation through the quotient uses each quotient edge once. No reduction factor is asserted unless the quotient is actually smaller than the original table. $\square$

Under the node-local presentation by the sets $\text{Good}_\varepsilon(h)$, there is no compatibility-only obstruction. Cover-incidence presentations can assign a cover-relative D_ε label, as in Theorem 11.22.

Definition 11.11 (Causal patches and partial strategies). Let $\text{Patch}(\mathcal{T})$ be the poset of nonempty prefix-closed node sets. Every patch contains the root; intersections of patches are therefore nonempty and prefix closed. A union cover is a family $\{U_i \subseteq U\}$ with $U = \bigcup_i U_i$. Pulling such a cover back along $V \subseteq U$ gives the cover $V = \bigcup_i (U_i \cap V)$.

Put

$$\mathcal{S}(U) = \prod_{h \in U \cap \mathcal{T}_P} C_h,$$

with coordinate restriction. Because the empty lower set is omitted, $\text{Patch}(\mathcal{T})$ is the basis of nonempty lower sets for the prefix order, equivalently Alexandroff opens under the order convention in which arrows point from larger patches to smaller patches, rather than the full open-set lattice.

Lemma 11.12 (Partial-strategy sheaf). *The assignment $U \mapsto \mathcal{S}(U)$ is a sheaf for union covers.*

Proof. A matching family agrees at every predictor node occurring in two patches. Define the glued value using any patch containing the node. Matching gives well-definedness, and coordinate-wise construction gives existence and uniqueness. $\square$

Remark 11.13 (Partial plans). A patch is an initial piece of the decision tree, and a section over a patch is a partial contingent plan. The sheaf condition says that compatible partial plans are not merely consistent pairwise; they glue uniquely to a plan on the union. Activated validity adds the operational convention that a loss obligation is checked only at nodes actually reached by the partial plan.

Definition 11.14 (Activated-validity presheaf). For a patch U define

$$\mathcal{V}_\varepsilon(U) = \{s \in \mathcal{S}(U) : \forall h \in U \cap \mathcal{T}_P \forall w \in W_h, \text{Reach}(w, h; s) \Rightarrow \lambda_h(w, s(h)) \leq \varepsilon\}.$$

A partial strategy may choose a successor outside its patch; obligations at nodes not activated by that choice are not evaluated. No hereditary implication from successor loss to predecessor loss is assumed or needed.

Theorem 11.15 (Activated validity is a subsheaf). *The assignment $\mathcal{V}_\varepsilon$ is a subsheaf of $\mathcal{S}$.*

Proof. If $V \subseteq U$ and $h \in V$, every ancestor of h lies in V , so the run to h computed from $s|_V$ is the run computed from s . Validity therefore restricts. For a matching family on a union cover, let s be its unique ambient glue. If s reaches h , choose a patch containing h . Prefix closure places all ancestors there, and matching makes the local and global runs to h identical. Local activated validity supplies the loss bound. $\square$

Corollary 11.16 (Full patch-site collapse). *For the full causal patch category of Section 11, arrows are restriction arrows from larger patches to smaller patches; with this convention the full tree is initial and activated validity is a subfunctor. Hence*

$$\Gamma_{\text{set}}(\mathcal{V}_\varepsilon) \cong \mathcal{V}_\varepsilon(\mathcal{T}),$$

and full-site D_ε is impossible.

Proof. Every patch receives the unique restriction arrow from the full tree. Apply Theorem 5.5 and Theorem 11.15. $\square$

In this presentation, “local” means local to an object of the patch category; the maximal full-tree patch is itself an object. Emptiness at that object is not the same notion as node-local emptiness.

Theorem 11.17 (Set-theoretic causal descent). *For $\Pi \subseteq \text{Strat}(\mathcal{T})$ write $\text{Cap}_\varepsilon^{\text{arena}}(\mathcal{T}, \Pi)$ when some $\pi \in \Pi$ satisfies $L_\mathcal{T}(\pi, w) \leq \varepsilon$ for every world. For the full patch $B = \mathcal{T}$,*

$$\text{Cap}_\varepsilon^{\text{arena}}(\mathcal{T}, \Pi) \iff \Pi \cap \mathcal{V}_\varepsilon(B) \neq \emptyset.$$

For every union cover of B , global valid strategies are in bijection with matching families of valid partial strategies.

Proof. The first statement expands the aggregate reached-node loss. The second is the sheaf equalizer of Theorem 11.15. $\square$

Proposition 11.18 (Finite effective gluing). *In a finite effectively tabulated arena with a finite effectively indexed cover, a uniformly computable matching family has a computable global glue.*

Proof. For each predictor node, search the finite cover for the first patch containing it and output that patch’s computable value. Matching makes the result independent of the selected patch. This finite coordinate procedure computes the unique glue. $\square$

Proposition 11.19 (Finite height-one recovery). *Let a static selector problem have finite $\text{im}(I)$ and finite choice set C . Form an arena with a nature root, nature moves $\text{im}(I)$, and $\nu_\varepsilon(w) = I(w)$. Its child h_z is a predictor node with choice set C and loss*

$$\lambda_{h_z}(w, c) = \begin{cases} 0, & V(w, c), \\ 1, & \neg V(w, c). \end{cases}$$

Then

$$\mathcal{V}_0(\mathcal{T}) \cong \{s : \text{im}(I) \rightarrow C : s(z) \in \text{Adm}_V(z) \ \forall z\}.$$

Proof. The root sends exactly the worlds in $I^{-1}(z)$ to h_z , so zero activated loss there is membership in $\text{Adm}_V(z)$. $\square$

11.1 Active sensing and presentation relativity

Active sensing is a compact case study in presentation relativity. The full patch site contains a decision skeleton whose validity set is empty, so the same incapability appears as local failure in that presentation. The branch cover omits that global object. Each branch can then be solved locally by choosing an action that leaves the branch, but the branch-local solutions cannot be matched: matching forces one common root action before the later observations are known.

Definition 11.20 (Two-bit sensing arena). Let $W = \{0, 1\}^2$. At the root choose sensor $a \in \{0, 1\}$ and set root loss identically to zero. The chosen edge leads to nature node n_a ; nature reveals $o = x_a$ and moves to predictor node $h_{a,o}$, where report $r \in \{0, 1\}$ is compared by zero-one loss with a specified target. Let $U_{a,o} = \downarrow h_{a,o}$ and let $B = \bigcup_{a,o} U_{a,o}$ be the decision skeleton. Terminal leaves carry no additional strategy coordinate or loss. This is a perfect-information arena; the four reporting nodes are distinct predictor information states. Figure 3 displays the arena and its branch incidence diagram. Restriction identifies $\mathcal{V}_0(\mathcal{T})$ with $\mathcal{V}_0(B)$.

Proposition 11.21 (Positive sensing). *For target x_a , the arena has a zero-loss strategy.*

Proof. Choose sensor 0 and report the observed bit at each reached node $h_{0,o}$. □

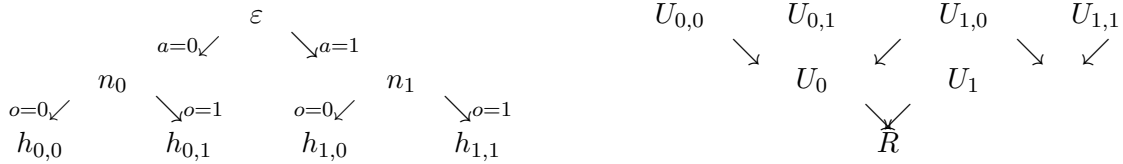


Figure 3: Active sensing. Left: the predictor selects a sensor and nature reveals the selected bit. Right: the seven-object branch-cover incidence diagram, with four branch patches, two nature-prefix intersections, and the root intersection.

Let $\mathcal{C}_{\text{cov}}$ be the category whose objects are all nonempty finite intersections of the four branch patches $U_{a,o}$, with one arrow from a larger intersection to each contained smaller intersection. Restrict $\mathcal{V}_0$ to this seven-object incidence category.

Theorem 11.22 (Full-site and cover-relative sensing outcomes). *For target x_{1-a} :*

- (i) *the full patch site has outcome L_0 ;*
- (ii) *the literal cover-incidence diagram has seven objects: the four branches $U_{a,o}$, the two same-sensor intersections $U_a = \{\varepsilon, n_a\}$, and the root intersection $R = \{\varepsilon\}$. Every local set is nonempty, but no matching family exists, so the cover-relative outcome is D_0 .*

The restriction $\mathcal{V}_0(U_a) \rightarrow \mathcal{V}_0(R)$ is a bijection on the root choice. Consequently the seven-object intersection diagram is naturally equivalent for matching-family calculations to the reduced five-object star with four branch objects and the root.

Proof. For (i), the decision skeleton B has empty activated-validity set. Either root choice activates a second-stage node whose information fixes only the sensed bit while the target unsensed bit takes both values. Since B is an object of the full patch site, the full-site outcome is local emptiness.

For (ii), on $U_{a,o}$ choose root action $1 - a$; then $h_{a,o}$ is unreachable, so either report gives a valid local strategy. The overlap objects contain no predictor node beyond the root and have identically zero root loss, hence are locally inhabited. A matching family agrees through the overlap arrows and therefore fixes one root action a^* . On each $U_{a^*,o}$ that action reaches $h_{a^*,o}$ for two worlds with opposite unsensed bits, so no report is valid for both. This contradicts local validity of the matching family. $\square$

Proposition 11.23 (Cover-relative boundary). *Let F be a restriction-stable validity subfunctor on a patch category with an initial full object B , and let $\mathcal{C}_{\text{cov}}$ be a cover-incidence subcategory not containing B . If $F(B) = \emptyset$, every object of $\mathcal{C}_{\text{cov}}$ has nonempty local set, and the matching-family space on $\mathcal{C}_{\text{cov}}$ is empty, then the full-site presentation has outcome $\mathbf{L}$ and the cover-incidence presentation has outcome $\mathbf{D}$.*

Proof. The full-site outcome is $\mathbf{L}$ because the initial object B has empty local set. On the cover-incidence subcategory, all local sets are nonempty by hypothesis, while the compatible-family space is empty; this is precisely the $\mathbf{D}$ outcome. $\square$

The cover-relative result is a commitment obstruction: matching forces a common root action before branch-local obligations are checked. It is obtained after deleting the full initial object and is not an independent failure of the full patch sheaf. Proposition 11.23 gives the general presentation-relative pattern.

11.2 Imperfect-information arenas and behavioural coordinates

Definition 11.24 (Finite imperfect-information arena). A finite imperfect-information arena is a finite arena together with a partition of predictor nodes into information sets $\mathcal{I}$. All nodes in $J \in \mathcal{I}$ have one common choice set C_J , and no information set contains two nodes on one root-to-leaf path. A pure strategy assigns one value $\pi(J) \in C_J$ to every information set. No-absentmindedness is the only information-set hypothesis used in Theorem 11.27; the stronger global recall conditions of Definition 11.32 are separate assumptions.

Lemma 11.25 (Imperfect-information path characterisation). *A node h is reached by (π, w) iff $w \in W_h$ and, at every predictor ancestor g of h , the path edge is labelled by $\pi(J_g)$, where J_g is the information set containing g .*

Proof. Induct along the unique path as in Lemma 11.3. The condition that one information set contains no two nodes on a path ensures the play to h never consults the current information set earlier on that path. $\square$

Definition 11.26 (Information-saturated patches). A patch U is information saturated when $h \in U \cap \mathcal{T}_P$ implies $J_h \subseteq U$. Intersections and unions of saturated prefix-closed patches remain saturated and prefix closed. Define

$$\mathcal{S}_{\mathcal{I}}(U) = \prod_{\substack{J \in \mathcal{I} \\ J \subseteq U}} C_J.$$

For $s \in \mathcal{S}_{\mathcal{I}}(U)$, define $\text{Reach}_{\mathcal{I}}(w, h; s)$ by the ordinary arena recursion, using choice $s(J_g)$ at each predictor ancestor g . Prefix closure and saturation ensure $J_g \subseteq U$. Put

$$\mathcal{V}_{\varepsilon, \mathcal{I}}(U) = \{s \in \mathcal{S}_{\mathcal{I}}(U) : \forall h \in U \cap \mathcal{T}_P \forall w \in W_h, \text{Reach}_{\mathcal{I}}(w, h; s) \Rightarrow \lambda_h(w, s(J_h)) \leq \varepsilon\}.$$

Theorem 11.27 (Imperfect-information activated subsheaf). *On information-saturated patches, $\mathcal{S}_{\mathcal{I}}$ is a sheaf for union covers and $\mathcal{V}_{\varepsilon, \mathcal{I}}$ is a subsheaf.*

Proof. A matching family assigns one value to each information set: if one node of an information set lies in a covering patch, saturation places the whole set there. Matching makes the value independent of the patch, giving unique coordinatewise gluing.

For restriction, prefix closure and saturation retain every information set consulted on the path to a retained node, so reachability and loss there are unchanged. For gluing, choose a patch containing a reached node. It contains all ancestor information sets; matching makes the local and glued runs coincide, and local activated validity supplies the loss bound. $\square$

Remark 11.28. The proof of Theorem 11.27 uses neither finiteness nor perfect recall; these restrictions are retained for the surrounding capability and complexity results.

Corollary 11.29 (Full-site collapse with imperfect information). *On the category of nonempty information-saturated prefix-closed patches, arrows are restriction arrows, and the full tree is initial, so*

$$\Gamma_{\text{set}}(\mathcal{V}_{\varepsilon, \mathcal{I}}) \cong \mathcal{V}_{\varepsilon, \mathcal{I}}(\mathcal{T}).$$

Thus the full site still has no D_{ε} outcome.

Proof. Apply Theorem 5.5 to Theorem 11.27. $\square$

Proposition 11.30 (Behavioural-coordinate sheaf). *Assume the information-set choice sets are finite. For an information-saturated patch U , put*

$$\mathcal{B}(U) = \prod_{\substack{J \in \mathcal{I} \\ J \subseteq U}} \Delta(C_J),$$

with restriction by dropping coordinates. Then $\mathcal{B}$ is a sheaf on the poset of information-saturated patches for union covers by information-saturated patches, and the Dirac embedding $\mathcal{S}_{\mathcal{I}} \hookrightarrow \mathcal{B}$ commutes with restriction.

Proof. A matching family assigns the same probability vector to an information set in every covering patch containing it. Choose that common vector for each information set in the union. This gives a unique global element of $\mathcal{B}(U)$ and restricts to every local member. Applying the Dirac map in each coordinate plainly commutes with deleting coordinates. $\square$

Remark 11.31. Proposition 11.30 is a coordinate-gluing statement and does not require perfect recall. A theorem about expected-loss validity or Kuhn realisation equivalence would additionally require player ownership, probabilities for nature, zero-reach conventions, and a specified expected-loss objective. In particular, a Bell coupling is shared randomisation over complete pure response tables and is not identified here with independent behavioural randomisation at information sets.

11.3 Information refinement and recall

Definition 11.32 (Information refinement and recall). Let two imperfect-information arenas have the same tree, nature maps, choice sets, and losses. Write $\mathcal{T} \preceq \mathcal{T}'$ when every information set of $\mathcal{T}'$ is contained in an information set of $\mathcal{T}$. The arena has *information recall* when any two nodes in one information set have the same ordered sequence of predecessor information sets. It has *action recall* when the corresponding predecessor predictor-edge labels also agree. Perfect recall

is the conjunction. This is a global predictor-level recall condition, stronger than the standard player-relative perfect-recall condition in multi-player extensive-form games; no player ownership relation is part of this definition.

For a no-absentminded arena $\mathcal{T}$, its action-recall refinement $\mathcal{T}^+$ splits each information set according to the complete sequence of predecessor predictor-edge labels, leaving every other arena field unchanged.

Proposition 11.33 (Capability monotonicity under information refinement). *If $\mathcal{T} \preceq \mathcal{T}'$, pure ε -capability of $\mathcal{T}$ implies pure ε -capability of $\mathcal{T}'$.*

Proof. Given a strategy on the coarser partition, assign its value at an original information set to every refined subset. Every world then follows the same predictor edges in the two arenas, so reached nodes and losses agree. $\square$

Proposition 11.34 (Reachable-action collapse criterion). *Let $\mathcal{T} \preceq \mathcal{T}'$ and let σ be a pure strategy on the refined arena. There is a coarse strategy π inducing the same play as σ at every world if and only if, for every original information set J , all refined classes $J' \subseteq J$ containing a σ -reachable node receive one common action. When this condition holds, all worldwide losses are preserved.*

Proof. Suppose first that the condition holds. For each original information set J with a σ -reachable node, let c_J be the common action on its reachable refined classes; choose an arbitrary element of C_J otherwise. Define $\pi(J) = c_J$. Induct down the play at any fixed world. The root and every nature move agree because the underlying arena data are unchanged. If the plays have agreed up to a reached predictor node $h \in J$, then the refined class J'_h contains a σ -reachable node and $\pi(J) = c_J = \sigma(J'_h)$. Thus the same predictor edge is selected. The full plays, and hence all reached losses, agree.

Conversely, suppose a coarse strategy π reproduces every σ -play. If a refined class $J' \subseteq J$ contains a node reached under σ at some world, equality of the two plays at that node forces $\sigma(J') = \pi(J)$. Therefore all reachable refined classes inside J have the common value $\pi(J)$. $\square$

Theorem 11.35 (Pure action-recall redundancy). *Let $\mathcal{T}$ be a finite no-absentminded arena with information recall in the global predictor-level sense of Definition 11.32. For every tolerance,*

$$\mathcal{T} \text{ is pure } \varepsilon\text{-capable} \iff \mathcal{T}^+ \text{ is pure } \varepsilon\text{-capable}.$$

Proof. The forward implication is Proposition 11.33. For the converse, let σ be a capable pure strategy on $\mathcal{T}^+$. Call a node σ -reachable when it is reached by (σ, w) for some world w . First prove by induction on the length of the predecessor information-set sequence that all σ -reachable nodes in one original information set J have the same predecessor action sequence.

The assertion is immediate when the sequence is empty. Suppose the common information-set sequence supplied by information recall is $J_1, \dots, J_t$, and let $h, h' \in J$ be σ -reachable. Their predecessor nodes in J_t are also reachable. By the induction hypothesis those nodes have the same action prefix and hence belong to one refined class inside J_t . The deterministic strategy σ assigns that class one action. Nature nodes between that last predictor ancestor and h or h' add no predictor-edge labels, so appending this common action proves that h and h' have the same predecessor action sequence. Thus all reachable nodes of J lie in one refined class J^+ .

The claim is exactly the reachable-action condition of Proposition 11.34. That proposition supplies a coarse strategy with the same play and losses at every world. It is therefore ε -capable. $\square$

Proposition 11.36 (Sharpness of information recall). *There is a finite no-absentminded arena without information recall which is not 0-capable although its action-recall refinement is 0-capable.*

Proof. Let $W = \{0, 1\}$. The nature root reveals w and moves to a predictor node a_w ; the two nodes a_0, a_1 are singleton information sets, have binary choices, and zero loss. Choice c leads to $b_{w,c}$. Put all four nodes $b_{w,c}$ in one information set J with binary report r , and set its loss to zero exactly when $r = w$.

A strategy in the original arena assigns one report to J and therefore fails on one world. In the action-recall refinement, J splits into classes J_0 and J_1 according to the first action. Choose action w at a_w and report c at J_c . The reached report equals the world, giving zero loss. Information recall fails because nodes over worlds 0 and 1 have predecessor information sets $\{a_0\}$ and $\{a_1\}$, respectively. $\square$

Proposition 11.37 (Strategic-mixed preservation). *Under the hypotheses of Theorem 11.35, every finite-support mixed strategy on the action-recall refinement $\mathcal{T}^+$ has a finite-support mixed strategy on $\mathcal{T}$ inducing the same distribution on terminal nodes for every world. Consequently any terminal expected-loss or expected-utility functional, once nature probabilities are supplied, has the same value under the two mixed strategies.*

Proof. The proof of Theorem 11.35 assigns to each pure strategy σ on $\mathcal{T}^+$ a pure strategy $\bar{\sigma}$ on $\mathcal{T}$ with the same reached terminal node at every world. Push a finite-support probability measure on pure strategies of $\mathcal{T}^+$ forward along $\sigma \mapsto \bar{\sigma}$. Conditional on each world, the terminal-node distribution is unchanged. Taking expectations against any specified nature law therefore preserves every payoff or loss depending only on the world and the terminal node. $\square$

Remark 11.38 (Boundary with perfect recall and Kuhn equivalence). Theorem 11.35 is a capability statement for pure strategies, and Proposition 11.37 extends it only to mixed strategies understood as ex ante randomisation over complete pure plans. This is not Kuhn’s theorem. Kuhn’s mixed-behavioural realisation equivalence is a player-relative theorem for finite extensive games with chance probabilities, player ownership, utilities, and standard perfect recall; see, for example, [22, 23]. The present arena formalism has no player ownership relation at this point, and its global predictor-level information recall is stronger in one direction and different in another: it can hold even when a standard player would have forgotten her own earlier action.

Accordingly, action recall is not redundant for independent behavioural randomisation at information sets. A minimal warning example has one decision maker, one initial binary move $b \in \{0, 1\}$, and a later information set J that merges the two histories following $b = 0$ and $b = 1$. The two nodes in J have the same predecessor information-set sequence, so global information recall holds, but action recall fails. Against an adversarially chosen target $t \in \{0, 1\}$, give success exactly when the terminal pair is $(b, r) = (t, t)$. In the action-recall refinement a behavioural strategy can choose b uniformly and then copy the recalled value of b , obtaining success probability $1/2$ against each target. In the original arena, a behavioural strategy chooses $b = 1$ with probability p and the later report $r = 1$ with an independent probability q at J , so the two target success probabilities are pq and $(1 - p)(1 - q)$; their maximum possible minimum is $1/4$. By contrast, an ex ante mixed strategy on the original arena can randomise equally between the pure plans $(b, r) = (0, 0)$ and $(1, 1)$ and again obtains $1/2$ against each target. Thus the redundancy theorem belongs to pure and strategic-mixed capability theory, not to behavioural-strategy equivalence.

A game-theoretic development of this observation would require a separate section or article with standard extensive-form data: players, chance laws, information partitions by player, mixed and behavioural strategies, expected utilities, zero-reach conventions, and equilibrium or max-min solution concepts. Its natural comparison point would be Kuhn’s theorem and the theory of imperfect recall, stochastic games, or team decision problems. In the present monograph the result

is best kept as a scoped structural lemma explaining why pure action labels need not be added to the interface when global information histories already determine all pure reachable action histories.

11.4 One-shot team arenas and blind-order normal forms

Definition 11.39 (One-shot team arena). Let a finite distributed selector problem be given and order its agents as $J = \{j_1, \dots, j_m\}$. Define a finite arena as follows.

- (i) The root is a nature node with move set W and nature map $\nu_\varepsilon(w) = w$; its child is denoted h_w and has $W_{h_w} = \{w\}$.
- (ii) For $1 \leq k \leq m$, each history $h_{w, \mathbf{c}_{<k}}$ is a predictor node for agent j_k , with choice set C_{j_k} and one child for each $c_k \in C_{j_k}$. At level $k = 1$ the preceding action tuple is empty.
- (iii) The information set associated with agent j_k and observation $z \in \text{im}(I_{j_k})$ is

$$\mathcal{I}_{j_k, z} = \{h_{w, \mathbf{c}_{<k}} : I_{j_k}(w) = z\}.$$

All nodes in this set have choice set C_{j_k} .

- (iv) Loss is zero at levels $k < m$. At the final predictor node,

$$\lambda_{h_{w, \mathbf{c}_{<m}}}(w, c_m) = \begin{cases} 0, & (c_1, \dots, c_m)|_{S(w)} \in V_w, \\ 1, & \text{otherwise.} \end{cases}$$

The final choice leads to a terminal child.

Every agent acts at every world, including worlds at which that agent is not in $S(w)$; terminal validity ignores coordinates outside $S(w)$. Earlier actions are not included in later agents' information labels.

Each distributed agent moves only once, so a player-relative perfect-recall condition would be automatic after adding player ownership. In the global single-predictor sense of Definition 11.32, however, the one-shot arena need not have information recall, since later information sets may identify histories with different predecessor information sets.

Theorem 11.40 (One-shot team-arena realisation). *For every finite distributed selector problem:*

- (i) *pure team strategies are in bijection with full local policy tuples*

$$\prod_{j \in J} C_j^{\text{im}(I_j)};$$

restriction to active observations gives the canonical CSP assignment;

- (ii) *a distributed strategy is valid iff the corresponding team strategy has zero worst-case terminal loss;*
- (iii) *the full information-saturated patch site has outcome L_0 if and only if the distributed problem is unsolvable; if every realised-scope relation is nonempty but no strategy exists, the canonical incidence diagram has outcome D_0 .*

Proof. A pure team strategy chooses one value at each information set $\mathcal{I}_{j,z}$, exactly a function $\text{im}(I_j) \rightarrow C_j$ for every agent. These functions form the full distributed policy tuple. Restricting them to active observations gives the CSP assignment, while inactive values do not enter terminal validity.

At world w , the action of agent j is $\pi_j(I_j(w))$. The terminal loss vanishes exactly when the tuple on active scope $S(w)$ lies in V_w , proving (ii). If the distributed problem is unsolvable, part (ii) makes the activated-validity set of the full tree empty, so the full-site outcome is L_0 by Corollary 11.29. Conversely, a valid distributed strategy gives a valid full-tree strategy by part (ii), and its restrictions inhabit every patch; hence the full site cannot have L_0 . The final assertion of (iii) is Theorem 7.4. $\square$

Definition 11.41 (Blind-order one-shot normal form). A finite imperfect-information arena is in blind-order one-shot normal form when it is equipped with a finite world set W , an ordered agent set $J = \{j_1, \dots, j_m\}$, finite choice sets C_j , finite observation sets Z_j , interfaces $I_j : W \rightarrow Z_j$, nonempty scopes $S(w) \subseteq J$, and relations $V_w \subseteq \prod_{j \in S(w)} C_j$, with the following structure. The root is a nature node selecting w . At level k , the predictor nodes are indexed by

$$(w, c_{<k}) \in W \times \prod_{\ell < k} C_{j_\ell},$$

and all nodes with the same value of $I_{j_k}(w)$ form one information set, independently of $c_{<k}$. Intermediate losses vanish; here terminal loss means the loss evaluated at the final predictor stage leading into terminal leaves. This final loss is zero exactly when

$$(c_j)_{j \in S(w)} \in V_w.$$

All choices lead to the corresponding action-prefix child. Isomorphisms of such arenas preserve the rooted labelled tree, node types, nature branches, information partition, choice sets, and losses.

Definition 11.42 (Distributed problem of a normal arena). For a blind-order one-shot arena $\mathcal{T}$, let $\text{Dist}(\mathcal{T})$ be the finite distributed selector problem given by its worlds, ordered agents, interfaces, choices, scopes, and terminal relations from Definition 11.41. Write $N(P)$ for the arena of Definition 11.39 constructed from a finite distributed problem P . Here finite means that the world, active observation, and choice sets and all relation tables are finite.

Theorem 11.43 (One-shot normal-form equivalence). *For every finite distributed selector problem P and every blind-order one-shot arena $\mathcal{T}$:*

(i) $N(P)$ is in blind-order one-shot normal form and

$$\text{Strat}(N(P)) \cong \prod_{j \in J} C_j^{\text{im}(I_j)};$$

(ii) this bijection preserves every worldwise zero-one final-stage loss and hence preserves and reflects capability;

(iii) $N(\text{Dist}(\mathcal{T}))$ is isomorphic to $\mathcal{T}$;

(iv) $\text{Dist}(N(P))$ recovers P up to restriction of each observation set to its realised image and renaming of observations and choices; the world set is recovered verbatim.

Thus finite distributed selectors and blind-order one-shot arenas are equivalent for pure-strategy capability and worldwise zero-one loss. This is not, by itself, a diagnostic equivalence between the canonical CSP incidence presentation and the full arena patch-site presentation. Morphisms and functoriality require additional structure beyond this object-level statement.

Proof. At each information set $\mathcal{I}_{j,z}$, a pure strategy chooses one element of C_j . Since these information sets are indexed exactly by $z \in \text{im}(I_j)$, choices at all information sets are precisely a tuple of maps $\pi_j : \text{im}(I_j) \rightarrow C_j$, proving (i).

Fix a world w . At level k , the reached node has action prefix equal to the previously selected values, and the next action is $\pi_{j_k}(I_{j_k}(w))$. The terminal tuple is therefore the distributed response tuple. By construction, terminal loss is zero exactly when its restriction to $S(w)$ belongs to V_w . This proves (ii).

For (iii), the normal arena already has one node $(w, c_{<k})$ for every world and action prefix. The operation **Dist** records the same worlds, order, interfaces, choices, scopes, and terminal relations; applying **N** therefore reconstructs the same predecessor and successor maps, information sets, and losses. The coordinatewise identification is the required arena isomorphism. Finally, applying **Dist** to **N**(P) reads back the same data used in the construction. Observations outside $\text{im}(I_j)$ never label an information set, giving exactly the harmless restriction stated in (iv). $\square$

11.5 World schedules

Definition 11.44 (World-scheduled one-pass arena). Let P be a finite distributed selector problem. A world schedule assigns to each w an ordering

$$\kappa_w = (j_{w,1}, \dots, j_{w,m})$$

of the full finite agent set, with every agent occurring once. The associated world-scheduled arena has a nature root selecting w , followed by predictor moves in the order κ_w . Agent j moves at the information set labelled $(j, I_j(w))$, independently of previous actions. Intermediate losses vanish, and terminal loss is zero exactly when the selected tuple restricted to $S(w)$ belongs to V_w . The schedule is part of the arena data.

Proposition 11.45 (World-scheduled distributed realisation). *For every finite distributed problem equipped with world schedules, pure strategies of the associated world-scheduled arena are canonically*

$$\prod_{j \in J} C_j^{\text{im}(I_j)},$$

and distributed validity is equivalent to zero worst-case terminal loss. The arena is reconstructed from the distributed datum together with the schedules, up to renaming of action-prefix nodes.

Proof. Every information set is indexed by one pair (j, z) , so a pure strategy is exactly one choice in C_j for each realised observation z . At world w , the schedule changes only the order in which the values $\pi_j(I_j(w))$ are selected; the terminal tuple contains exactly those values. The terminal loss condition is therefore the distributed validity condition. The distributed data and κ_w determine every world branch, action-prefix child, information-set label, and terminal loss, proving reconstruction. $\square$

Remark 11.46. Proposition 11.45 is a sufficient structure-preserving normal form, strictly allowing world-dependent orders. It is not a converse for arbitrary imperfect-information arenas: ordinary distributed data do not record schedules, and strategy-dependent signaling can still be represented only by the extensional compilation of Theorem 11.47.

11.6 Distributed compilation of finite arenas

Theorem 11.47 (Finite pure-strategy distributed compilation). *Let $\mathcal{T}$ be any finite imperfect-information arena and fix a tolerance ε . There is a finite distributed selector problem $P_{\mathcal{T},\varepsilon}$ whose strategy space is canonically the pure-strategy space of $\mathcal{T}$ and such that distributed capability is equivalent to ε -capability of the arena.*

Proof. Use one distributed agent a_J for every information set $J \in \mathcal{I}$. Give it the singleton observation space $Z_{a_J} = \{*\}$ and choice set C_J . Use the same world set W , take every agent active at every world, and define

$$V_w^\varepsilon = \{(c_J)_{J \in \mathcal{I}} : L_{\mathcal{T}}((c_J)_{J \in \mathcal{I}}, w) \leq \varepsilon\}.$$

A distributed strategy chooses one value in C_J for every singleton interface, exactly a pure arena strategy. Membership in V_w^ε is, by definition, the arena loss bound at w . Universal validity over worlds is therefore equivalent to ε -capability. $\square$

Remark 11.48. Theorem 11.47 is extensional. Its world relations can mention the complete pure-strategy tuple and can be exponentially large under explicit tabulation. It preserves strategy space, worldwide validity, and capability, but not temporal locality, the original arena tree, or the diagnostic outcome attached to a chosen patch presentation. By contrast, Theorem 11.43 characterises the restricted class for which the distributed presentation reconstructs the arena itself.

11.7 One-shot Bell realisation

Theorem 11.49 (Distributional one-shot realisation of Bell locality). *For a finite Bell scenario, let $\mathcal{T}_{\text{Bell}}$ be the one-shot information arena whose parties are agents, local settings are observations, and local outcomes are choices. For every finite behaviour p , the following are equivalent:*

- (i) p has a local hidden-variable model;
- (ii) $\text{Coupl}(p) \neq \emptyset$;
- (iii) there is $\mu \in \Delta(\text{Strat}(\mathcal{T}_{\text{Bell}}))$ such that evaluation at every setting tuple has distribution $p_{\mathbf{x}}$;
- (iv) the Bell star diagram has a compatible global family.

Proof. Theorem 9.24 proves the equivalence of (i), (ii), and (iv). By Theorem 11.40(i),

$$\text{Strat}(\mathcal{T}_{\text{Bell}}) \cong \prod_i O_i^{X_i} = \Omega.$$

Under this bijection, evaluation of a pure strategy at $\mathbf{x}$ is exactly $\pi_{\mathbf{x}}$. The induced affine bijection between the two probability simplices therefore sends the distribution of mixed-strategy evaluation to $(\pi_{\mathbf{x}})_*\mu$. The marginal equations in (iii) are consequently exactly the defining equations of $\text{Coupl}(p)$, proving (ii) $\Leftrightarrow$ (iii). $\square$

Remark 11.50. Theorem 11.49 concerns mixed-strategy marginal realisation, not the worst-case capability predicate. The deterministic CHSH winning loss and a Bell behaviour use the same one-shot information skeleton and deterministic response tables but impose different, respectively support-level and probability-level, conditions.

Corollary 11.51 (One-shot deterministic CHSH arena). *The private-information team problem of Corollary 7.14 has a finite one-shot imperfect-information arena realisation. Nature chooses (x, y) , the first agent observes only x , the second only y , and terminal loss vanishes exactly when $a \oplus b = x \wedge y$. These are the support constraints of the CHSH winning predicate; the statement concerns impossibility of a deterministic strategy that wins every input.*

Proof. Apply Theorem 11.40 to the two-agent distributed selector. $\square$

11.8 Explicit-table complexity boundary

Proposition 11.52 (Explicit-table complexity boundary). *On finite perfect-information arenas, backward induction performs $O(\sum_{h \in \mathcal{T}_P} |W_h| |C_h|)$ predictor loss-table checks and runs in time linear in the explicit game description, including nature routing tables. Capability for explicitly represented finite imperfect-information arenas is NP-complete.*

Proof. Membership in NP follows by guessing one choice per information set and checking all explicitly represented world-node loss constraints. For hardness, reduce the NP-complete variant [29] of 3SAT in which each clause has three distinct variables. Nature selects a clause. The arena then queries, in a fixed order, the information sets corresponding to the variables occurring in that clause; all occurrences of the same Boolean variable in different clauses share one binary information set, whose chosen value is interpreted as the truth value of that variable. The terminal loss is zero exactly when the selected clause is satisfied. This arena has size polynomial in the formula, because each branch contains only the variables of one clause. A pure strategy is a truth assignment, and it is zero-capable exactly when it satisfies every clause. Thus the explicit perfect-information regime is polynomial-time by backward induction, whereas the explicit imperfect-information regime is NP-complete. The generic blind-order normal-form construction of Definition 11.39 may be exponential in the number of distributed agents and is not used for this explicit-tree hardness reduction; for compressed blind-order input data, capability is exactly the finite distributed-selector problem of Theorem 7.12. $\square$

11.9 Static compilation

Definition 11.53 (Static compilation). A static compilation consists of a compiled game with world set $W^\sharp$, a surjection $j : W \twoheadrightarrow W^\sharp$, and a strategy transformation $K : \text{Strat}(\mathcal{T}) \rightarrow \Pi^\sharp$ whose image is the declared compiled policy class and which preserves aggregate losses:

$$L^\sharp(K(\pi), j(w)) = L_{\mathcal{T}}(\pi, w).$$

Thus the loss profile of every source strategy must factor through j after applying K ; if $j(w_0) = j(w_1)$, then any strategy in the domain of the compilation has equal transported losses at w_0 and w_1 . This factorisation is part of the well-formedness of the compilation data.

Proposition 11.54 (Compiled selector semantics). *Under a static compilation, sequential and compiled capability are equivalent.*

Proof. A sequential witness maps to a compiled policy. For $u \in W^\sharp$, choose w with $j(w) = u$ and use loss preservation. Conversely, every declared compiled policy lies in the image of K ; choose a preimage strategy and use loss preservation at every world. $\square$

Proposition 11.55 (Composition of static compilations). *Static compilations compose when the declared policy class of a composite is the image of its composite strategy transformation. If $(j_1, K_1) : A \rightarrow B$ and $(j_2, K_2) : B \rightarrow C$ preserve the corresponding worldwide aggregate losses, then*

$$(j_2 \circ j_1, K_2 \circ K_1) : A \rightarrow C$$

is a static compilation with declared compiled policy class $\text{im}(K_2 \circ K_1)$. Identity maps give identity compilations, so static compilations form a category with this image-as-declared-class convention.

Proof. The composite of surjections is a surjection, and the declared policy class of the composite is, by convention, the image of the composite strategy map. For every strategy π and world w ,

$$L_C(K_2(K_1(\pi)), j_2(j_1(w))) = L_B(K_1(\pi), j_1(w)) = L_A(\pi, w).$$

The identity case and associativity are ordinary identities and associativity of function composition. If one fixes target policy classes independently of the maps, closure under composition additionally requires $\text{im}(K_2 \circ K_1)$ to equal the predeclared target class. $\square$

Remark 11.56. A static compilation is a capability equivalence. Preservation of the diagnostic outcome additionally requires transport of the declared local objects, admissibility sets, and realised global-family spaces as part of the compilation data.

11.10 Operational guarded feedback and universal concurrency

A finite causal poset represents universally executed cuts whose order is partial. Protocols in which an action determines whether a later cut occurs are instead represented by the branching arenas of Section 11.

Definition 11.57 (Finite-poset guarded protocol). A finite-poset guarded protocol consists of a finite strict poset $(T, <)$, a finite nonempty world set W , and, for each $\tau \in T$, finite nonempty sets $E_\tau, Z_\tau, R_\tau, A_\tau, M_\tau, H_\tau$. Put $C_\tau = R_\tau \times A_\tau$ and

$$\mathbf{Hist}_{<\tau} = \prod_{\sigma < \tau} (Z_\sigma \times M_\sigma \times A_\sigma \times H_\sigma),$$

where this predecessor-indexed product is a set of functions on the strict past, and is a singleton when the past is empty. The structural maps are

$$Q_\tau : W \rightarrow E_\tau, \quad I_\tau : E_\tau \times \mathbf{Hist}_{<\tau} \rightarrow Z_\tau, \quad D_\tau : R_\tau \rightarrow M_\tau,$$

$$U_\tau : W \times \mathbf{Hist}_{<\tau} \times M_\tau \times A_\tau \rightarrow H_\tau,$$

$$\lambda_\tau : W \times \mathbf{Hist}_{<\tau} \times C_\tau \rightarrow [0, \infty].$$

The independently supplied map Q_τ is the accessible exogenous record at cut τ . Strict-past typing is the guard: no cut reads its own or a future record. Every cut occurs in every execution.

For an arbitrary complete choice profile $c = (c_\sigma)_\sigma \in \prod_\sigma C_\sigma$, its syntactic record at τ is defined by well-founded recursion, writing $c_\sigma = (r_\sigma, a_\sigma)$:

$$z_\tau^c(w) = I_\tau(Q_\tau(w), \mathbf{h}_{<\tau}^c(w)), \quad m_\tau^c = D_\tau(r_\tau), \quad h_\tau^c(w) = U_\tau(w, \mathbf{h}_{<\tau}^c(w), m_\tau^c, a_\tau),$$

where $\mathbf{h}_{<\tau}^c(w)$ collects the evaluated predecessor tuples $(z_\sigma^c(w), m_\sigma^c, a_\sigma, h_\sigma^c(w))_{\sigma < \tau}$. For such a complete profile also define

$$L_{\text{op}}(c, w) = \max(\{\lambda_\tau(w, \mathbf{h}_{<\tau}^c(w), c_\tau) : \tau \in T\} \cup \{0\}).$$

Define

$$Z_\tau^{\text{act}} = \{z_\tau^c(w) : w \in W, c \in \prod_\sigma C_\sigma\}.$$

A policy is a family $\pi = (\pi_\tau)_\tau$ with $\pi_\tau : Z_\tau^{\text{act}} \rightarrow C_\tau$. It records the realised report–action pair. The larger two-component policy space $R_\tau^{Z_\tau} \times A_\tau^{Z_\tau \times R_\tau}$ has this space as its observational quotient; counterfactual actions at unchosen reports are not part of the operational semantics. If report-conditional counterfactual actions are required, a cut should be split into a report decision followed by an action decision, or the action carrier should be replaced by a function carrier $R_\tau \rightarrow A_\tau$.

Definition 11.58 (Poset execution and capability). For a policy π and world w , an execution is a family $(z_\tau, c_\tau, m_\tau, h_\tau)_{\tau \in T}$ satisfying, with $c_\tau = (r_\tau, a_\tau)$ and $\mathbf{h}_{<\tau} = (z_\sigma, m_\sigma, a_\sigma, h_\sigma)_{\sigma \prec \tau} \in \mathbf{Hist}_{<\tau}$,

$$\begin{aligned} z_\tau &= I_\tau(Q_\tau(w), \mathbf{h}_{<\tau}), & c_\tau &= \pi_\tau(z_\tau), & m_\tau &= D_\tau(r_\tau), \\ h_\tau &= U_\tau(w, \mathbf{h}_{<\tau}, m_\tau, a_\tau). \end{aligned}$$

Its loss is

$$L_{\text{op}}(\pi, w) = \max(\{\lambda_\tau(w, \mathbf{h}_{<\tau}, c_\tau) : \tau \in T\} \cup \{0\}).$$

The added value 0 fixes the empty-poset convention. For $\Pi \subseteq \prod_\tau C_\tau^{Z_\tau^{\text{act}}}$, the protocol is ε -capable in Π when some $\pi \in \Pi$ has $L_{\text{op}}(\pi, w) \leq \varepsilon$ for every world.

Theorem 11.59 (Unique extension-independent execution). *Every policy and world determine a unique execution. Evaluating its equations in any linear extension of $(T, \prec)$ gives the same T -indexed family and the same aggregate loss.*

Proof. Apply Proposition A.1 to the finite strict poset. At τ , every coordinate of $\mathbf{h}_{<\tau}$ has already been determined. The resulting observation lies in Z_τ^{act} because the already determined finite predecessor choice profile extends to a complete choice profile using arbitrary elements of the nonempty later choice sets. Thus $\pi_\tau(z_\tau)$ is defined, and the displayed equations determine successively $z_\tau, c_\tau, m_\tau, h_\tau$. Well-founded induction gives uniqueness. A linear extension orders these evaluations; since the equation at τ reads exactly the strict-predecessor coordinates, its value is the unique recursively determined value in every extension. The aggregate loss is a function of that indexed family and is therefore unchanged. $\square$

Remark 11.60 (Partial order of cuts). A finite cut poset records concurrency without making execution nondeterministic. Incomparable cuts may be evaluated in either order, but each cut reads only its strict causal past. The resulting execution is therefore indexed by the poset, not by the chosen linear extension used to compute it. Figure 4 illustrates a four-cut poset and two of its linear-extension evaluations.

Definition 11.61 (Order-ideal configuration system). Let $\mathcal{I}(T)$ be the finite lattice of order ideals of T . For fixed (π, w) , a configuration over an order ideal $U \in \mathcal{I}(T)$ records the execution tuples at cuts in U . A cut $\tau \notin U$ is enabled when every predecessor of τ lies in U ; the transition $U \rightarrow U \cup \{\tau\}$ computes its tuple from the recorded strict past.

Proposition 11.62 (Concurrency diamond and confluence). *If incomparable cuts τ and σ are both enabled at a configuration over an order ideal $U \in \mathcal{I}(T)$, then executing τ followed by σ or σ followed by τ produces the same configuration over $U \cup \{\tau, \sigma\}$. Every maximal path through the configuration system therefore yields the unique execution of Theorem 11.59.*

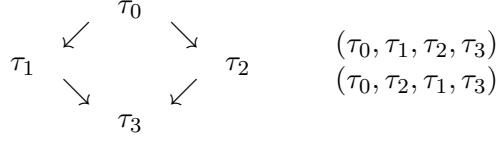


Figure 4: A finite cut poset and two linear-extension evaluations. Incomparable cuts may be evaluated in either order; confluence says the operational record and aggregate loss are independent of that order.

Proof. Simultaneous enablement implies that neither cut is a predecessor of the other. The computation at τ reads only coordinates indexed by $\{\xi : \xi \prec \tau\} \subseteq U$ and writes only the τ coordinate; the analogous statement holds for σ . Thus the first transition changes no input read by the second, and the two computed tuples are identical in either order. Adjacent swaps of simultaneously enabled incomparable cuts connect the maximal paths associated with linear extensions; repeated use of the diamond property gives the same final record. Equivalently, confluence follows from the uniqueness theorem. $\square$

Definition 11.63 (Linear-extension arena). Fix a linear extension $L = (\tau_1, \dots, \tau_m)$. The arena $\mathcal{A}_L(P)$ has a nature root selecting $w \in W$. At stage k it has a predictor node $h_{w, c_{<k}}^k$ for every world and every prefix $c_{<k} \in \prod_{i < k} C_{\tau_i}$, with choice set C_{τ_k} . Here $h_{w, c_{<k}}^k$ denotes an arena tree node, distinct from the stage output values $h_\tau \in H_\tau$. Compute the causal-predecessor record from w and the prefix, ignoring earlier incomparable cuts, and label the node by

$$z_{\tau_k} = I_{\tau_k}(Q_{\tau_k}(w), \mathbf{h}_{<\tau_k}).$$

Two stage- k nodes are in the same information set exactly when these labels agree. The choice c_k leads to $h_{w, c_{\leq k}}^{k+1}$ for $k < m$, and to a terminal leaf $h_{w, c_{\leq m}}^{m+1}$ at $k = m$, and node loss is

$$\lambda_{h_{w, c_{<k}}^k}(w, c_k) = \lambda_{\tau_k}(w, \mathbf{h}_{<\tau_k}, c_k).$$

All worlds and prefixes are included before a policy is selected.

Theorem 11.64 (Operational arena realisation). *For every linear extension L , $\mathcal{A}_L(P)$ is a finite imperfect-information arena, independent of the policy, and*

$$\text{Strat}(\mathcal{A}_L(P)) \cong \prod_{\tau \in T} C_\tau^{\text{Z}_{\tau}^{\text{act}}}.$$

Under this bijection, arena play and operational execution agree at every cut,

$$L_{\mathcal{A}_L(P)}(\pi, w) = L_{\text{op}}(\pi, w),$$

and capability is preserved and reflected for every transported policy class. Different linear extensions need not give isomorphic trees, but they realise the same operational policy space and worldwide loss functional. On the information-saturated patch site of each arena, activated validity is a subsheaf and the full tree is initial, so full-site $\mathbf{D}_\varepsilon$ is impossible.

Proof. Finiteness and no-absentmindedness follow because the data and prefix products are finite and each information set lies at one stage. The arena contains all syntactic prefixes, including

prefixes that may be unreachable under a fixed observation-indexed policy. Every label in Z_τ^{act} occurs at a node: a complete syntactic choice profile witnessing the label restricts to a prefix in any linear extension, and incomparable earlier choices are ignored. Conversely every node label is in the defined active image. Hence a pure strategy is exactly one map $Z_\tau^{\text{act}} \rightarrow C_\tau$ per cut.

Fix a policy and world. Induct along L . All causal predecessors of τ_k occur earlier, and by induction their arena records equal their operational records. The node label is therefore the operational observation, and the selected edge is the operational choice. The stage record and loss then agree by definition. Taking maxima proves loss equality, and quantifying over worlds proves capability equivalence. Extension independence follows from Theorem 11.59. The patch statement is Theorem 11.27 and Corollary 11.29. $\square$

Proposition 11.65 (Size of a linear-extension arena). *For $L = (\tau_1, \dots, \tau_m)$, the arena of Definition 11.63 has*

$$1 + |W| \sum_{k=1}^{m+1} \prod_{i < k} |C_{\tau_i}|$$

nodes: the root, all predictor nodes, and all terminals; the term $k = m + 1$ counts terminal leaves. Thus explicit arena synthesis can be exponential in m . Different linear extensions can have different numbers of intermediate predictor nodes, although Theorem 11.64 identifies their operational policy spaces and worldwide loss functionals.

Proof. At stage $k \leq m$, a predictor node is determined by a world and one choice for each of the $k - 1$ earlier cuts in L , giving $|W| \prod_{i < k} |C_{\tau_i}|$ nodes. The same product with $k = m + 1$ counts the terminal leaves after a complete choice profile. Adding the nature root proves the formula. For $m = 0$, the empty product is understood as 1; the formula counts a nature root with $|W|$ terminal leaves and no predictor information sets, which may be identified without changing capability. The products can grow exponentially with the number of cuts. Their partial sums can depend on the order of unequal choice-alphabet sizes, while the operational equivalence is already proved in Theorem 11.64. $\square$

Proposition 11.66 (Perfect-recall decoding criterion). *In the global arena-level sense of Definition 11.32, the arena $\mathcal{A}_L(P)$ has perfect recall exactly when, for every cut τ and realised observation z , all syntactic nodes in the information set (τ, z) have the same preceding sequence of predictor information sets and choices. Equivalently, for each realised label there is a decoding map from (τ, z) to that common sequence.*

Proof. The preceding information-set/action sequence is precisely the predictor's own history in the synthesized arena. Perfect recall says this sequence is constant on each current information set. Constancy defines the decoding map, and a decoding map implies constancy. The existence of the observation map alone is insufficient. $\square$

Remark 11.67 (Concurrency and recall). When a linear extension orders two incomparable cuts τ_1 and τ_2 , the observation at the later cut does not read the earlier incomparable choice. If τ_1 has at least two possible choices and τ_2 is placed after it, the linear-extension arena therefore contains syntactic nodes at τ_2 with the same current observation but different preceding action histories. In the global arena-level sense of Definition 11.32, such a sequentialisation fails perfect recall whenever such an earlier incomparable cut has at least two choices. Thus Proposition 11.66 is nontrivially satisfied only when the cut order is total along the relevant prefixes, or when all earlier incomparable cuts have singleton choice sets.

Corollary 11.68 (Effective operational evaluation). *For effectively tabulated finite protocol data and a computable policy, the unique execution and every linear-extension arena are computable uniformly from the supplied finite codes. Here effectively tabulated means that finite carrier sets are given by finite tagged codes with decidable equality and that structural maps, losses, nature maps, and restrictions are given by finite lookup tables or primitive-recursive indices on the declared finite domains.*

Proof. Well-founded evaluation makes finitely many table lookups. Arena synthesis enumerates finite world and choice-prefix products and performs the same lookups. $\square$

11.11 World-only decoupling

Theorem 11.69 (Operational decoupling). *Suppose each observation is independent of the interaction history:*

$$I_\tau(Q_\tau(w), \mathbf{h}) = \bar{I}_\tau(Q_\tau(w)).$$

Then the protocol has an equivalent finite distributed selector with agents T , interfaces $w \mapsto \bar{I}_\tau(Q_\tau(w))$, choices C_τ , and world relation

$$V_w^\varepsilon = \{c \in \prod_{\tau \in T} C_\tau : L_{\text{op}}(c, w) \leq \varepsilon\},$$

where $L_{\text{op}}(c, w)$ is the complete-profile loss defined in Definition 11.57. This preserves and reflects ε -capability.

If, in addition, there are predicates $V_{\tau, w}^\varepsilon \subseteq C_\tau$ such that operational aggregate validity is exactly

$$L_{\text{op}}(c, w) \leq \varepsilon \iff c_\tau \in V_{\tau, w}^\varepsilon \text{ for every } \tau,$$

then the relation factors as $V_w^\varepsilon = \prod_{\tau} V_{\tau, w}^\varepsilon$ and capability is simultaneous fibrewise selection. In the corresponding discrete fibre-indexed packaging over cuts and observations, local nonemptiness implies a global family by finite choice (provable in ZF), so there is no D_ε outcome in that presentation.

Proof. Under the hypothesis, a policy supplies at world w exactly the profile $(\pi_\tau(\bar{I}_\tau(Q_\tau(w))))_\tau$. By definition, that profile belongs to V_w^ε exactly when its operational loss is at most ε . This is the distributed validity condition. Under the additional equivalence, membership in the world relation is coordinatewise, giving the displayed product. Grouping the unary constraints by each interface fibre gives the ordinary fibrewise selector conditions; the discrete compatibility result then excludes D_ε . $\square$

11.12 Observation-preserving compilation

Definition 11.70 (Observation-preserving static compilation). An observation-preserving static compilation of a guarded protocol is a static compilation in the sense of Definition 11.53, together with a policy-independent interface $I^\sharp : W^\sharp \rightarrow Z^\sharp$ and maps $q_\tau : Z^\sharp \rightarrow Z_\tau$ satisfying

$$z_\tau(\pi, w) = q_\tau(I^\sharp(j(w)))$$

for every cut, policy, and world.

Proposition 11.71 (Disclosure obstruction to observation compilation). *If the domain of the strategy transformation contains two policies that at one world produce different observations at some cut, then no observation-preserving static compilation exists on that domain. This does not preclude a loss-preserving static compilation.*

Proof. For fixed w the value $q_\tau(I^\sharp(j(w)))$ is independent of the policy, so it cannot equal two distinct operational observations. The existing static compilation definition requires only aggregate-loss preservation and therefore is not contradicted. $\square$

Remark 11.72 (Active sensing as guarded operational feedback). The sensing arena of Definition 11.20 is the two-cut chain $\tau_0 \prec \tau_1$. At τ_0 the observation is trivial and the physical action chooses $a \in \{0, 1\}$. At τ_1 the observation generator reads that action from the predecessor record and returns x_a ; the current choice is the binary report. Hence the two stage observations can differ at one world under different first-stage policies, so Proposition 11.71 excludes an observation-preserving static compilation. A loss-preserving contingent-plan compilation is not excluded. The full-site L_0 and cover-relative D_0 outcomes remain those of Theorem 11.22.

11.13 Trace-relative past factorisation

Proposition 11.73 (Trace-relative past factorisation). *Fix a cut τ and a syntactically valid choice assignment $\mu = (c_\sigma)_{\sigma \prec \tau}$ on its causal past. Recursively evaluate the predecessor history $\mathbf{h}_{<\tau}^\mu(w)$ using these fixed choices and put*

$$\begin{aligned} P_\tau^\mu &= E_\tau \times \mathbf{Hist}_{<\tau}, & P_\tau^\mu(w) &= (Q_\tau(w), \mathbf{h}_{<\tau}^\mu(w)), \\ J_\tau^\mu(e, \mathbf{h}) &= I_\tau(e, \mathbf{h}), & I_\tau^\mu(w) &= I_\tau(Q_\tau(w), \mathbf{h}_{<\tau}^\mu(w)). \end{aligned}$$

Then

$$I_\tau^\mu = J_\tau^\mu \circ P_\tau^\mu.$$

The map J_τ^μ is the original observation interface I_τ on the past-record space; the superscript records the fixed predecessor trace used to form P_τ^μ . The construction is independent of a policy after μ is fixed.

Proof. The strict past of τ is downward closed, so fixed predecessor choices and well-founded recursion uniquely determine every predecessor record. The factorisation equation is the definition of the three displayed maps. The exogenous map Q_τ was specified independently of the observation interface and of every policy, so this is a past factorisation of the form required in Definition 4.10. $\square$

Remark 11.74. The same idea applies to a fixed branch of a contingent arena after restricting to worlds that reach that branch; no separate formal object called a contingent tree protocol is used here.

Corollary 11.75 (Trace-slice obstruction). *Fix τ and μ . If two worlds $w_0, w_1 \in W$ satisfy $P_\tau^\mu(w_0) = P_\tau^\mu(w_1)$ and*

$$\bigcap_{i=0}^1 \{c \in C_\tau : \lambda_\tau(w_i, \mathbf{h}_{<\tau}^\mu(w_i), c) \leq \varepsilon\} = \emptyset,$$

then no policy whose execution realises μ on both worlds is ε -capable.

Proof. Equal past records give one common observation. Such a policy must therefore choose one common $c \in C_\tau$ at the two worlds. The empty intersection says that every choice violates the stage bound at one of them, and aggregate loss is at least that stage loss. $\square$

Remark 11.76 (Guardedness boundary). The poset protocol models acyclic, universally executed dependencies; contingent occurrence is represented by the arena tree. Cyclic simultaneous equations are not operational tree executions. The Scott-continuous least-fixed-point model in Appendix A gives a separate semantics from the operational poset results.

11.14 A bounded-height countable effectivity instance

The preceding arena theory is finite. The following narrowly bounded extension has countably many nodes but no infinite plays and requires no limit-payoff or transfinite semantics.

Definition 11.77 (Countable bounded-height arena). A countable bounded-height imperfect-information arena has the data of Definitions 11.1 and 11.24, except that the rooted tree, world set, and branching sets may be countable and there is a fixed finite bound on the length of every root-to-leaf path. Predictor choice sets remain finite. Reachability is defined recursively along each finite path, and aggregate loss is the maximum of the finitely many reached-node losses. Patches are nonempty information-saturated prefix-closed node sets.

Proposition 11.78 (Bounded-height activated subsheaf). *On a countable bounded-height arena, $\mathcal{S}_{\mathcal{I}}$ is a sheaf and $\mathcal{V}_{\epsilon, \mathcal{I}}$ is a subsheaf for union covers.*

Proof. A matching family gives one value to every information set occurring in the union: choose a node of the information set that lies in the union, then choose a covering patch containing that node. Saturation places the whole information set in that patch, and matching makes the value independent of the choice. This defines the unique glued strategy.

If a partial strategy is restricted to a smaller saturated prefix-closed patch, every information set consulted on a path to a retained node is retained, so reachability and loss at that node are unchanged. For a glued matching family and a reached node h , choose a cover member containing h . Prefix closure and saturation place every consulted ancestor information set there; matching makes the local and global runs identical. Local validity therefore supplies the loss bound. Neither argument uses finiteness of the total node set. $\square$

Definition 11.79 (Effective cover representation). Information sets are indexed by natural numbers, and finite choice-alphabet elements are represented by fixed finite codes. A cover $(U_i)_{i \in \mathbb{N}}$ is effectively represented when the predicate $J \subseteq U_i$ is decidable uniformly in (i, J) . A locator is a total computable map L satisfying $J \subseteq U_{L(J)}$. A matching family is uniformly computable when a partial recursive evaluator $\Phi(i, J)$ halts with value $s_i(J)$ on the represented domain $J \subseteq U_i$.

Theorem 11.80 (Uniform effective gluing with a locator). *Let a countable bounded-height arena have computably indexed information sets and finite choice alphabets. Let $(U_i)_{i \in \mathbb{N}}$ be a computably indexed cover by information-saturated patches. Suppose there is a computable locator L such that $J \subseteq U_{L(J)}$ for every information set J , and suppose a matching family $(s_i)_i$ is uniformly computable in the sense that $(i, J) \mapsto s_i(J)$ is computable whenever $J \subseteq U_i$. Then its unique global glue is computable. If every s_i is activated-valid, so is the glue.*

Proof. Define

$$s(J) = s_{L(J)}(J).$$

The locator and the uniform evaluator compute this value; the decidability of the cover predicate is part of the representation convention and is not needed once such a locator is supplied. If another cover member contains J , matching says that its value agrees with the displayed one, so s is the unique coordinatewise glue. Activated validity follows from Proposition 11.78. $\square$

Definition 11.81 (Finite invalidity witness). Index binary strategy coordinates by $\mathbb{N}$ and give $2^{\mathbb{N}}$ its standard product topology. A finite binary string σ is an invalidity witness when every total strategy in the basic clopen cylinder $[\sigma]$ fails zero-loss activated validity. A represented arena has uniformly c.e. finite invalidity when such strings can be enumerated uniformly from its name and their cylinders cover the entire invalid-strategy set.

Proposition 11.82 (WKL upper bound for represented causal selection). *Consider represented countable bounded-height arenas with computably indexed binary information-set choices and uniformly c.e. finite invalidity witnesses. On the promised domain where a zero-loss global strategy exists, selecting one is Weihrauch-reducible to WKL. Uniformly finite alphabets admit a fixed computable coding into Cantor space; the binary hypothesis avoids that coding.*

Proof. A total strategy is a point of $2^{\mathbb{N}}$. Enumerating the witness strings gives a negative name for the valid set

$$\mathcal{V}_{0,\mathcal{I}}(\mathcal{T}) = 2^{\mathbb{N}} \setminus \bigcup_{\sigma \text{ enumerated}} [\sigma],$$

a co-c.e. closed subset of Cantor space. The promise says this closed set is nonempty. One invocation of closed choice $\mathsf{C}_{2^{\mathbb{N}}}$ returns a valid strategy, and Theorem 10.2 gives $\mathsf{C}_{2^{\mathbb{N}}} \equiv_{\mathsf{W}} \mathsf{WKL}$. $\square$

Remark 11.83. Proposition 11.82 is an upper bound. The node-local height-two arena of Theorem 11.85 enforces coordinatewise DNR conditions and does not encode arbitrary prefix correlations, so no matching WKL-hardness claim is made for that bounded-height class.

Definition 11.84 (Height-two halting-stage arena). Let $W = \mathbb{N} \times \mathbb{N}$. The root, denoted ε , is a nature node with moves $n \in \mathbb{N}$ and $\nu_{\varepsilon}(n, s) = n$. Its child h_n is a predictor node with choice set $\{0, 1\}$ and singleton information set; each choice leads to a terminal node. Put

$$\lambda_{h_n}((n, s), c) = \begin{cases} 0, & \neg \text{Comp}(n, s, c), \\ 1, & \text{Comp}(n, s, c). \end{cases}$$

This is a primitive-recursively presented countable arena of height two.

Theorem 11.85 (Causal halting-stage effectivity obstruction). *For the height-two halting-stage arena:*

(i) *its zero-loss global strategy space is exactly the class $\mathcal{S}$ of Theorem 10.11; hence it is nonempty and effectively closed but has no computable member;*

(ii) *for the branch cover*

$$U_n = \{\varepsilon, h_n, h_n \cdot 0, h_n \cdot 1\},$$

there is a matching family of activated-valid partial strategies such that each individual section is computable, but no activated-valid matching family is uniformly computable;

(iii) *the arena has an X -computable zero-loss strategy exactly when X has PA degree. In particular, it has low and hyperimmune-free strategies.*

Proof. A global strategy is a binary function g . At node h_n , zero loss for every world (n, s) is equivalent to

$$\forall s \neg \text{Comp}(n, s, g(n)),$$

which is exactly the defining condition for $g \in \mathcal{S}$. Part (i) follows from Theorem 10.11.

The patches U_n cover the tree and distinct patches intersect only at the nature root, so their strategy restrictions have no common predictor coordinate. Use the classical selector f from Theorem 10.10 and let the section on U_n choose the single bit $f(n)$. Each such finite constant section is a computable object, and the family is matching and valid. If an activated-valid matching family were uniformly computable, evaluating its section at h_n uniformly in n would produce a computable member of $\mathcal{S}$, contrary to part (i). This proves (ii). Part (iii) is Corollary 10.12, since the global strategy condition is the same class $\mathcal{S}$. $\square$

Remark 11.86. Theorem 11.80 isolates the required uniformity. Each branch-local section in Theorem 11.85 is finite and therefore computable as an individual object. What fails is a single computable procedure that supplies the correct finite section for every branch. The example is a bounded-height countable instance, not a general complexity classification of infinite arenas. In particular, the full patch is computable as a set of nodes but has no computable valid section.

Corollary 11.87 (Division of labour). *Within the proved framework:*

- (i) *single-observer canonical diagrams have no D;*
- (ii) *perfect-information arenas are decided by backward induction and have no full-site D under node-local nonemptiness;*
- (iii) *finite distributed selectors canonically generate CSP compatibility obstructions, while nested countable distributed selectors can exhibit the effectivity obstruction of Theorem 10.16;*
- (iv) *every finite distributed instance has a blind-order one-shot realisation, and supplied world schedules permit world-dependent move orders;*
- (v) *the explicit finite perfect-information and one-shot imperfect-information regimes are separated by the complexity boundary of Proposition 11.52;*
- (vi) *universally executed partially ordered cuts have a confluent operational semantics and linear-extension-relative arena realisations, while contingent stage occurrence remains tree-indexed.*

Proof. Clause (i) is Corollary 7.11 together with discrete compatibility. Clause (ii) is Theorem 11.6, Corollary 11.7, and full-site collapse. Clause (iii) combines Theorem 7.4 with Theorem 10.16. Clause (iv) is Theorems 11.40 and 11.43 together with Proposition 11.45. The complexity statement in (v) is proved in Proposition 11.52 by backward induction and a direct polynomial 3SAT reduction for explicit imperfect-information arenas. Clause (vi) is Theorems 11.59 and 11.64 together with Proposition 11.62. $\square$

12 Functorial and compositional effective causal descent

The preceding sections establish object-level presentations. This section develops the corresponding morphism-level theory in the cases where the morphisms are specified explicitly. The guiding separation is

$$\text{capability preservation} \neq \text{diagnostic preservation} \neq \text{certificate preservation}.$$

A map may preserve the yes–no capability predicate while losing the local incidence data that distinguish L from D, and a diagnostic isomorphism may still fail to transport a particular parity, affine, holonomy, or Farkas certificate unless the algebraic structure used by that certificate is also preserved.

12.1 A category of constrained presentations over a fixed base

Fix a context category $\mathcal{C}$ and a tolerance level ε . A presentation object over $\mathcal{C}$ consists of an assignment functor $\mathcal{A}^P : \mathcal{C} \rightarrow \mathbf{Set}$, local admissibility sets $\mathcal{A}_{\text{loc},P}(c) \subseteq \mathcal{A}^P(c)$, and effective and bounded realised images inside the ambient family product, as in Definition 5.7. Write

$$\Gamma(P) = (\Gamma_{\text{set}}(P) \supseteq \Gamma_{\text{eff}}(P) \supseteq \Gamma_{\mathcal{B}}(P)).$$

Let **Nested** be the category whose objects are nested triples $X_0 \supseteq X_1 \supseteq X_2$ and whose morphisms are functions $h : X_0 \rightarrow Y_0$ satisfying $h(X_1) \subseteq Y_1$ and $h(X_2) \subseteq Y_2$.

Definition 12.1 (Presentation morphism over a fixed base). For presentations P, Q over $\mathcal{C}$, a morphism $P \rightarrow Q$ consists of a natural transformation $\alpha : \mathcal{A}^P \Rightarrow \mathcal{A}^Q$ such that

$$\alpha_c(\mathcal{A}_{\text{loc},P}(c)) \subseteq \mathcal{A}_{\text{loc},Q}(c) \quad (c \in \mathcal{C}),$$

and such that componentwise application of α carries the effective and bounded realised images of P into those of Q . Composition is composition of natural transformations.

Theorem 12.2 (Global-family functor). *The assignment $P \mapsto \Gamma(P)$ extends to a functor*

$$\Gamma : \mathbf{Pres}_{\mathcal{C}} \longrightarrow \mathbf{Nested}.$$

On a morphism represented by α , the map is

$$(\Gamma(\alpha)s)_c = \alpha_c(s_c).$$

Proof. By the hypothesis on α , Proposition 6.4 applies and makes componentwise application a well-defined map on the set-theoretic global spaces, preserving the effective and bounded components under the image hypotheses. Identities and composition are inherited from natural transformations, giving a functor. $\square$

Remark 12.3 (Direction of preservation). A presentation morphism transports positive success forward: nonempty local sets, compatible global families, effective realisations, and bounded realisations map to corresponding witnesses in the target. Failure labels are not functorial in this forward direction. A target presentation may contain additional local elements, global families, or implementations not lying in the image of the source. Preservation and reflection of failure therefore require isomorphism-strength data.

12.2 Changing context bases

The preceding functoriality result fixes the context category. Structural comparisons also require controlled changes of base: a presentation may be renamed, subdivided, or transported along an isomorphic context category without changing the section problem.

Definition 12.4 (Base isomorphism of presentations). Let P be a presentation over $\mathcal{C}$ and Q a presentation over $\mathcal{D}$. A base isomorphism $P \cong Q$ consists of an isomorphism of categories $F : \mathcal{C} \rightarrow \mathcal{D}$, a natural isomorphism

$$\alpha : \mathcal{A}^P \Rightarrow \mathcal{A}^Q \circ F,$$

objectwise equalities

$$\alpha_c(\mathcal{A}_{\text{loc},P}(c)) = \mathcal{A}_{\text{loc},Q}(Fc),$$

and bijections of effective and bounded implementation images compatible with the induced map on global families.

Unpacking the naturality of α : for every arrow $f : c \rightarrow c'$ of $\mathcal{C}$, the square

$$\begin{array}{ccc} \mathcal{A}^P(c) & \xrightarrow{\alpha_c} & \mathcal{A}^Q(Fc) \\ \mathcal{A}^P(f) \downarrow & & \downarrow \mathcal{A}^Q(Ff) \\ \mathcal{A}^P(c') & \xrightarrow{\alpha_{c'}} & \mathcal{A}^Q(Fc') \end{array}$$

commutes.

Theorem 12.5 (Invariance under base isomorphism). *A base isomorphism of presentations induces an isomorphism of obstruction data. Consequently it preserves and reflects*

$$L_\varepsilon, \quad D_\varepsilon, \quad K_\varepsilon, \quad R_\varepsilon, \quad \checkmark_\varepsilon.$$

Proof. Precomposition with an isomorphism of categories only renames the compatibility equations. Componentwise application of the natural isomorphism α then carries compatible families of P bijectively to compatible families of Q . The displayed equality of local sets preserves objectwise local nonemptiness and local admissibility. The implementation-image hypotheses identify the effective and bounded subspaces. Thus the local-success bit and all three nested global spaces are preserved and reflected. $\square$

Remark 12.6 (Refinements versus base isomorphisms). A base isomorphism is a safe change of coordinates. Noninvertible refinements are more delicate. Adding objects that are determined by existing restrictions can preserve the section problem, while deleting an initial object or quotienting contexts can change the diagnostic. Such refinements require their own cofinality, density, or certificate-preservation hypotheses.

12.3 Capability, diagnostic, and certificate morphisms

For behaviourally annotated presentations, a capability morphism preserves the policy or strategy space together with worldwide loss, possibly up to a declared loss-overhead map. This is enough to transport capability, but it need not preserve local incidence. A diagnostic morphism preserves the obstruction datum of Definition 6.5. A certificate morphism additionally preserves the local objects, local admissibility sets, restriction maps, and any algebraic or convex structure used to express a certificate.

Definition 12.7 (Morphism levels). A *capability isomorphism* identifies strategy or policy spaces and worldwide loss functionals (Definition 6.6). A *diagnostic isomorphism* is an isomorphism of obstruction data (Definition 6.5). A *certificate isomorphism* is a diagnostic isomorphism induced by an isomorphism of the underlying local-incidence presentation, including any linear, affine, convex, torsor, or order structure used by the certificate language.

Proposition 12.8 (Strictness of morphism levels). *Capability equivalence need not imply diagnostic equivalence, and diagnostic equivalence need not imply certificate equivalence.*

Proof. For the first assertion, take a compatibility-failing presentation and its terminal compression. The nested global spaces, and hence capability at the set-theoretic, effective, and bounded levels, are preserved, but the outcome changes from D to L by Theorem 6.10. For the second assertion, duplicate a constraint occurrence in a finite incidence diagram. The local-success bit and all nested global spaces can remain unchanged, while the incidence structure and minimal finite certificates change. A diagnostic datum records emptiness and nonemptiness of the distinguished spaces, not the multiplicity or shape of local constraint occurrences. $\square$

Theorem 12.9 (Certificate transport under structured isomorphism). *A certificate isomorphism transports every certificate construction defined from the preserved structure. In particular, it transports finite local certificates, finite incompatibility subdiagrams, parity certificates, affine inconsistency certificates, holonomy fixed-point certificates, and Farkas certificates for finite polyhedral systems whenever the corresponding finite, algebraic, torsor, or convex structure is included in the isomorphism data.*

Proof. Finite incidence certificates transport by relabelling the referenced objects, arrows, and local elements. In an affine presentation, a certificate of inconsistency is a finite linear combination of equations with zero left-hand side and nonzero right-hand side; affine coordinate and equation isomorphisms preserve that identity. In a bijective network, an isomorphism conjugates the loop holonomies, and conjugation preserves common fixed-point emptiness. In a finite polyhedral presentation, a Farkas certificate is a dual vector satisfying linear equalities and inequalities; declared affine coordinate changes, constraint permutations, and positive row scalings preserve those relations. The same argument applies to any certificate scheme functorially defined from the preserved local structure. $\square$

Remark 12.10. Noninvertible maps require separate soundness theorems for each certificate language. A map into a more permissive system can send an unsatisfiable source to a satisfiable target. Conversely, a quotient can hide the local structure needed to identify a minimal certificate. Certificate preservation is therefore a stronger requirement than diagnostic preservation.

12.4 Terminal compression as an idempotent functor

At a fixed tolerance, define a terminal-presentation functor $\text{Term} : \mathbf{Nested} \rightarrow \mathbf{Pres}_1$ by sending $X_0 \supseteq X_1 \supseteq X_2$ to the terminal-category presentation with assignment set X_0 , sole local admissibility set X_0 , and realised effective and bounded images X_1 and X_2 .

Theorem 12.11 (Terminal retraction). *At fixed tolerance,*

$$\Gamma \circ \text{Term} = \text{Id}_{\mathbf{Nested}}.$$

Consequently $\text{ExtComp} = \text{Term} \circ \Gamma$ is idempotent whenever it is regarded as an endofunctor on a category containing the relevant context shapes:

$$\text{ExtComp}^2 = \text{ExtComp}.$$

For a fixed nonterminal base $\mathcal{C}$, the correctly typed statement is $\text{ExtComp}_1 \circ \text{ExtComp}_{\mathcal{C}} = \text{ExtComp}_{\mathcal{C}}$.

Proof. A terminal-category diagram has no nontrivial matching equation, so its set-theoretic global-family space is its sole local set, namely X_0 . Its effective and bounded realised images are X_1 and X_2 by construction. Thus $\Gamma(\text{Term}(X_0 \supseteq X_1 \supseteq X_2)) = X_0 \supseteq X_1 \supseteq X_2$. Idempotence follows by substitution:

$$(\text{Term}\Gamma)(\text{Term}\Gamma) = \text{Term}(\Gamma\text{Term})\Gamma = \text{Term}\Gamma.$$

The final assertion records that terminal compression changes the base category from $\mathcal{C}$ to $\mathbf{1}$. $\square$

The two identities admit the standard retraction picture. On a nested triple $N = X_0 \supseteq X_1 \supseteq X_2$, the identity $\Gamma \circ \text{Term} = \text{Id}$ is the retraction square

$$\begin{array}{ccc} N & \xrightarrow{\text{Term}} & \text{Term}(N) \\ \parallel & & \downarrow \Gamma \\ N & \xlongequal{\quad} & N \end{array}$$

and the idempotence $\text{ExtComp}^2 = \text{ExtComp}$ is the triangle

$$\begin{array}{ccc} & P & \\ \text{ExtComp} \swarrow & & \searrow \text{ExtComp} \\ \text{ExtComp}(P) & \cdot \text{ExtComp} \cdot & \text{ExtComp}(P) \end{array}$$

whose bottom edge acts as the identity on the compressed image, as the typed identity of the theorem records.

Proposition 12.12 (Decompression fibres of spectra). *Let a compressed spectrum have the form (S_D, S_D, S_K, S_R) . At the spectral level, its possible decompressions are exactly the quadruples*

$$(S_L, S_D, S_K, S_R)$$

where S_L is upward closed and $S_D \subseteq S_L$. Thus the forgotten spectral datum is precisely the possible D-gap $S_L \setminus S_D$.

Proof. Corollary 6.12 shows that terminal compression replaces T_L by T_D and leaves the other three success spectra unchanged. Conversely, the nesting theorem requires only that S_L be an upward-closed superset of S_D at the level of spectra. Certificate-level decompression is strictly richer, because many nonisomorphic local-incidence diagrams can realise the same success spectra. $\square$

Theorem 12.13 (Realisation of abstract success spectra). *Let E be a tolerance poset and let*

$$S_R \subseteq S_K \subseteq S_D \subseteq S_L$$

be upward-closed subsets of E . In the unrestricted set-based category of tolerance-filtered presentations, there is a presentation P with

$$(T_L(P), T_D(P), T_K(P), T_R(P)) = (S_L, S_D, S_K, S_R).$$

Proof. Use the cospan category $c_0 \rightarrow u \leftarrow c_1$. Let

$$\mathcal{A}(c_0) = \{\ell_0, d, k, r\}, \quad \mathcal{A}(c_1) = \{\ell_1, d, k, r\}, \quad \mathcal{A}(u) = \{0, 1, d, k, r\},$$

with restrictions $\ell_0 \mapsto 0$, $\ell_1 \mapsto 1$, and d, k, r mapped to the same-named elements of $\mathcal{A}(u)$. At tolerance ε , put

$$\begin{aligned} \mathcal{A}_{\text{loc}, \varepsilon}(c_0) &= \{\ell_0 : \varepsilon \in S_L\} \cup \{d : \varepsilon \in S_D\} \cup \{k : \varepsilon \in S_K\} \cup \{r : \varepsilon \in S_R\}, \\ \mathcal{A}_{\text{loc}, \varepsilon}(c_1) &= \{\ell_1 : \varepsilon \in S_L\} \cup \{d : \varepsilon \in S_D\} \cup \{k : \varepsilon \in S_K\} \cup \{r : \varepsilon \in S_R\}, \\ \mathcal{A}_{\text{loc}, \varepsilon}(u) &= \{0, 1 : \varepsilon \in S_L\} \cup \{d : \varepsilon \in S_D\} \cup \{k : \varepsilon \in S_K\} \cup \{r : \varepsilon \in S_R\}. \end{aligned}$$

Here $\{x : \phi\}$ denotes $\{x\}$ when ϕ holds and $\emptyset$ otherwise, and $\{0, 1 : \phi\}$ denotes $\{0, 1\}$ when ϕ holds and $\emptyset$ otherwise. All local sets are nonempty exactly on S_L . If $\varepsilon \in S_L \setminus S_D$, the only available special choices are $\ell_0, \ell_1, 0, 1$, and they cannot form a matching family because the two restrictions require the u -component to be both 0 and 1. If $\varepsilon \in S_D$, the matching family (d, d, d) exists. Thus set-theoretic success is exactly S_D .

Let the effective realised image be declared to be exactly the subset consisting of the matching families (k, k, k) and (r, r, r) , and let the bounded realised image be declared to be exactly (r, r, r) . In particular, the set-theoretic family (d, d, d) is not in the effective image unless it is explicitly included. Since $S_R \subseteq S_K \subseteq S_D$, effective success occurs exactly on S_K and bounded success exactly on S_R . The local sets are monotone because all four S -sets are upward closed. $\square$

Remark 12.14. Theorem 12.13 is deliberately set-based. In represented or finite-parametric subcategories, only spectra admitting the corresponding descriptions are realisable. Thus the result identifies the abstract degrees of freedom forgotten by compression; it is not a uniform effective construction for arbitrary unrepresented subsets of a tolerance poset.

12.5 Empirical-support packaging as a functor on isomorphisms

Definition 12.15 (Empirical-support isomorphism). An isomorphism between finite empirical support models consists of a measurement bijection carrying maximal contexts to maximal contexts, outcome bijections on measurements, and exact preservation of the support relation in every maximal context.

Theorem 12.16 (Functorial support packaging on the support groupoid). *The downward-closed packaging of Definition 9.5 is functorial on the groupoid of finite empirical support models and support isomorphisms. It preserves and reflects global supported assignments and therefore preserves and reflects strong contextuality.*

Proof. A support isomorphism sends every context D in the downward closure to its image context and acts coordinatewise on assignments. Coordinatewise maps commute with restriction, and support preservation on maximal contexts implies support preservation on all restricted supports. Hence the packaging is carried to an isomorphic constrained diagram. A global assignment g is supported on every maximal context exactly when the transported global assignment is supported on every transported maximal context. Strong contextuality is the nonexistence of such a global assignment with all local supports nonempty. $\square$

12.6 Noninvertible empirical reindexing doctrines

Support isomorphisms preserve all structure. Noninvertible maps do not have a single canonical variance. The three basic doctrines are direct image, inverse image, and saturation. Each preserves a different part of the local-to-global problem.

Definition 12.17 (Outcome coarse-graining and direct image). Fix a finite measurement scenario with contexts M , let $X = \bigcup_{C \in M} C$ be the measurement set, and let $q_x : O_x \rightarrow O'_x$ be outcome maps. For a support model $S = (S_C)_{C \in M}$, define the direct image support model by

$$(q_* S)_C = q_C(S_C) \subseteq \prod_{x \in C} O'_x,$$

where $q_C = \prod_{x \in C} q_x$.

Proposition 12.18 (Direct image preserves global sections forward). *If $g \in \prod_x O_x$ is a global supported assignment for S , then $q_X(g) \in \prod_x O'_x$ is a global supported assignment for $q_* S$. Thus direct image preserves noncontextuality forward. It need not preserve strong contextuality.*

Proof. For every context C , $g|_C \in S_C$. Applying q_C gives $q_X(g)|_C = q_C(g|_C) \in q_C(S_C) = (q_* S)_C$. Hence $q_X(g)$ is global for the direct image. For failure of preservation of strong contextuality, take any strongly contextual support model and map every outcome set to a singleton. The direct image then has the unique global assignment. $\square$

Definition 12.19 (Inverse-image reindexing). For a target support model $T = (T_C)_C$ on outcome sets O'_x and outcome maps $q_x : O_x \rightarrow O'_x$, define

$$(q^* T)_C = q_C^{-1}(T_C) \subseteq \prod_{x \in C} O_x.$$

Proposition 12.20 (Inverse image and global sections). *There is an equality of global-section sets*

$$\Gamma(q^*T) = q_X^{-1}(\Gamma(T)).$$

*Consequently, if T is strongly contextual then q^*T has no global section. If all q_x are surjective and the local supports of T are nonempty, then the local supports of q^*T are nonempty as well, so strong contextuality of T pulls back to a $\mathbf{D}$ outcome for q^*T .*

Proof. A global assignment $g \in \prod_x O_x$ lies in $\Gamma(q^*T)$ exactly when $g|_C \in q_C^{-1}(T_C)$ for every context C , equivalently when $q_X(g)|_C = q_C(g|_C) \in T_C$ for every C . This is exactly $q_X(g) \in \Gamma(T)$. Surjectivity of each q_x makes each product map q_C surjective, so preimages of nonempty local supports are nonempty. $\square$

Definition 12.21 (Global-section saturation). For a support model S , define its global-section saturation by

$$S_C^{\text{sat}} = \{g|_C : g \in \Gamma(S)\}.$$

Proposition 12.22 (Saturation preserves global sections and erases certificates). *For every finite support model,*

$$\Gamma(S^{\text{sat}}) = \Gamma(S), \quad (S^{\text{sat}})^{\text{sat}} = S^{\text{sat}}.$$

If S is strongly contextual, then every nonempty-context support in S^{sat} is empty. Thus saturation preserves global-section emptiness but changes a support-level $\mathbf{D}$ presentation into a local failure.

Proof. Every $g \in \Gamma(S)$ restricts into S_C^{sat} for each context, so $\Gamma(S) \subseteq \Gamma(S^{\text{sat}})$. Conversely, $S_C^{\text{sat}} \subseteq S_C$ whenever S is the downward support packaging, so a global assignment supported by S^{sat} is supported by S . The same equality of global sections gives idempotence. If $\Gamma(S) = \emptyset$, then no restrictions $g|_C$ exist, so each nonempty-context saturated support is empty. $\square$

Remark 12.23. Direct image, inverse image, and saturation are all natural operations, but they answer different questions. Direct image models coarse-graining of recorded outcomes; inverse image models refinement of outcome labels; saturation records only what is extendable to a global assignment. None is the canonical noninvertible morphism of empirical models without an explicitly chosen semantics.

12.7 Distributed selectors and blind-order arenas as a groupoid equivalence

Let $\mathbf{Dist}_{\text{red}}$ be the groupoid of finite distributed selector problems whose observation sets are the active observation images, with isomorphisms given by bijections of worlds, agents, observations, and choices preserving interfaces, scopes, and relations. Let $\mathbf{Blind}$ be the groupoid of finite blind-order one-shot arenas in strict normal form, with isomorphisms preserving the rooted tree, world branches, level order, information sets, choice sets, and loss tables.

Theorem 12.24 (Reduced distributed selectors and blind-order arenas). *The constructions $\mathbf{N}$ and $\mathbf{Dist}$ of Theorem 11.43 restrict to an equivalence of groupoids*

$$\mathbf{N} : \mathbf{Dist}_{\text{red}} \rightleftarrows \mathbf{Blind} : \mathbf{Dist}.$$

They preserve pure strategies, worldwide zero-one loss, and pure-strategy capability.

Proof. In a reduced distributed problem, every observation labels an information set in $N(P)$. A pure strategy in the arena is therefore exactly one map $\text{im}(I_j) \rightarrow C_j$ for each agent. Terminal loss was defined to be the distributed validity predicate. Reading a strict normal-form arena back recovers its worlds, agent order, observation labels, choices, scopes, and terminal relations. Structure-preserving isomorphisms act by the same relabellings on both sides, so the object-level bijections assemble into an equivalence of groupoids. $\square$

Remark 12.25. This is a capability and loss equivalence. It is not automatically a diagnostic or certificate equivalence between an incidence presentation and a full arena patch presentation. Such preservation requires transporting the local-incidence presentation as additional data.

12.8 Arena patch sheaves under arena isomorphisms

Theorem 12.26 (Patch-sheaf functoriality under arena isomorphisms). *An isomorphism of finite imperfect-information arenas preserving worlds, rooted tree structure, node types, nature maps, information sets, choice sets, and loss tables induces an isomorphism of information-saturated patch sites and an isomorphism of the strategy sheaves carrying $\mathcal{V}_{\varepsilon, \mathcal{I}}$ onto $\mathcal{V}'_{\varepsilon, \mathcal{I}'}$. Hence it preserves and reflects global activated-valid strategies and pure ε -capability.*

Proof. The node bijection carries prefix-closed information-saturated patches to such patches and commutes with intersections and union covers. Choice-set bijections act coordinatewise on partial strategies and commute with restriction. Because worlds, nature maps, selected edges, and losses are preserved, a node is reached under a partial strategy exactly when its image is reached under the transported partial strategy, and the loss values agree. Thus activated validity is preserved and reflected. The inverse arena isomorphism supplies the inverse sheaf map. $\square$

12.9 Constraint-system composition

Proposition 12.27 (Conjunction and disjoint union). *For finite CSP presentations over a common variable set, occurrence-preserving conjunction has solution set*

$$\text{Sol}(P \wedge Q) = \text{Sol}(P) \cap \text{Sol}(Q).$$

For presentations over disjoint variable sets,

$$\text{Sol}(P \sqcup Q) \cong \text{Sol}(P) \times \text{Sol}(Q).$$

If effective and bounded implementation objects factor as products in the disjoint-union case, the same product law holds at the effective and bounded levels.

Proof. A valuation satisfies the occurrence-preserving conjunction exactly when it satisfies both original lists of constraints. For disjoint union, a valuation on the union of variables is uniquely a pair of valuations, one on each side, and constraints do not mix the sides. The implementation-level statement is the same argument applied to the assumed product realisation maps and product resource filters. $\square$

Proposition 12.28 (Shared interfaces and hiding). *In this proposition, $\text{Sol}(P)$ denotes the set-theoretic global-family space $\Gamma_{\text{set}}(P)$ of a constraint presentation. If presentations P and Q have solution maps to a shared interface presentation I and the pullback presentation is formed semantically, then*

$$\text{Sol}(P \times_I Q) \cong \text{Sol}(P) \times_{\text{Sol}(I)} \text{Sol}(Q).$$

If a set of internal variables is hidden, the visible solution relation is the projection of the full solution relation. Hiding preserves set-level nonemptiness but may erase diagnostics and certificates, and implementation-level equality requires uniform witness lifting.

Proof. The pullback presentation imposes exactly the condition that a solution of P and a solution of Q have the same interface value; this is the defining property of the fibre product. Hiding existentially quantifies the internal variables, hence takes the image of the full solution relation under projection. Projection sends nonempty sets to nonempty sets, but it need not preserve local incidence, minimal unsatisfiable subdiagrams, or computable access to a hidden witness. $\square$

12.10 Parallel, sequential, and guarded-feedback composition

Proposition 12.29 (Parallel capability under independent implementations). *Suppose $P \otimes Q$ is formed with independent worlds, disjoint context components, componentwise success, and an implementation-pairing operation whose realisation is the product of realised families. If P and Q are effectively capable, then $P \otimes Q$ is effectively capable. If the paired implementation has resource cost bounded by $b_P \oplus b_Q$, bounded capability composes at the combined budget.*

Proof. Pair the two successful implementations. The realised global family is valid on each component, hence valid for the componentwise success predicate of the parallel presentation. The resource claim is exactly the assumed cost inequality. $\square$

Remark 12.30. The converse and exact product laws require projection of joint implementations back to component implementations. Shared memory, shared randomness, preprocessing, amortisation, or nonfactorising budgets can invalidate such projections. Parallel composition is therefore doctrine-dependent at the implementation and resource levels.

12.11 Resource-overhead calculus

Let $(B, \leq, \oplus, 0)$ be an ordered resource monoid. A budgeted implementation doctrine assigns monotone implementation classes

$$\text{Impl}_b(P) \subseteq \text{Impl}_{\text{eff}}(P) \quad (b \in B),$$

and corresponding realised spaces

$$\Gamma_b^\varepsilon(P) = \Gamma_{\text{set}}^\varepsilon(P) \cap \sigma_{\text{eff}}(\text{Impl}_b(P)).$$

The bounded success spectrum is then the upward-closed subset

$$T_R(P) = \{(\varepsilon, b) : \Gamma_b^\varepsilon(P) \neq \emptyset\}.$$

Proposition 12.31 (Spectrum transport with overhead). *Suppose a presentation morphism $F : P \rightarrow Q$, with assignment components $F_c = \alpha_c$, has tolerance overhead ρ , an unbudgeted effective implementation translator, and resource overhead H , meaning that*

$$\alpha_c(\mathcal{A}_{\text{loc}, P}^\varepsilon(c)) \subseteq \mathcal{A}_{\text{loc}, Q}^{\rho(\varepsilon)}(c),$$

there is a translator $\text{trans}_{\text{eff}}$ satisfying

$$\sigma_{\text{eff}}^Q(\text{trans}_{\text{eff}}(\iota)) = \alpha_*(\sigma_{\text{eff}}^P(\iota)),$$

and implementations of budget b in P are translated, with the same realisation compatibility, to implementations of budget $H(b)$ in Q . Then

$$\varepsilon \in T_\chi(P) \implies \rho(\varepsilon) \in T_\chi(Q) \quad (\chi \in \{\mathbf{L}, \mathbf{D}, \mathbf{K}\}),$$

and

$$(\varepsilon, b) \in T_{\mathbf{R}}(P) \implies (\rho(\varepsilon), H(b)) \in T_{\mathbf{R}}(Q).$$

Proof. The proof transports witnesses. Objectwise local witnesses are mapped into the larger-tolerance local sets of Q . Compatible global families are transported by the global-family functor. Effective witnesses are transported by the effective implementation translator. A budget- b witness is transported to a budget- $H(b)$ witness by the resource-overhead hypothesis. $\square$

Proposition 12.32 (Parallel resource convolution). *Assume $P \otimes Q$ has independent implementations and cost aggregation by $\oplus$: pairing a budget- b_P implementation of P with a budget- b_Q implementation of Q gives a budget- $b_P \oplus b_Q$ implementation of $P \otimes Q$. For a componentwise success predicate,*

$$\bigcup_{b_P \oplus b_Q \leq b} \Gamma_{b_P}^\varepsilon(P) \times \Gamma_{b_Q}^\varepsilon(Q) \subseteq \Gamma_b^\varepsilon(P \otimes Q).$$

If every budget- b implementation of $P \otimes Q$ projects to component implementations with budgets b_P, b_Q satisfying $b_P \oplus b_Q \leq b$, then the inclusion is equality.

Proof. The inclusion is the stated pairing operation and the componentwise success criterion. For equality, take a budget- b composite witness and project it to component witnesses. By hypothesis their budgets combine to at most b , and the componentwise success predicate gives membership in the corresponding product on the left. $\square$

Remark 12.33. The operation $\oplus$ is doctrine-specific. It may be addition for total work, maximum for ideal parallel time, a communication-cost sum, or a vector-valued resource operation. The abstract framework supplies the monotone transport law; resource additivity must be proved in the chosen implementation model.

12.12 Minimax thresholds and resource margins

Loss-induced presentations give a quantitative version of the four-level diagnostic. Suppose each local condition is generated by a loss map $L_c : \mathcal{A}(c) \rightarrow [0, \infty]$ via

$$\mathcal{A}_{\text{loc}, \varepsilon}(c) = \{a \in \mathcal{A}(c) : L_c(a) \leq \varepsilon\}.$$

Let $\mathcal{S}_{\text{set}}$ be the structurally compatible family space, and let $\mathcal{S}_{\text{eff}}$ and $\mathcal{S}_{\mathcal{B}}$ be its intersections with the effective and bounded realised images. Define

$$\theta_{\mathbf{L}} = \sup_c \inf_{a \in \mathcal{A}(c)} L_c(a),$$

and, for $X \in \{\text{set}, \text{eff}, \mathcal{B}\}$,

$$\theta_X = \inf_{s \in \mathcal{S}_X} \sup_c L_c(s_c),$$

with the convention $\inf \emptyset = +\infty$.

Proposition 12.34 (Minimax tower). *For a loss-induced presentation,*

$$\theta_{\mathbf{L}} \leq \theta_{\text{set}} \leq \theta_{\text{eff}} \leq \theta_{\mathcal{B}}.$$

Moreover, endpoint membership is governed by attainment: if the relevant local or global infimum is attained at value at most ε , then ε lies in the corresponding success set; if no minimiser exists, the threshold may be an infimum not belonging to the success set.

Proof. For every compatible family s , the component s_c is a candidate in $\mathcal{A}(c)$, so

$$\inf_{a \in \mathcal{A}(c)} L_c(a) \leq L_c(s_c)$$

for every c . Taking the supremum over c and then the infimum over compatible families gives $\theta_{\mathbf{L}} \leq \theta_{\text{set}}$. The remaining inequalities follow from $\mathcal{S}_{\mathcal{B}} \subseteq \mathcal{S}_{\text{eff}} \subseteq \mathcal{S}_{\text{set}}$. The statement about endpoints is the distinction between an infimum and a realised minimiser. $\square$

Remark 12.35 (Interpreting the gaps). The differences

$$\theta_{\text{set}} - \theta_{\mathbf{L}}, \quad \theta_{\text{eff}} - \theta_{\text{set}}, \quad \theta_{\mathcal{B}} - \theta_{\text{eff}}$$

when finite measure, respectively, the tolerance cost of gluing, effective realisation, and bounded realisation. They are threshold gaps, not an additive physical decomposition. Additivity requires an additional theorem in the chosen loss and resource doctrine.

Remark 12.36 (Relation to the single-observer degrees). The quantities $d_{\text{pt}}, d_{\text{obs}}, d_{\text{eff}}, d_{\mathcal{B}}$ of Definition 8.1 are the single-observer prediction-game form of this minimax tower. In that special case the local pointwise optimum is d_{pt} , the set-theoretic observation-uniform optimum is d_{obs} , and the realised optima are d_{eff} and $d_{\mathcal{B}}$. The symbols θ are used here for the more general multi-context threshold presentation.

Proposition 12.37 (Affine overhead composition). *If two morphisms have affine tolerance overheads*

$$\rho_F(\varepsilon) = L_F \varepsilon + \eta_F, \quad \rho_G(\varepsilon) = L_G \varepsilon + \eta_G,$$

with $L_F, L_G \geq 0$, then the composite overhead is

$$\rho_{G \circ F}(\varepsilon) = L_G L_F \varepsilon + L_G \eta_F + \eta_G.$$

The same formula applies to scalar affine cost overheads.

Proof. The composite overhead is the composite function $\rho_G \circ \rho_F$:

$$\rho_G(\rho_F(\varepsilon)) = L_G(L_F \varepsilon + \eta_F) + \eta_G = L_G L_F \varepsilon + L_G \eta_F + \eta_G.$$

$\square$

Proposition 12.38 (Sequential section composition). *Let P be a first-stage presentation and let $Q[y]$ be continuation presentations indexed by realised first-stage interface values. If s is a valid section of P and t_y is a compatible valid continuation section for each reachable y , then the sequential composite has a valid section. Effective sequential capability additionally requires a uniformly effective map $y \mapsto t_y$ and an implementation language closed under the corresponding bind operation.*

Proof. The set-theoretic composite section is the dependent pair consisting of the first-stage section and the family of continuation sections on the interface values reached by it. Compatibility is imposed by the definition of the sequential composite. For the effective statement, the first-stage realiser computes the interface value and then invokes the uniformly supplied continuation realiser; this is exactly closure under bind. $\square$

Proposition 12.39 (Guarded feedback). *Let $(J, \prec)$ be a finite well-founded dependency order. Suppose each component j has a nonempty set of admissible outputs for every valid predecessor record. Then a global feedback section exists. If each admissible output set is a singleton, the section is unique. Effective and bounded versions require, respectively, uniform effective selection from the local output sets and a resource recurrence bounding the well-founded evaluation.*

Proof. Choose any linear extension of the finite order. At stage j , all predecessor outputs have already been chosen, and the local admissible set is nonempty by hypothesis; choose an element of it. This constructs a global section by finite induction. If every local set is a singleton, the choice at each stage is forced. The effective and bounded statements add precisely the stated effective selector and cost recurrence to the same induction. $\square$

12.13 Finite tabular verifier games

The certificate transport of Theorem 12.9 also covers the finite tabular certificates; the following representation turns any finite tabular constrained presentation into a finite verifier game, so that certificates and refutations become winning strategies and their absence.

Theorem 12.40 (Finite verifier-game representation). *Every finite tabular constrained presentation has a finite verifier-game presentation with the same set-theoretic global-family space. If the effective and bounded realised images are transported as declared subsets of the same strategy space, the three nested global spaces are preserved.*

Proof. Let the original finite presentation have context objects c , assignment sets $\mathcal{A}(c)$, local sets $\mathcal{A}_{\text{loc}}(c)$, and restriction maps $\mathcal{A}(f) : \mathcal{A}(c) \rightarrow \mathcal{A}(c')$. The verifier game has one pure strategy coordinate $x_c \in \mathcal{A}(c)$ for every context object. It has one local test world for each object c , accepting exactly when $x_c \in \mathcal{A}_{\text{loc}}(c)$, and one compatibility test world for each arrow $f : c \rightarrow c'$, accepting exactly when $\mathcal{A}(f)(x_c) = x_{c'}$. A pure strategy wins all verifier worlds exactly when the family $(x_c)_c$ is locally admissible and satisfies every restriction equation, i.e. exactly when it is an element of Γ_{set} for the original presentation. Transporting the effective and bounded images as subsets of this common strategy space preserves the nested intersections defining Γ_{eff} and $\Gamma_{\mathcal{B}}$. $\square$

Remark 12.41. The construction is a finite tabular representation theorem, not a physical realisation theorem. It preserves prediction-relevant finite data by explicitly turning local admissibility and restriction equations into verifier tests. It may change certificate size or presentation geometry unless the incidence decoration is retained.

12.14 Source-domain placement principle

Definition 12.42 (Source-domain placement). A placement of a source-domain theorem in the ECD taxonomy consists of a source class, a target presentation class, a translation from source objects to target presentations, declared preservation properties, and a proof that the translated object has the asserted diagnostic outcome. Without these data, no diagnostic placement is asserted. Table 7 records the current status of the representative source-domain placements.

Table 7: Status of representative source-domain placements in the monograph.

Source domain	Status here	Additional structure needed for broader claims
Finite Bell scenarios	finite coupling, Farkas, robustness, and repair theory developed	fixed scenario, setting distribution for finite-sample claims, declared repair model
Finite empirical support models	support packaging and support-isomorphism functoriality developed	noninvertible reindexing doctrine for broader functoriality
Finite source integration	distributed-selector representation and $\#P$ counting lower bound developed	chosen repair/edit-cost doctrine for database repair claims
Protocol complexes	finite section representation developed	process model, topology, carrier theorem, and protocol legality for impossibility placements
Decentralised control	representation target	observation histories, measurability, controller architecture, and performance objective
Gödel-style diagonals	theoremhood selector placement developed	truth variants require soundness/Tarski assumptions and internal representability data
No-cloning/no-broadcasting	finite-dimensional quantum placement developed	broader process theories require tensor, states, channels, and success predicates
Quantum instruments	representation target	completely positive maps, POVMs/instruments, compatibility, and tensor/discarding structure

12.15 Representation targets and boundaries

Programme 12.43 (Representation and application targets). The following directions are mathematically meaningful once their source-domain semantics and morphisms are fixed.

- (i) Extensions of the finite verifier-game representation theorem to richer classes of represented games and operational systems.
- (ii) Functorial probabilistic and quantum variants using affine or process-theoretic assignment categories, with physical realisability separated from set-theoretic existence.
- (iii) Topological impossibility placements for protocol-complex tasks, building on the section representation theorem by adding the process model, carrier theorem, and protocol legality.
- (iv) Decentralised-control and synthesis presentations with observation histories as contexts, nonanticipation as structural admissibility, and controller languages as implementation images.
- (v) Further Gödel-style diagonal placements beyond theoremhood, formulated as reductions from candidate deciders with explicit distinctions between truth, consistency, soundness, and internal representability.
- (vi) Process-theoretic extensions of the finite-dimensional no-cloning and no-broadcasting placements, with physical channels forming an implementation or physical-realisation layer rather than a local fixed-point contradiction.
- (vii) Reverse-mathematical and Weihrauch classifications for represented causal selection problems beyond the bounded-height examples, with the input representation fixed in each statement.

12.16 Structural spine theorem

The monograph-level structure can be summarised as a single spine: sections are functorial, diagnostics require the obstruction datum, certificates require local incidence, and composition lifts only under explicit implementation operations.

Theorem 12.44 (Structural spine). *Within the classes and hypotheses fixed in this section:*

- (i) *admissibility-preserving natural transformations induce morphisms of nested global-family triples;*
- (ii) *base isomorphisms of presentations preserve and reflect the complete first-failure diagnostic;*
- (iii) *certificate isomorphisms transport finite local, parity, affine, holonomy, and Farkas certificates when the relevant structure is included;*
- (iv) *terminal compression is an idempotent compression of presentations onto terminal presentations and replaces the local-success spectrum by the set-theoretic success spectrum;*
- (v) *support packaging, blind-order normal forms, and arena patch sheaves are functorial under their stated isomorphism notions;*
- (vi) *conjunction, disjoint union, pullback, hiding, parallel composition, sequential composition, and guarded feedback obey the section laws stated above, with effective and bounded versions exactly when the required realiser operations and resource recurrences are part of the doctrine.*

Proof. Clause (i) is Theorem 12.2. Clause (ii) is Theorem 12.5. Clause (iii) is Theorem 12.9. Clause (iv) is Theorem 12.11 together with Corollary 6.12. Clause (v) combines Theorems 12.16, 12.24, and 12.26. Clause (vi) is the collection of Propositions 12.27, 12.28, 12.29, 12.38, and 12.39. $\square$

Remark 12.45. The theorem is structural rather than classificatory. It identifies the maps and operations under which the section spaces, diagnostics, certificates, and implementation filters are transported. It also marks the boundary of the abstract theory: whenever a composition law mentions effective or bounded success, the corresponding operation on realisers and its cost law must be specified.

12.17 Finite protocol-complex section representation

A finite chromatic complex can be read as a presentation of distributed views. Let K be a finite simplicial complex whose vertices are coloured by process identities, with at most one vertex of each colour in each simplex. Let O be a finite output complex with the same colour set. A task carrier assigns to each simplex $\sigma \in K$ a subcomplex $\Delta(\sigma) \subseteq O$ on the same set of colours, monotone under faces.

Definition 12.46 (Protocol-complex presentation). The context category is the face poset of K , with arrows from a simplex to its faces. For a simplex σ , let $\mathcal{A}(\sigma)$ be the set of colour-preserving vertex maps from σ to O . Restriction is restriction of maps to faces. The local admissibility set $\mathcal{A}_{\text{loc}}(\sigma)$ consists of those maps whose image simplex lies in $\Delta(\sigma)$.

Theorem 12.47 (Protocol-complex section theorem). *Compatible admissible global families of the protocol-complex presentation are in canonical bijection with chromatic simplicial maps*

$$f : K \longrightarrow O$$

carried by Δ , i.e. satisfying $f(\sigma) \subseteq \Delta(\sigma)$ for every simplex $\sigma \in K$.

Proof. A compatible global family gives, on each vertex v of K , an output vertex of the same colour: take the component assigned on the singleton simplex $\{v\}$. Compatibility forces the assignment on any simplex σ to be the restriction of these vertex assignments, so the vertex map extends to a chromatic simplicial map. Local admissibility at σ is exactly the carrier condition $f(\sigma) \subseteq \Delta(\sigma)$. Conversely, any carried chromatic simplicial map restricts to an admissible local assignment on each simplex, and these restrictions are compatible on faces. $\square$

Proposition 12.48 (Functoriality under chromatic carrier isomorphisms). *An isomorphism of finite protocol-complex tasks, consisting of chromatic simplicial isomorphisms of input and output complexes that commute with the task carrier, induces a certificate isomorphism of the associated presentations and therefore preserves the existence or nonexistence of carried decision maps.*

Proof. The simplicial isomorphism sends faces to faces and commutes with taking faces, so it is an isomorphism of context categories. The output isomorphism transports colour-preserving local maps bijectively and preserves the carrier subcomplexes by hypothesis. Hence local admissibility, restrictions, and global sections are transported isomorphically. $\square$

Remark 12.49. Topological impossibility theorems for consensus and set agreement become certificates for absence of carried chromatic maps only after the process model, protocol complex, carrier map, and protocol legality have been fixed. For a fixed finite complex, all candidate maps are finite tables; effectivity and resource placements require a uniform protocol family or a restricted implementation doctrine.

12.18 Gödel theoremhood selector placement

The theoremhood placement is an effectivity obstruction, not a local contradiction. The local choices are singletons, so a set-theoretic selector exists. What fails is computable realisation.

Definition 12.50 (Theoremhood selector). Let T be a computably axiomatised theory whose theorem set

$$\text{Thm}_T = \{n : n \text{ codes a theorem of } T\}$$

is not decidable. The theoremhood selector has observations $n \in \mathbb{N}$, choice set $\{0, 1\}$, and local admissibility sets

$$\mathcal{A}_{\text{loc}}(n) = \{\chi_{\text{Thm}_T}(n)\}.$$

Effective policies are total computable binary functions.

Theorem 12.51 (Theoremhood selector has outcome $\mathbf{K}$). *For every computably axiomatised theory T with undecidable theorem set, the theoremhood selector has a set-theoretic global section and no computable global section. Hence its diagnostic outcome is $\mathbf{K}$ for the computable realisation model.*

Proof. Every local set is the singleton containing the theoremhood bit, so the characteristic function χ_{Thm_T} is a set-theoretic global section. If a computable global section existed, it would compute χ_{Thm_T} and decide the theorem set of T , contrary to the hypothesis. $\square$

Remark 12.52 (Assumption hygiene). Theorem 12.51 concerns theoremhood, not truth. It uses an external undecidability hypothesis about the theorem set. For standard consistent effectively axiomatised extensions of enough arithmetic, this is the usual undecidability theorem. The local singleton predicate is not decidable in this presentation; the primitive-recursive halting-stage construction of Theorem 10.32 gives a different effective syntax when decidable structural maps are required. Truth in the standard model requires different assumptions, and internal representability of a candidate decider is distinct from external correctness. Gödel, Rosser, and Tarski-style arguments should be placed only after specifying which predicate is being selected and which consistency, soundness, or representability assumptions are being used.

12.19 Quantum process restrictions: no-cloning and no-broadcasting

Quantum process restrictions require a process doctrine. In finite-dimensional quantum theory the assignments are states or channels, while physically realised implementations are completely positive trace-preserving maps. The diagnostic placement depends on which data are treated as local constraints and which are treated as implementation restrictions.

Definition 12.53 (Quantum cloning presentation). Let $\mathcal{H}$ be a finite-dimensional Hilbert space and let S be a finite set of states. A channel-level cloning requirement asks for one quantum channel

$$\Phi : \mathcal{D}(\mathcal{H}) \rightarrow \mathcal{D}(\mathcal{H} \otimes \mathcal{H})$$

such that $\Phi(\rho) = \rho \otimes \rho$ for every $\rho \in S$.

Theorem 12.54 (Two-state pure no-cloning). *No quantum channel exactly clones two distinct nonorthogonal pure states [25].*

Proof. Let $0 < |\langle \psi, \phi \rangle| < 1$ and suppose a channel clones both pure states. Choose a Stinespring isometry for the channel. Since the required output on each input is pure, the environment must factor, so for some unit vectors e_ψ, e_ϕ ,

$$V\psi = \psi \otimes \psi \otimes e_\psi, \quad V\phi = \phi \otimes \phi \otimes e_\phi.$$

Taking inner products and using that V is an isometry gives

$$\langle \psi, \phi \rangle = \langle \psi, \phi \rangle^2 \langle e_\psi, e_\phi \rangle.$$

Taking absolute values yields $c \leq c^2$ for $c = |\langle \psi, \phi \rangle|$, a contradiction when $0 < c < 1$. $\square$

Proposition 12.55 (Two ECD placements of no-cloning). *The no-cloning obstruction has at least two natural ECD presentations.*

- (i) *If arbitrary set maps on states are allowed at the set level and quantum channels form the physical or effective implementation image, then exact cloning fails at that implementation layer.*
- (ii) *If the assignment space consists of quantum channels and separate local contexts require cloning of individual states, then each one-state local context is inhabited, but the nonorthogonal two-state family has no common global channel; this is a compatibility presentation.*

Proof. For (i), the set map $\rho \mapsto \rho \otimes \rho$ exists, but Theorem 12.54 excludes its realisation by a quantum channel on a nonorthogonal pair. For (ii), a single state ρ is cloned by the constant channel $\sigma \mapsto \rho \otimes \rho$, so every one-state local context is inhabited. A global section is one channel satisfying all state constraints; Theorem 12.54 says no such section exists for two nonorthogonal pure states. $\square$

Theorem 12.56 (No-broadcasting placement). *In finite-dimensional quantum theory, a family of density matrices is exactly broadcastable by one channel if and only if the density matrices commute pairwise [26]. Thus noncommuting families give the same kind of implementation or compatibility obstruction, depending on whether physical channels are placed in the implementation layer or in the assignment space of a common-channel presentation.*

Proof. The equivalence is the standard no-broadcasting theorem. The ECD placement then follows as in Proposition 12.55: either a set-theoretic broadcasting map is compared with the image of physical channels, or local single-state broadcasting requirements are intersected through one common channel variable. $\square$

Remark 12.57. These placements do not identify no-cloning with recursion-theoretic diagonal failure. They identify which layer changes when the presentation changes. A separate physical stratum $\Gamma_{\text{set}} \supseteq \Gamma_{\text{phys}} \supseteq \Gamma_{\text{eff}} \supseteq \Gamma_{\mathcal{B}}$ can be introduced when physical realisability should be distinguished from computability.

12.20 Topological-order development principle

The extension programme is governed by a simple priority rule. Dependency order is a constraint: no module should assert an invariance, comparison, complexity classification, or empirical advantage before the source doctrine on which that claim depends has been fixed. Feasibility then orders the remaining work: among claims whose prerequisites are available, the preferred next claim is the one with the shortest path to a checkable theorem, certificate, or reproducible computation.

In practice this gives the following topological layers.

- (i) *Source doctrine.* Specify the objects under discussion: arenas, empirical models, protocol complexes, quantum channels, represented spaces, finite mechanical discretisations, or resource monoids.
- (ii) *Presentation.* Define the local data, interfaces, covers, and global section space before assigning an ECD diagnostic label.
- (iii) *Morphisms and invariants.* State functoriality, transport, or compression only after the direction of maps and the preserved data are fixed.
- (iv) *Certificates.* Every obstruction or repair claim should name the certificate transported or verified by the theorem.
- (v) *Complexity and effectivity.* Complexity claims require an input representation; computability claims require represented spaces or indices; resource claims require an ordered cost doctrine.
- (vi) *Empirical performance.* Practical advantage claims require an artifact, baselines, benchmark instances, and a rule separating certified success, certified failure, and inconclusive computation.

This order is not merely expository. It prevents a presentation-relative theorem from being read as an invariant of a bare capability problem, and it prevents a certificate theorem from being promoted to a performance claim before the corresponding computations have been carried out.

12.21 Dependency ledger for remaining foundational modules

The following ledger fixes where the remaining foundational items may safely be developed. It is ordered by dependency first and feasibility second.

Module	Missing prerequisite	Safe next claim
Repeated-move games	Player ownership, chance laws, utilities, and mixed or behavioural strategy semantics	Keep Theorem 11.35 as a pure and strategic-mixed capability theorem; develop Kuhn-style comparisons only in a game-theoretic source doctrine.
Noninvertible arena simulations	Direction of simulation, preservation of nature maps, information, choices, and losses	First define simulation morphisms; only then ask whether causal-patch sheaves preserve or reflect global capability.
Empirical reindexing	Chosen class of noninvertible maps and whether support, possibility, or probability data are primary	Use the already proved direct image, inverse image, and saturation operations as basic constructors; broader functoriality must state which global-support property is preserved.
Protocol complexes	Chromatic complexes, carrier maps, task legality, and process-model implementation layer	Prove only the representation theorem that carried chromatic decision maps are global sections; impossibility theorems require the process model.
Measurable policies and kernels	Measurable spaces, admissible kernels, loss measurability, and selection theorem hypotheses	State measurable descent only after the sigma-algebras and selection doctrine are fixed.
Quantum process theories	State spaces, channels, tensor product, discarding, and exact or approximate success equation	Retain finite no-cloning and no-broadcasting placements; add broader process restrictions only as source-domain theorems.
Resource calculus	Ordered resource monoid, attainment convention, and composition law for realisers	Apply the existing minimax and overhead lemmas only after resource addition, hiding, and parallel composition are specified.
Infinite arenas	Representation of branching, strategies, termination, and limit payoff	Keep bounded-height WKL results; do not assert an infinite-arena hierarchy before the representation is fixed.

This ledger also determines placement. General source-domain entries remain in this foundational monograph. Finite explicit encodings, proof objects, width algorithms, and mechanical relation generation belong to the companion algorithmic monograph (version 11, with its artifact appendix, version 8). A module may move from this ledger into the theorem body only when its prerequisite column has become part of the formal source doctrine.

12.22 Finite-algorithmic handoff

The finite algorithmic theory uses the same section language but has a different source doctrine. Once a finite instance is declared by variables, finite domains, explicit relation tables, observation

maps, implementation catalogues, costs, and budgets, the following statements become companion-theorem rather than foundational-theorem obligations.

- (i) A proposed global section is verified by checking each finite factor.
- (ii) A one-shot observation fibre fails exactly when the intersection of its scenario-admissible action sets is empty; a conflict certificate has at most one scenario for each action.
- (iii) Finite history-tree safety compiles to forbidden tuples by adding one forbidden tuple for each reachable unsafe path and discarding paths inconsistent with observation-uniformity.
- (iv) Explicit-table complexity statements are relative to explicit relation encoding; succinct generators, circuits, or finite-element preprocessors are separate source representations.
- (v) Width and weighted-width algorithms are certificate-producing algorithms for the finite source doctrine, not new presentation-invariant classifications of bare decision problems.

This handoff is an application of presentation relativity. The foundational monograph supplies the invariants and transport vocabulary; the companion algorithmic monograph supplies the input format, verifier grammar, algorithms, and mechanics front end. In particular, a finite solver returning a timeout supplies no ECD classification unless a checked certificate is also returned.

Proposition 12.58 (Finite certificate conservativity). *Let a finite companion instance be translated into the section presentation whose variables are the declared policy variables and whose local constraints are the declared finite factors. Any verified policy certificate is a global section of that presentation. Any verified factor-wise refutation certificate for the whole finite table instance is an obstruction to such a global section. These conclusions are relative to the declared finite source representation.*

Proof. A policy certificate assigns one declared domain element to each variable and is verified by testing that every factor sees an allowed tuple. This is precisely the definition of a global section of the associated finite presentation. A refutation certificate is required, by its verifier, to prove that every total assignment violates at least one declared factor, either directly or through a sound dynamic-programming, join-tree, or exhaustive certificate. Hence no global section exists in the declared finite presentation. The final sentence records that changing the source representation, for example by replacing exact tables with conservative finite-element inner or outer tables, changes the presentation being certified. $\square$

Remark 12.59 (Certificates as verifier-game strategies). Theorem 12.40 and Proposition 12.58 carry one shared reading. A verified policy certificate is exactly a winning pure strategy in the tabular verifier game of the cited theorem: the certificate prescribes the strategy, verification is play acceptance, and the checked factorwise tests are the verifier’s accepting worlds. A verified refutation certificate records that no pure strategy wins — every candidate loses some verifier test. The status “inconclusive” records that no winning strategy for either side was produced within the declared limits; it is the absence of a game-outcome certificate, not a third outcome. The companion’s typed certificate grammar and four-status discipline make this reading constructive, and it explains the handoff discipline above: an unfinished play decides nothing, so a bare solver timeout supplies no ECD classification.

Proposition 12.60 (Inner–outer finite-model handoff). *Suppose a source-domain computation supplies finite relations $R_i^- \subseteq R_i^{\text{true}} \subseteq R_i^+$ for every declared factor scope. Let P^- , P^{true} , and P^+ be the corresponding finite presentations. Then*

$$\Gamma(P^-) \subseteq \Gamma(P^{\text{true}}) \subseteq \Gamma(P^+).$$

Consequently an inner-model policy is a certified policy for the true finite model, while an outer-model refutation is a certified refutation for the true finite model. The mixed case $\Gamma(P^-) = \emptyset$ and $\Gamma(P^+) \neq \emptyset$ is inconclusive.

Proof. The inclusions are factorwise. If an assignment satisfies each R_i^- , then the same local row lies in R_i^{true} for every factor, so the assignment is a true finite-model section. If an assignment is a true finite-model section, each local row lies in R_i^+ , so it is an outer-model section. The two certificate consequences are the contrapositive directions appropriate to the inclusions. The final case is inconclusive because the inclusions give neither a true section nor a true refutation. $\square$

The companion algorithmic monograph proves the mechanics front-end instance of this proposition, with certified finite rows in place of the relations R_i^- , R_i^{true} , and R_i^+ , as its inner–outer soundness proposition.

Remark 12.61 (Mechanics boundary). The preceding proposition is a finite-model statement. In a mechanics source doctrine, residual or interval computations may certify rows of a discrete algebraic model. They do not by themselves certify continuum modelling adequacy, mesh convergence, or unmodelled uncertainty. Those require additional source-domain hypotheses before the finite certificate can be transported beyond the declared finite model.

Proposition 12.62 (Artifact status separation). *For a finite computational implementation of a declared ECD presentation, the statuses “certified satisfiable”, “certified unsatisfiable”, “certified optimal”, and “inconclusive” are logically distinct. Only the first three transport to mathematical claims about the declared finite presentation. The status “inconclusive”, including timeout, failed certificate verification, or unresolved numerical rows, transports no existence or nonexistence claim.*

Proof. A certified satisfiable status includes a checked section; a certified unsatisfiable status includes a checked refutation; and a certified optimal status includes both a feasible object and a checked proof that no better object exists in the declared finite search space. These are precisely the data needed to assert the corresponding finite mathematical claim. An inconclusive status is the absence of such data. Since both satisfiable and unsatisfiable instances may lead to timeouts, failed searches, or unresolved numerical rows, inconclusiveness has no sound implication in either direction. $\square$

Remark 12.63 (Baseline comparisons). A solver comparison is an empirical claim about implementations and encodings, not a theorem of effective causal descent. Such a comparison becomes meaningful only after the finite instances, baseline encodings, hardware, resource limits, and certificate-verification criteria are fixed. Without these data, the foundational theory supports certificate transport and width-sensitive algorithms but not a claim of practical superiority over SAT, CP-SAT, MILP, or other baselines.

12.23 Further development modules

The remaining modules require additional source-domain doctrines rather than only abstract section data.

Protocol complexes. A protocol-complex chapter should fix chromatic complexes, carrier maps, and task legality. The target theorem is that carried chromatic decision maps are exactly global sections of the associated presentation. Consensus and set-agreement impossibilities can then be placed only after the process model and protocol legality are included in the implementation layer.

Quantum process restrictions. The finite-dimensional quantum placement above treats exact no-cloning and no-broadcasting. Broader process theories require the same ingredients to be specified explicitly: states, channels, tensor product, discarding, and a success equation. A separate physical-realisation stratum Γ_{phys} may be appropriate between Γ_{set} and Γ_{eff} .

Logical diagonalisation. The theoremhood selector gives a scoped K placement. Further Gödel-style placements should distinguish theoremhood from truth, consistency from soundness, and external correctness from internal representability. The safe form is a reduction theorem from a candidate total decider satisfying the stated representability and correctness assumptions.

Reverse mathematics and Weihrauch degrees. The existing WKL results give exact degrees for binary fibre selection and nested prefix selection. Broader arena and patch classes require fixed input representations, cover locators, and validity encodings before exact degrees can be stated.

General minimax and resource calculus. The minimax tower and resource-overhead laws above apply once a loss doctrine, attainment convention, ordered resource monoid, and realiser operations are specified. Further domain-specific work is needed for additive physical decompositions, amortised costs, parallel repetition, and memory-sensitive resource laws.

Remark 12.64. The results in this section develop the companion algorithmic programme at the level of specified isomorphisms, fixed-base morphisms, and compositional operations with explicit hypotheses. Broader noninvertible simulations, measurable or quantum representation theorems, and resource-specific parallel laws remain source-doctrine questions rather than consequences of the abstract framework alone.

13 Conjectural extensions

The conjectures below are development targets, not asserted consequences of the preceding theorems. They are to be read under the topological-order principle of Subsection 12.20: first fix the source doctrine and morphisms, then prove preservation or reflection, and only then add complexity, effectivity, or performance claims.

Conjecture 13.1 (Structure-preserving repeated-move representation). Theorem 11.47 gives an extensional distributed compilation of every finite pure-strategy imperfect-information arena, while Proposition 11.45 gives a structure-preserving sufficient class with supplied schedules. Characterise broader repeated-move perfect-recall protocols admitting such presentations, develop expected-loss behavioural semantics only after player and nature probabilities are specified, and establish functoriality only after explicit arena and distributed-problem morphisms have been fixed.

Conjecture 13.2 (Functoriality of contextual packaging). Beyond the support-isomorphism theorem of Theorem 12.16, specify useful noninvertible morphisms of finite empirical support models and constrained diagrams for which downward-closed packaging is functorial and preserves or reflects the intended global-support criterion.

Conjecture 13.3 (Arena functoriality). Beyond the arena-isomorphism theorem of Theorem 12.26, identify noninvertible arena simulations that preserve nature dynamics, predictor information, choices, and losses in a direction strong enough to induce functorial maps of causal-patch sheaves and to reflect or preserve global capability.

Conjecture 13.4 (Effective gluing beyond bounded height). Theorem 11.80 treats bounded-height arenas with a computable cover locator and a uniformly computable matching family. Extend the result to a specified class of computably infinite, locally finite arenas with represented strategy spaces and an explicit no-infinite-play or limit-payoff semantics.

Remark 13.5 (No infinite-arena hierarchy claimed). Except for the bounded-height countable results of Definitions 11.77–11.84, the arena theorems in Section 11 are finite. Arenas with unbounded finite plays or infinite plays require termination or limit-payoff semantics, representations of branching and strategies, and separate choice and effectivity analyses. No arithmetical, hyperarithmetical, or analytic complexity classification of such arenas is asserted.

14 Conclusion

The static tier separates observation-uniform choice, distributed compatibility, effective realisation, and resource bounds. Initial-object collapse explains why a single-observer canonical presentation has no compatibility level, while distributed selectors generate CSP incidence diagrams in which D outcomes can occur. The extensional obstruction datum transports the diagnostic outcome when local success and all nested global spaces are preserved. Terminal compression shows that global spaces alone cannot preserve the distinction between L and D. Isomorphism of the full relational constraint datum preserves the solution space. Compactness and Helly theory give finite local certificates; join trees give a positive local-to-global theorem; finite source integration instantiates the compatibility diagnostic and has a #P-complete counting problem; the affine cokernel and bijective-network fixed-point criteria give complete certificates on their stated subclasses. At the probabilistic level, Bell coupling failure is equivalent to a Farkas dual certificate. The halting-stage selector space is a nonempty Π_1^0 class without computable members, with oracle capability exactly at the PA degrees. Countably nested distributed selectors realise arbitrary binary-tree path spaces; when the tree has no computable path they give an overlapping-scope K obstruction. Their input-uniform selection problem, and negatively represented binary fibre selection, are Weihrauch-equivalent to WKL. Recursive capability is Σ_3^0 -complete for the concrete primitive-recursive game language of Definition 10.31. Computational-class separation and deterministic-time resource separation are stated distinctly.

The causal tier fixes the arena independently of the strategy. Backward induction characterises finite perfect-information capability, while patch sheaves represent contingent partial strategies. Active sensing demonstrates that obstruction labels depend on whether one uses the full patch site or a branch cover. Blind-order one-shot arenas and finite distributed selectors are equivalent for pure-strategy capability and worldwide loss, but not necessarily for the presentation-relative diagnostic; supplied world schedules extend this to world-dependent move orders. More general finite pure-strategy imperfect-information arenas admit an extensional distributed compilation, although a general structure-preserving converse remains open. Simplex-valued behavioural coordinates glue as a sheaf, without being identified with shared-randomness Bell mixtures. On the Bell one-shot information skeleton, mixed-strategy marginal realisation is exactly local coupling. A finite causal poset of universally executed cuts has a unique confluent operational execution; each linear extension gives a policy-independent arena preserving aggregate loss, while contingent occurrence remains natively tree-indexed. World-only observations give an extensional distributed represen-

tation, and trace-relative records instantiate past factorisation. Under global information recall, pure action recall is capability-redundant; finite acyclic record congruences permit quotient backward induction. Finally, the height-two halting-stage arena shows that pointwise computable local sections need not form a uniformly computable family; with a computable locator and uniform local evaluator, effective gluing is computable. The common lesson is that the location of failure is part of the presentation: the same yes–no capability question may conceal local infeasibility, incompatibility of local data, noncomputability, or resource inadequacy, and the declared diagram determines which distinction is visible.

A final tier records what survives change of presentation. Capability, diagnostic, and certificate morphisms separate the three levels of comparison; terminal compression is an idempotent functor erasing precisely the D-gap; distributed and one-shot arenas agree up to groupoid equivalence; and finite tabular verifier games identify certificates and refutations with winning strategies, so that an unfinished play, like a bare solver timeout, transports no claim. Source-domain placements follow a dependency ledger under the topological-order development principle: dependency order is a constraint, and feasibility orders the remaining work. Under that principle the finite algorithmic theory is delegated rather than duplicated: the companion algorithmic monograph (version 11, with its artifact appendix, version 8) supplies the input format, typed certificate verifiers, width-based dynamic programs, and the recorded artifact.

A Domain-theoretic feedback

Proposition A.1 (Well-founded guarded recursion). *Let $(T, \prec)$ be well founded. For each $\tau \in T$, let X_τ be a set, let $\text{Dep}(\tau) \subseteq \{\sigma : \sigma \prec \tau\}$, and let*

$$F_\tau : W \times \prod_{\sigma \in \text{Dep}(\tau)} X_\sigma \rightarrow X_\tau.$$

For every world w there is a unique family $(x_\tau)_{\tau \in T}$ satisfying

$$x_\tau = F_\tau(w, (x_\sigma)_{\sigma \in \text{Dep}(\tau)}).$$

Proof. Well-founded recursion applies because every dependency of τ precedes τ . Uniqueness follows by well-founded induction from the determinism of the maps F_τ . $\square$

Unguarded feedback motivates the following domain-theoretic formulation.

Definition A.2 (Parameterized feedback operator). Let D be a pointed dcpo. A parameterized feedback operator is a map

$$\Phi : \text{Strat} \times W \times D \rightarrow D$$

such that $F_{\pi,w}(d) = \Phi(\pi, w, d)$ is Scott-continuous for every (π, w) .

Proposition A.3 (Least fixed-point semantics). *Every $F_{\pi,w}$ has the least fixed point*

$$\mu F_{\pi,w} = \bigvee_{n \geq 0} F_{\pi,w}^n(\perp).$$

Proof. This is the Kleene fixed-point theorem [2]. Scott continuity preserves the supremum of the ascending iteration chain. If x is any fixed point, induction gives $F_{\pi,w}^n(\perp) \sqsubseteq x$ for every n , so the supremum is least. $\square$

Proposition A.4 (Query-invariant feedback semantics). *Let $Q : D \rightarrow Y$ and suppose all fixed points of $F_{\pi,w}$ have the same Q -value. Then $q_{\Phi}(\pi,w) = Q(\mu F_{\pi,w})$ is independent of fixed-point choice.*

Proof. The least fixed point is itself a fixed point, and all fixed points have the same Q -image by hypothesis. $\square$

Remark A.5. With an interface and decoder, this query value defines a static validity relation to which the selector lemma applies. No computability of the least fixed point is asserted without effective representations of D and Φ .

Formal verification

The formal artifact associated with this monograph contains a 36-part reproduction gate: twenty-four core Lean files and twelve deterministic Python experiments. The Lean artifacts check the discrete selector, robust-selector, section, world-pair, obstruction, discrete-ECD, finite-representation, parity, and L/D/K/R priority-classification cores. The file `theorem_D_tri.lean` proves the parity contradiction underlying Theorem 9.2; it does not formalise the six-object category. The separate Mathlib artifact `research/mathlib/k_game.lean` verifies a co-c.e. family with nonempty fibres and no total computable selector; it is not part of the core gate.

The compactness/Helly certificates, initial-object collapse, distributed-selector and NP-completeness theorems, nested distributed effectivity, constraint-datum invariance and saturated-data recovery, obstruction transport and compression boundary, join-tree, source-integration and counting lower-bound, affine cokernel, and bijective-network holonomy theorems, categorical packaging, the binary fibre-selection and nested prefix Weihrauch classifications, finite Bell coupling and its Farkas certificate, Π_1^0 /PA-degree formulation, REC hardness reduction, Ackermann and deterministic-time separations, extensive-form backward induction, perfect- and imperfect-information subsheaf theorems, sensing, one-shot normal form, general finite pure-strategy compilation, world-scheduled realisation, causal-poset operational execution and linear-extension arena realisation, trace-relative factorisation, refinement-collapse, action-recall, arena-size, and quotient-backward-induction results, behavioural-coordinate gluing, mixed Bell realisation, bounded-height effective gluing and WKL upper bound, height-two causal effectivity, private-information team and division-of-labour results, stochastic-kernel Bell construction, and static-compilation results are proved in the text without dedicated Lean artifacts. Object-level encodings and conjectures are likewise outside the machine-checked claim. The command `make reproduce` runs the available gate. The recorded environment uses Lean 4.33.0 and Python 3.13 on Linux; the separate K-game requires the Mathlib project described in `research/mathlib/README.md`. Successful execution verifies the formal statements in the source files. It does not verify the modelling adequacy of the monograph’s additional constructions.

Artifact availability. The Lean and Mathlib sources described above—the twenty-four core Lean files, `theorem_D_tri.lean`, the separate `research/mathlib/k_game.lean` project, and the twelve deterministic Python experiments, together with the recorded Lean 4.33.0 and Python 3.13 environment—did not persist and are not part of the repository deposit. The gate was executed and its checks passed when the results were verified: the listed formal statements are therefore machine-checked, but the verifying sources are no longer available for independent re-execution or audit. The reproduction entry point available in the present repository, <https://github.com/MIKEAA2020/halting-physics>, is the companion artifact’s `make reproduce` gate, the finite-ECD

prototype of the companion algorithmic monograph (harness Python 3.12.14); it is a separate artifact, verifying the finite feasibility pipeline and its certificates, and does not re-run the Lean checks. The proofs in the text are self-contained and do not depend on the lost sources.

Manuscript content	Artifact	Scope
selector/robust/section cores	<code>theorem_S.lean</code> , <code>theorem_SE.lean</code> , <code>theorem_ESS.lean</code>	checked core
triangle	<code>theorem_D_tri.lean</code>	parity core only
finite representation	<code>theorem_Rep_fin.lean</code>	combinatorial core
co-c.e. selector	<code>mathlib/k_game.lean</code>	separate Mathlib check
extensive causal tier	none	manuscript proof
REC and time hierarchy	none	manuscript proof

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